
Stochastic Models

The previous chapter outlined likelihood analysis of some standard models. Here we turn to data in which the dependence among the observations is more complex. We start by explaining how our earlier discussion extends to Markov processes in discrete and continuous time. We then extend this to more complex indexing sets and in particular to Markov random fields, in which basic concepts from graph theory play an important role. A special case is the multivariate normal distribution, an important model for data with several responses. We give some simple notions for time series, a very widespread form of dependent data, and then turn to point processes, describing models for rare events in passing.

6.1 Markov Chains

In certain applications interest is focused on transitions among a small number of states. A simple example is rainfall modelling, where a sequence $\dots 010011 \dots$ indicates whether or not it has rained each day. Another is in *panel studies* of employment, where many individuals are interviewed periodically about their employment status, which might be full-time, part-time, home-worker, unemployed, retired, and so forth. Here interest will generally focus on how variables such as age, education, family events, health, and changes in the job market affect employment history for each interviewee, so that there are many short sequences of state data taken at unequal intervals, unlike the single long rainfall sequence. In each case, however, the key aspect is that transitions occur amongst discrete states, even though these typically are crude summaries of reality.

Example 6.1 (DNA data) When the double helix of deoxyribonucleic acid (DNA) is unwound it consists of two oriented linked sequences of the bases adenine (A), cytosine (C), guanine (G), and thymine (T). Just one chain determines a DNA sequence, because A in one sequence is always linked to T on the other, and likewise with C and G. An example is Table 6.1, which shows the first 150 bases from a sequence of

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GTATTAAATCCGTAGTCTCGAACTAACATA
TCAATATGGTTGGAATAAAGCCTGTGAAAA
CTATGATTAGTGAATAAGGTCTCAGTAATT
TAGAATAAATATTCTGCACAATGATCAAAT
GTTTAAAGTATCCTTGTGATAAAAGCAGAC
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Table 6.1 First 150 bases of the first intervening sequence of the human preproglucagon gene (Avery and Henderson, 1999). To be read across rows.

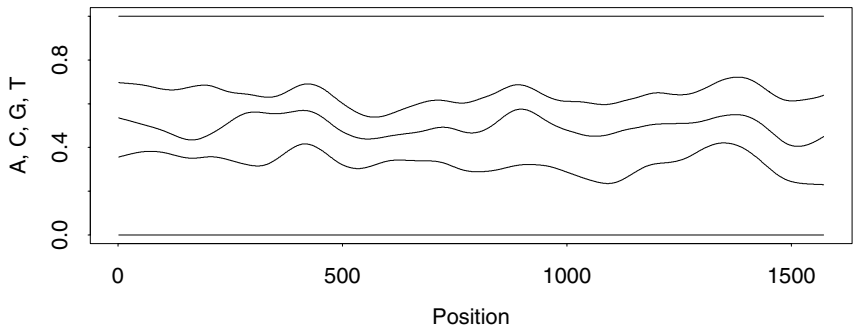


Figure 6.1 Estimated proportions of bases A, C, G and T in the first intervening sequence of the human preproglucagon gene. At a point t on the x -axis, the vertical distances between the lines above correspond to the proportions of times the bases appear in a window of width 100 centred at t .

1572 bases found in the human preproglucagon gene. Figure 6.1 shows proportions of the different bases along the sequence, smoothed using a form of moving average. Roughly speaking, the number of times each base occurs in a window of width 100 centred at t has been counted, giving estimated proportions $(\hat{\pi}_A, \hat{\pi}_C, \hat{\pi}_G, \hat{\pi}_T)$. These are plotted at t , and then the procedure is repeated at $t + 1$, and so forth. Although there is local variation, the proportions seem fairly stable along the sequence, with A occurring about 30 times in every 100, C about 15 times, and so forth.

Certain sequences of bases such as CTGAC — known as *words* — are of biological interest. If they occur very often in particular stretches of the entire sequence, it may be supposed that they serve some purpose. But before trying to see what that purpose might be, it is necessary to see if they have occurred more often than chance dictates, for example by comparing the actual number of occurrences with that expected under a model. It is simplest to suppose that bases occur randomly, but the code of life is not so simple. Table 6.2 contains observed frequencies for pairs of consecutive bases. The pair AA occurs 185 times, AC 74 times, and so forth. The lower table shows corresponding proportions, obtained by dividing the frequencies by their row totals. About 80% of the bases following a C are A or T, while a G is rare; Gs occur much more frequently after A, G, or T. The sequence does not seem purely random. ■

Example 6.2 (Breast cancer data) Table 6.3 gives data on 37 women with breast cancer treated for spinal metastases at the London Hospital. Their ambulatory status — defined as ability to walk unaided or not — was recorded before treatment began, as it started, and then 3, 6, 12, 24, and 60 months after treatment. The three states are: able to walk unaided (1); unable to walk unaided (2); and dead (3). Thus a sequence 111113 means that the patient was able to walk unaided each time she was seen, but was dead five years after the treatment began. She may have been unable to walk for periods between the times at which her state was recorded. This is illustrated in

Table 6.2 Observed frequencies of the 16 possible successive pairs of bases in the first intervening sequence of the human preproglucagon gene. There are 1571 such pairs. The lower table shows the proportion of times the second base follows the first.

First base	Frequencies for second base				Total
	A	C	G	T	
A	185	74	86	171	516
C	101	41	6	115	263
G	69	45	34	78	226
T	161	103	100	202	566
Total	516	263	226	566	1571

First base	Proportion for second base				Total
	A	C	G	T	
A	0.359	0.143	0.167	0.331	1.0
C	0.384	0.156	0.023	0.437	1.0
G	0.305	0.199	0.150	0.345	1.0
T	0.284	0.182	0.177	0.357	1.0
Overall	0.328	0.167	0.144	0.360	1.0

Table 6.3 Breast cancer data (de Stavola, 1988). The table gives the initial and follow-up status for 37 breast cancer patients treated for spinal metastases. The status is able to walk unaided (1), unable to walk unaided (2), or dead (3), and the times of follow-up are 0, 3, 6, 12, 24, and 60 months after treatment began. Woman 24 was alive after 6 months but her ability to walk was not recorded.

Initial			Follow-up			Initial			Follow-up		
1	1	111113	13	2	23	25	1	11113			
2	1	1113	14	2	1113	26	2	22223			
3	2	23	15	2	2	27	2	12223			
4	2	121113	16	2	23	28	2	11113			
5	1	111123	17	1	1113	29	2	1223			
6	1	1113	18	2	223	30	2	1123			
7	1	12113	19	1	13	31	2	1222			
8	2	123	20	1	12223	32	1	11223			
9	1	1111	21	2	23	33	2	1223			
10	2	23	22	2	11111	34	1	1113			
11	2	23	23	2	23	35	1	113			
12	1	1113	24	1	12?3	36	2	23			
						37	2	23			

the left panel of Figure 6.2, which shows a possible sample path for a woman with sighting history 111223. Although there is a visit to state 1 between 12 and 24 months, it is unobserved, and the data suggest that her sojourn in state 2 is uninterrupted. The number of sightings varies from woman to woman; case 9, for example, was able to walk when seen after 12 months, but her later history is unknown.

One aspect of interest here is whether inability to walk always precedes death, while another is whether a woman's state before treatment affects her subsequent history. Although no explanatory variables are available here, their effect on the transition probabilities would often be of importance in practice. ■

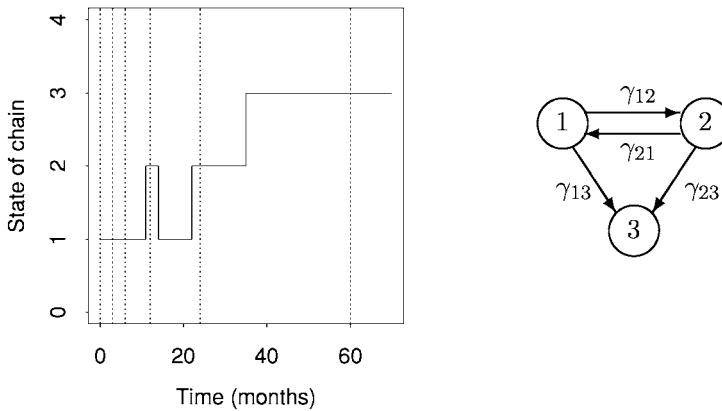


Figure 6.2 Markov chain model for breast cancer data. Left: possible sample path (solid) for a woman with states 111223 observed at 0, 3, 6, 12, 24, 60 months shown by the dotted lines. Right: parameters for possible transitions among the states.

Let X_t denote a process taking values in the *state space* $\mathcal{S} = \{1, \dots, S\}$, where S may be infinite. For general discussion we call the quantity t on which X_t depends time, and suppose that our data have form $X_0 = s_0, X_{t_1} = s_1, \dots, X_{t_k} = s_k$, where $0 < t_1 < \dots < t_k$. In the DNA example t is in fact location, $k = 1571$, and $\mathcal{S} = \{1, 2, 3, 4\} \equiv \{A, C, G, T\}$. In the breast cancer example there are $S = 3$ states, $k = 5$ at most, and $t_0 = 0, t_1 = 3, t_2 = 6, t_3 = 12, t_4 = 24$, and $t_5 = 60$ months.

Let $X_{(t_j)} = s_{(j)}$ denote the composite event $X_{t_j} = s_j, \dots, X_0 = s_0$, for $j = 0, \dots, k - 1$. Then the joint density of the data may be written

$$\Pr(X_0 = s_0, \dots, X_{t_k} = s_k) = \Pr(X_0 = s_0) \prod_{j=1}^k \Pr(X_{t_j} = s_j \mid X_{(t_{j-1})} = s_{(j-1)});$$

using the prediction decomposition (4.7). The conditional probabilities may be complicated, but modelling is greatly simplified if the process has the *Markov property*

$$\Pr(X_{t_j} = s_j \mid X_0 = s_0, \dots, X_{t_{j-1}} = s_{j-1}) = \Pr(X_{t_j} = s_j \mid X_{t_{j-1}} = s_{j-1}).$$

Thus the ‘future’ X_{t_j} is independent of the ‘past’ $X_{t_{j-2}}, \dots, X_0$, given the ‘present’ $X_{t_{j-1}}$ — all information available about the future evolution of X_t is contained in its current state. If so, then

$$\Pr(X_0 = s_0, \dots, X_{t_k} = s_k) = \Pr(X_0 = s_0) \prod_{j=1}^k \Pr(X_{t_j} = s_j \mid X_{t_{j-1}} = s_{j-1}). \quad (6.1)$$

Matters simplify further if the process is *stationary*, for then the conditional probabilities in (6.1) depend only on differences among the t_j . Thus

$$\Pr(X_t = s \mid X_u = r) = \Pr(X_{t-u} = s \mid X_0 = r),$$

and we assume this to be the case below. These simplifications yield a rich structure with many important and interesting models, in which these *transition probabilities* play a central role. They determine the likelihood (6.1), apart from the initial term

Andrei Andreyevich Markov (1856–1922) studied with Chebyshev in St Petersburg and initially worked on pure mathematics. His study of dependent sequences of variables stemmed from an attempt to understand the Central Limit Theorem.

Some authors use the term *homogeneous* rather than *stationary*.

$\Pr(X_0 = s_0)$. If k is large this term usually contains little information and can safely be dropped, but it may be important to include it when k is small; see Example 6.10.

6.1.1 Markov chains

We call a Markov model observed at discrete equally-spaced times a *Markov chain*. In this section we consider inference for simple Markov chain models, but in Section 11.3.3 we describe the use of Markov chains for inference. As the following outline of their properties serves both purposes, it is slightly more detailed than immediately required.

A stationary chain X_t on the countable set $\mathcal{S}$ of size S observed at equally-spaced times $t = 0, 1, \dots, k$ has properties determined by the transition probabilities

$$p_{rs} = \Pr(X_1 = s \mid X_0 = r), \quad r, s \in \mathcal{S},$$

which form the $S \times S$ *transition matrix* P whose (r, s) element is p_{rs} . The elements of P are non-negative and the fact that $\sum_s p_{rs} = 1$ implies that $P \mathbf{1}_S = \mathbf{1}_S$, so P is a *stochastic matrix*. If the r th element of the $S \times 1$ vector p is the initial probability $p_r = \Pr(X_0 = r)$, then the s th element of $p^T P$ is $\Pr(X_1 = s) = \sum_r p_r p_{rs}$. Iteration shows that the density of X_n is given by $p^T P^n$, so the (r, s) element of P^n is the n -step transition probability $p_{rs}(n) = \Pr(X_n = s \mid X_0 = r)$. Hence properties of X_t are governed by P . The probability of a *run* of $m \geq 1$ successive visits to state s is $p_{ss}^{m-1}(1 - p_{ss})$; this is the geometric density with mean $(1 - p_{ss})^{-1}$ (Exercise 6.1.8).

Classification of chains

It is useful to classify the states of a chain. A state s is *recurrent* if

$$\Pr(X_t = s \text{ for some } t > 0 \mid X_0 = s) = 1,$$

meaning that having started from s , eventual return is certain; s is *transient* if this probability is strictly less than one. If $T_{rs} = \min\{t > 0 : X_t = s \mid X_0 = r\}$ is the *first-passage time* from r to s , then $E(T_{ss})$ is the *mean recurrence time* of state s ; we set $E(T_{ss}) = \infty$ if s is transient, and say that a recurrent state is *positive recurrent* if $E(T_{ss}) < \infty$; otherwise it is *null recurrent*. The *period* of s is $d = \gcd\{n : p_{ss}(n) > 0\}$, the greatest common divisor of those times at which return to s is possible; s is *aperiodic* if $d = 1$, and *periodic* otherwise.

We now classify chains themselves. We say that r *communicates with* s , $r \rightarrow s$, if $p_{rs}(n) > 0$ for some $n > 0$, and that r and s *intercommunicate*, $r \leftrightarrow s$, if $r \rightarrow s$ and $s \rightarrow r$. It may be shown that two intercommunicating states have the same period, while if one is transient so is the other, and similarly for null recurrence. A set $\mathcal{C}$ of states is *closed* if $p_{rs} = 0$ for all $r \in \mathcal{C}$, $s \notin \mathcal{C}$, and *irreducible* if $r \leftrightarrow s$ for all $r, s \in \mathcal{C}$; a closed set with just one state is called *absorbing*. It may be proved that $\mathcal{S}$ may be partitioned uniquely as $\mathcal{T} \cup \mathcal{C}_1 \cup \mathcal{C}_2 \cup \dots$, where $\mathcal{T}$ is the set of transient states and the $\mathcal{C}_i$ are irreducible closed sets of recurrent states; if $\mathcal{S}$ is finite, then at least one state is recurrent, and all recurrent states are positive. A chain is called aperiodic, positive recurrent, and so forth if its states all share the corresponding property. An aperiodic irreducible positive recurrent chain is *ergodic*.

If infinite matrices worry you, think of $\mathcal{S}$ as finite.

$\mathbf{1}_S$ is the $S \times 1$ vector of ones.

Some authors use the terms persistent and non-null rather than recurrent and positive.

Example 6.3 (Breast cancer data) Here $\mathcal{T} = \{1, 2\}$ contains the two transient states with the patient alive, while $\mathcal{C} = \{3\}$, death, is absorbing. ■

Example 6.4 (DNA data) As transitions occur between every pair of states, $\mathcal{C} = \{A, C, G, T\}$ is an irreducible aperiodic closed set of states, all recurrent and hence all positive recurrent. This chain is ergodic. ■

Each of the properties of an ergodic chain is important. Irreducibility means that any state is accessible from any other. Positive recurrence implies that the chain has at least one stationary distribution with probability vector π such that $\pi^T P = \pi^T$, and the mean recurrence time for state s is $E(T_{ss}) = \pi_s^{-1} < \infty$. There is a unique stationary distribution when the chain is both irreducible and positive recurrent. In this case each state is visited infinitely often as $t \rightarrow \infty$, but the chain need not be stationary because it might oscillate among states. Aperiodicity stops this.

When S is infinite and the chain has all three properties, the transition probabilities $p_{rs}(n) \rightarrow \pi_s$ as $n \rightarrow \infty$: the chain converges to its stationary distribution whatever the initial state. Moreover, if $m(X_t)$ is such that $E_\pi \{|m(X_t)|\} = \sum_r \pi_r |m(r)| < \infty$, then

$$\Pr \left\{ n^{-1} \sum_{t=1}^n m(X_t) \rightarrow \sum_{r=1}^S \pi_r m(r) \text{ as } n \rightarrow \infty \right\} = 1 : \quad (6.2)$$

starting from any X_0 , the average of $m(X_t)$ converges almost surely to the mean $E_\pi \{m(X_t)\}$ of $m(X_t)$ with respect to π . This ensures the convergence of so-called *ergodic averages* $n^{-1} \sum_{t=1}^n m(X_t)$ and is crucial to the use of Markov chains for inference. When S is finite, an irreducible aperiodic chain is automatically positive recurrent and hence ergodic.

If S is finite then P is an $S \times S$ matrix, whose eigenvalues $l_1, \dots, l_S$ are roots of its characteristic polynomial $\det(P - \lambda I_S)$. If the l_r are distinct, then

$$P = E^{-1} L E, \quad (6.3)$$

where $L = \text{diag}(l_1, \dots, l_S)$, the r th row e_r^T of the $S \times S$ matrix E is the left eigenvector of P corresponding to l_r and the r th column e_r' of E^{-1} is the right eigenvector of P corresponding to l_r . The l_r are complex numbers with modulus no greater than unity, but as P is real, any complex roots of its characteristic polynomial occur in conjugate pairs $a \pm ib$. For some real $r > 0$,

Here $i = \sqrt{-1}$.

$$(a \pm ib)^n = r^n \exp(\pm in\omega) = r^n (\cos n\omega \pm i \sin n\omega).$$

As P^n is a real matrix, it may be better to express its elements in terms of sines and cosines when P has complex eigenvalues.

If S is finite and the chain is irreducible with period d , then the d complex roots of unity $l_1 = \exp(2\pi i/d), \dots, l_d = \exp\{2\pi i(d-1)/d\}$ are eigenvalues of P and $l_{d+1}, \dots, l_S$ satisfy $|l_s| < 1$. If the chain is irreducible and aperiodic, then $l_1 = 1$, and $|l_s| < 1$ for $s = 2, \dots, S$. Now $\pi^T P = \pi^T$ and $P 1_S = 1_S$, so if X_t has stationary distribution π , then π^T and 1_S are the left and right eigenvectors of P corresponding

to $l_1 = 1$, that is, $e_1 = \pi$ and $e'_1 = 1_S$. The convergence of an ergodic chain with distinct eigenvalues is obvious, because

$$P^n = (E^{-1}LE)^n = E^{-1}L^nE = \sum_{r=1}^S l_r^n e'_r e_r^T \rightarrow e'_1 e_1^T = 1_S \pi^T \quad \text{as } n \rightarrow \infty :$$

the (r, s) element of P^n , $p_{rs}(n)$, tends to π_s . Moreover, if $p(0)$ is the probability vector of X_0 then X_n has distribution $p(0)^T P^n$, which converges to $p(0)^T 1_S \pi^T = \pi^T$ whatever the initial vector $p(0)$.

If S is infinite and the chain ergodic, its first eigenvalue l_1 equals 1 and corresponds to the unique stationary distribution π , but the second eigenvalue l_2 need not exist. If l_2 exists and $|l_2| < 1$, then $|l_2|$ controls the rate at which the chain approaches its stationary distribution. More precisely, the chain is *geometrically ergodic* if there exists a function $V(\cdot) > 1$ such that

$$\sum_s |p_{rs}(n) - \pi_s| \leq V(r) |l_2|^n \quad \text{for all } n; \quad (6.4)$$

$|l_2|$ is then the *rate of convergence* of the chain.

An irreducible chain is *reversible* if there exists a π such that

$$\pi_r p_{rs} = \pi_s p_{sr}, \quad \text{for all } r, s \in S; \quad (6.5)$$

the chain is then positive recurrent with stationary distribution π . Another way to express the *detailed balance condition* (6.5) is

$$\Pr(X_t = r, X_{t+1} = s) = \Pr(X_t = s, X_{t+1} = r), \quad \text{for all } r, s \in S,$$

or $\Pi P = P \Pi$, where Π is the $S \times S$ diagonal matrix whose elements are the components of the stationary distribution π .

Decomposition (6.3) applies to reversible chains, whose eigenvalues and eigenvectors l_r and e_r are real. Chains that fail to be geometrically ergodic have an infinite number of eigenvalues in any open interval containing one of ± 1 , but those that are geometrically ergodic have all their eigenvalues but l_1 uniformly bounded in modulus below unity.

Example 6.5 (Two-state chain) Consider the chain for which

$$P = \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix}, \quad 0 \leq p, q \leq 1.$$

When $p = q = 0$, there are two absorbing states $\mathcal{C}_1 = \{1\}$ and $\mathcal{C}_2 = \{2\}$ and the chain is entirely uninteresting. When both p and q are positive it is clearly irreducible and $\pi^T P = \pi^T$, where $\pi^T = (p+q)^{-1}(q, p)$. The chain is then positive recurrent with $E(T_{11}) = (p+q)/q$ and $E(T_{22}) = (p+q)/p$.

When $p = q = 1$, X_t takes values $\dots, 1, 2, 1, 2, 1, \dots$ and is periodic with period two, so $T_{11} = T_{22} = 2$ with probability one. If $p(0) = \pi$, then X_t has this distribution for all t , but if not, then the fact that $P^2 = I_2$ implies that $X_0, X_1, X_2, \dots$ have distributions $p(0)^T, p(0)^T P, p(0)^T, \dots$; the chain cycles among these and never reaches stationarity.

The eigenvalues of P are $l_1 = 1, l_2 = 1 - p - q$. Its eigendecomposition is

$$\begin{pmatrix} 1 & p \\ 1 & -q \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 - p - q \end{pmatrix} \cdot \frac{1}{p + q} \begin{pmatrix} q & p \\ 1 & -1 \end{pmatrix}.$$

If $|l_2| < 1$, then $0 < p < 1$ or $0 < q < 1$ or both, the chain is aperiodic and

$$P^n = \frac{1}{p + q} \begin{pmatrix} q + pl_2^n & p - pl_2^n \\ q - ql_2^n & p + ql_2^n \end{pmatrix} \rightarrow \frac{1}{p + q} \begin{pmatrix} q & p \\ q & p \end{pmatrix} = 1_2 \pi^T \quad \text{as } n \rightarrow \infty.$$

■

Example 6.6 (Five-state chain) The state space of the chain with

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{2} \end{pmatrix}$$

decomposes as $\mathcal{C}_1 \cup \mathcal{C}_2 \cup \mathcal{T}$, where $\mathcal{C}_1 = \{1, 2\}$, $\mathcal{C}_2 = \{3, 4\}$ and $\mathcal{T} = \{5\}$. Evidently $\mathcal{C}_1$ is a special case of the previous example, so it is ergodic. The set $\mathcal{C}_2$ is closed and irreducible, but it is periodic because $X_t = X_{t+2} = X_{t+4} = \dots$. The set $\mathcal{T}$ is transient: at each step the probability of leaving it is $\frac{1}{2}$, with equal probabilities of landing in $\mathcal{C}_1$ and $\mathcal{C}_2$. Although $\mathcal{C}_1$ is ergodic, the chain as a whole is not.

Owing to the presence of two irreducible sets, one with period two, the eigenvalues include $l_1 = 1, l_2 = 1$ and $l_3 = -1$. The repeated eigenvalue means that the eigendecomposition of P is not unique. One version is

$$\begin{pmatrix} 1 & 1 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & \frac{1}{2} \\ 1 & -1 & -6 & 0 & 0 \\ 1 & -1 & 6 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} \frac{1}{6} & \frac{1}{3} & \frac{1}{4} & \frac{1}{4} & 0 \\ \frac{1}{6} & \frac{1}{3} & -\frac{1}{4} & -\frac{1}{4} & 0 \\ 0 & 0 & -\frac{1}{12} & \frac{1}{12} & 0 \\ \frac{1}{2} & -1 & -\frac{1}{6} & -\frac{1}{3} & 1 \\ -\frac{2}{3} & \frac{2}{3} & 0 & 0 & 0 \end{pmatrix}.$$

For large n we have approximately

$$P^n = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{2}{3} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2}\{1 + (-1)^n\} & \frac{1}{2}\{1 + (-1)^{n+1}\} & 0 \\ 0 & 0 & \frac{1}{2}\{1 + (-1)^{n+1}\} & \frac{1}{2}\{1 + (-1)^n\} & 0 \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} & 2^{-n} \end{pmatrix}.$$

If $X_0 \in \mathcal{C}_1$, the stationary distribution of X_t is $(\frac{1}{3}, \frac{2}{3}, 0, 0, 0)^T$ and the states have mean recurrence times 3 and $\frac{3}{2}$. If $X_0 = 3$, then $X_2 = X_4 = \dots = 3$ and $X_1 = X_3 = \dots = 4$, while the converse is true if $X_0 = 4$; X_t oscillates within $\mathcal{C}_2$ but has a stationary distribution only if the initial probability vector is $(0, 0, \frac{1}{2}, \frac{1}{2}, 0)^T$. If $X_0 = 5$, the probability that $X_n = 5$ is essentially zero for large n and the process is equally likely to end up in $\mathcal{C}_1$ or $\mathcal{C}_2$. ■

Likelihood inference

We now consider inference from data $s_0, s_1, \dots, s_k$ at times $0, 1, \dots, k$ from a stationary discrete-time Markov chain X_t with finite state space. The likelihood is

$$\begin{aligned} \Pr(X_0 = s_0, \dots, X_k = s_k) &= \Pr(X_0 = s_0) \prod_{t=0}^{k-1} \Pr(X_{t+1} = s_{t+1} \mid X_t = s_t) \\ &= \Pr(X_0 = s_0) \prod_{t=0}^{k-1} p_{s_t s_{t+1}} \\ &= \Pr(X_0 = s_0) \prod_{r=1}^S \prod_{s=1}^S p_{rs}^{n_{rs}}, \end{aligned} \quad (6.6)$$

where n_{rs} is the observed number of transitions from r to s . Apart from the first term in (6.6), the log likelihood is

$$\ell(p) = \sum_{r=1}^S \sum_{s=1}^S n_{rs} \log p_{rs}, \quad (6.7)$$

so the $S \times S$ table of transition counts n_{rs} is a sufficient statistic; see Table 6.2. As $\sum_r p_{sr} = 1$ for each s , (6.7) sums log likelihood contributions from S separate multinomial distributions $(n_{r1}, \dots, n_{rS})$ whose denominators n_r equal the row sums $n_{r1} + \dots + n_{rS}$ and whose probability vectors $(p_{r1}, \dots, p_{rS})$ correspond to transitions out of state r ; see (4.45). As $\sum_s p_{rs} = 1$ for each r , this model has $S(S-1)$ parameters.

The results of Section 4.5.3 imply that p_{rs} has maximum likelihood estimate $\hat{p}_{rs} = n_{rs}/n_r$. Standard likelihood asymptotics will apply if $0 < p_{rs} < 1$ for all r and s and if the denominators $n_r \rightarrow \infty$ as $k \rightarrow \infty$. Now n_r is the number of visits the chain makes to state r during the period $1, \dots, k$, and if the chain is ergodic r is visited infinitely often as $k \rightarrow \infty$. The $\hat{p}_{rs}$ then have an approximate joint normal distribution with covariances estimated by

$$\text{cov}(\hat{p}_{rs}, \hat{p}_{tu}) \doteq \begin{cases} \hat{p}_{rs}(1 - \hat{p}_{rs})/n_r, & r = t, s = u, \\ -\hat{p}_{rs}\hat{p}_{ru}/n_r, & r = t, s \neq u, \\ 0, & \text{otherwise.} \end{cases}$$

The above discussion ignores the first term in (6.6). If k is large it will add only a small contribution to $\ell(p)$ and can safely be dropped, but if k is small it might be replaced by the stationary probability π_{s_0} , found from the elements of P . In general the log likelihood must then be maximized numerically.

An alternative asymptotic scenario is that m independent finite segments of Markov chains having the same parameters are observed, and $m \rightarrow \infty$. The overall information in the initial terms of the segments is then $O(m)$ and retrieval of it may be worthwhile, particularly if the segments are short. Below we continue to suppose that there is a single chain of length k .

In simpler models the p_{rs} might depend on a parameter with dimension smaller than $S(S-1)$. For instance, setting $p = q$ in Example 6.5 gives a one-parameter

First base	Observed frequency				Expected frequency			
	A	C	G	T	A	C	G	T
A	185	74	86	171	169.5	86.4	74.2	185.9
C	101	41	6	115	86.4	44.0	37.8	94.8
G	69	45	34	78	74.2	37.8	32.5	81.4
T	161	103	100	202	185.9	94.8	81.4	203.9

Table 6.4 Fit of independence model to DNA data: observed and fitted frequencies of one-step transitions.

model. If the chain is ergodic, likelihood inference for such models will be regular under the usual conditions on the parameter space.

Thus far transition probabilities have depended only on the current state, so our chains have been *first-order*. The simpler *independence model* posits transition probabilities independent of the current state, $p_{rs} \equiv p_s$; this *zeroth-order* chain has just $S - 1$ parameters. Row and column classifications in the table of counts n_{rs} are then independent, (6.7) reduces to $\sum n_{.s} \log p_s$, and $\hat{p}_s = n_{.s}/n_{..}$, where $n_{.s} = n_{1s} + \cdots + n_{Ss}$ and $n_{..} = \sum_s n_{.s}$. Thus the likelihood ratio statistic for comparison of the zeroth- and first-order chains is

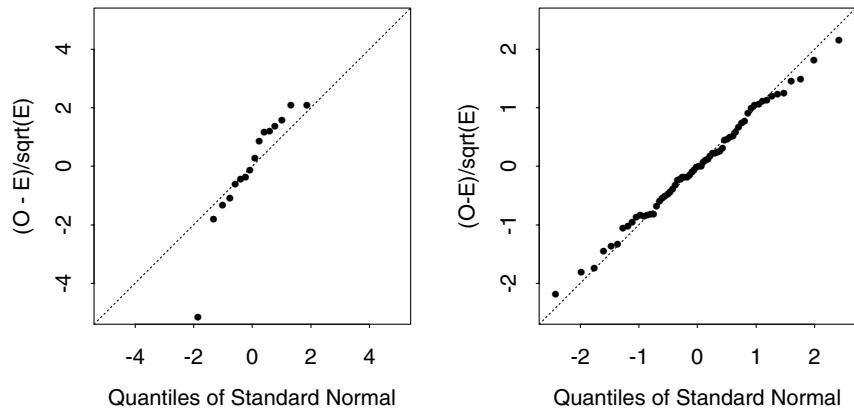
$$W = 2 \sum_{r,s} n_{rs} \log \left(\frac{\hat{p}_{rs}}{\hat{p}_s} \right) = 2 \sum_{r,s} n_{rs} \log \left(\frac{n_{rs} n_{..}}{n_{r.} n_{.s}} \right);$$

this is the likelihood ratio statistic for testing row-column independence in the square table of counts n_{rs} . Under the zeroth-order chain the rows of P all equal $(p_1, \dots, p_S)$, row and column classifications are independent, and W is a natural statistic to assess this; its asymptotic distribution is chi-squared with $S(S - 1) - (S - 1) = (S - 1)^2$ degrees of freedom. As we saw in Section 4.5.3, W approximately equals Pearson's statistic $P = \sum (O - E)^2/E$, where O and E denote the observed count n_{rs} and its expected counterpart $n_{r.} n_{.s}/n_{..}$ under the independence model and the sum is over the cells of the table. The quantities $(O - E)/E^{1/2}$ squared give the contribution of each cell to P .

Example 6.7 (DNA data) The lowest line of Table 6.2 gives maximum likelihood estimates for the zeroth-order independence model, while the four previous lines give estimates for the first-order model. For the independence model we have $\hat{p}_A = 516/1571 = 0.328$ and $\hat{p}_C = 263/1571 = 0.167$, for example, while under the first-order model $\hat{p}_{AA} = 185/516 = 0.359$, $\hat{p}_{AC} = 74/516 = 0.143$, $\hat{p}_{CG} = 6/263 = 0.023$ and so forth. If the independence model was correct, $W = 2 \sum_{r,s} n_{rs} \log \{n_{rs}/(n_{r.} \hat{p}_s)\}$ would have a χ^2_9 distribution, but the observed value $w = 64.45$ makes this highly implausible. The value of P is 50.3.

Table 6.4 shows the counts n_{rs} and the fitted values $n_{r.} n_{.s}/n_{..}$ under the independence model. The largest discrepancy is for the CG cell, for which $(O - E)/E^{1/2} = -5.18$, so this cell contributes 26.79 to the value of P . The normal probability plot of

Figure 6.3 Fit of zeroth- and first-order Markov chains to the DNA data. The panels show normal probability plots of the signed contributions $(O - E)/E^{1/2}$ made by the 16 cells of the two-way table under the independence model (left) and the 64 cells of the three-way table under the first-order model (right). The large negative value on the left is due to the CG cell. The dots show the null line $x = y$.



the $(O - E)/E^{1/2}$ in the left panel of Figure 6.3 shows that the other cells contribute much less. The values of W and P remain large even if this cell is dropped from the table, however, so it is not the sole cause of the poor fit of the independence model. ■

Higher-order models

First-order Markov chains extend to chains of order m , where the probability of transition into s depends on the m preceding states. One way to think of this is that the state of the chain is augmented from X_j to $Y_j = (X_j, X_{j-1}, \dots, X_{j-m+1})$ and the transition probabilities change to

$$\begin{aligned} \Pr(Y_j = y_j \mid Y_{j-1} = y_{j-1}) &= \Pr(X_j = s \mid X_{j-1} = s_{j-1}, \dots, X_{j-m} = s_{j-m}) \\ &= p_{s_{j-m}s_{j-m+1}\dots s_{j-1}s}, \end{aligned}$$

say. Thus the ‘current’ state $Y_{j-1} = (s_{j-1}, \dots, s_{j-m})$ contains information not only from time $j - 1$ but also from the $m - 1$ previous times. Whereas with $m = 1$ the properties of the chain were determined by the S vectors of transition probabilities $(p_{r1}, \dots, p_{rS})$, there are now S^m such vectors, so much more data is needed in order to get reliable estimates of the transition probabilities. A compromise is a *variable-order* chain, the simplest example of which is when $m = 2$ and $S = 2$, so that the chain of order two is determined by the probabilities $p_{111}, p_{121}, p_{211}$ and p_{221} , giving the transition probabilities π_{sur} from (s, u) to r . A simple variable-order chain is obtained by specifying $\pi_{111} = \pi_{211}$, that is, given that $u = 1$, the transition probabilities do not depend on s . This chain is first-order when $u = 1$, but not when $u = 2$. In this case the number of parameters only diminishes by one, but in general the reduction might be much larger.

Likelihood ratio statistics or criteria such as AIC enable systematic comparison of Markov chains of different orders, but care is needed when computing them. Suppose that we fit models of orders up to m to a sequence of length k . There are $k - 1$ successive pairs, $k - 2$ triplets and so forth, so the fit of the m th-order model is based

Table 6.5 Observed transition counts for second-order Markov chain for DNA data.

First base	Second base	Frequencies for third base				Total
		A	C	G	T	
A	A	81	22	29	53	185
	C	30	7	2	35	74
	G	29	18	11	27	86
	T	54	23	33	61	171
C	A	30	20	15	36	101
	C	15	2	1	23	41
	G	2	1	0	3	6
	T	28	26	20	41	115
G	A	30	3	14	22	69
	C	18	10	1	16	45
	G	12	5	10	7	34
	T	27	11	12	27	77
T	A	44	29	28	60	160
	C	38	22	2	41	103
	G	26	21	13	40	100
	T	51	43	35	73	202

on the $k - m$ successive $(m + 1)$ -tuples from which the transition probabilities and maximized log likelihood are computed, treating the last $k - m$ of the k observations as responses. Standard likelihood methods presuppose that the same responses are used throughout, so fits for chains of smaller order must also treat only the last $k - m$ observations as responses.

Example 6.8 (DNA data) We compare models of order up to $m = 3$. The preceding discussion implies that as the data in Table 6.1 begin GTAT. . ., the first response is the second T, so the initial GTA, GT and G should be ignored when fitting the zeroth-, first- and second-order models respectively. The frequencies for the $k - m = 1572 - 3 = 1569$ triplets of transition counts in our sequence are shown in Table 6.5. The implied numbers of TA and GT transitions, $54 + 28 + 27 + 51 = 160$ and $27 + 3 + 7 + 40 = 77$, are smaller than the numbers 161 and 78 in Table 6.2 which include such transitions in the initial GTAT.

Estimates under the second-order model are obtained as before, by dividing each row by its total, giving $\hat{p}_{AAA} = 81/185$, $\hat{p}_{AAC} = 22/185$, $\hat{p}_{ACA} = 30/74$ and so forth. Evidently estimates such as $\hat{p}_{CGA} = 2/6$ are very unreliable.

Estimates under the first-order model are computed from the two-way table of counts obtained by collapsing the table over the first base, giving a 4×4 table whose top left (AA) element is $81 + 30 + 30 + 44 = 185$, whose CG element is $2 + 1 + 1 + 2 = 6$ and so forth. For estimates under the independence model we use the 1×4 table from a further collapse over the second base; both sets of estimates are essentially unaffected by dropping the first few bases.

The maximized log likelihoods for the zeroth-, first-, second- and third-order models are -2058.44 , -2026.02 , -1998.41 , and -1923.25 on 3, 12, 48, and 192 degrees

of freedom, so the AIC values are 4122.9, 4076.0, 4092.8, and 4230.5 and the likelihood ratio statistics for comparison of each model with the next are 64.8, 55.2, and 150.3, on 9, 36, and 144 degrees of freedom. There is strong evidence for first-order dependence compared to independence, while as $\Pr(\chi_{36}^2 > 55.2) \doteq 0.02$ and $\Pr(\chi_{144}^2 > 150.3) \doteq 0.34$ the evidence for second- compared to first-order dependence is weaker, and there is no suggestion of third-order dependence. The AIC values clearly indicate the first-order model.

The signed contributions $(O - E)/E^{1/2}$ to Pearson's statistic under the first-order model can be obtained using Table 6.5. The contribution for the AAA cell, for example, is $(81 - E)/E^{1/2}$, where $E = 185\hat{p}_{AA}$, with $\hat{p}_{AA}$ calculated under the first-order model. The value of Pearson's statistic is 52.84. The right panel of Figure 6.3 shows no highly unusual cells and apparently good fit.

The eigenvalues for the observed first-order matrix of transition probabilities $\hat{P}$ are 1, $-0.0147 \pm 0.0704i$ and 0.0524. The small absolute values of the last three suggest that the chain is close to independence, and indeed the rows of $\hat{P}^4$ are essentially equal: four steps are (almost) enough to forget the past.

Our earlier discussion suggested that the main departures from independence occur after C, suggesting taking a model where $p_{rs} = \psi_s$ whenever $r \neq C$ and $p_{Cs} = \phi_s$. That is, for each s we have

$$\Pr(X_{t+1} = s \mid X_t = A) = \Pr(X_{t+1} = s \mid X_t = G) = \Pr(X_{t+1} = s \mid X_t = T),$$

but these do not equal p_{Cs} . This model has six independent parameters and as its log likelihood $\sum_s (\sum_{r \neq C} n_{rs}) \log \psi_s + \sum_s n_{Cs} \log \phi_s$ is of multinomial form, their estimates are readily obtained. The maximized log likelihood is -2031.0 , so AIC = 4074.0 is lower than for the full first-order chain and this model seems marginally preferable. See Exercise 6.1.7 for further details. ■

We have presumed above that X_t is stationary. If instead the transition probabilities are of form $p_{rs}(t; \theta)$, dependent on a parameter θ , then the likelihood $\Pr(X_0 = s_0; \theta) \prod_{t=0}^{k-1} p_{s_t s_{t+1}}(t; \theta)$ is found by the argument leading to (6.6). In many cases the initial probability $\Pr(X_0 = s_0; \theta)$ may be unknown, and if the series is long little will be lost by ignoring it. If the transition probabilities do not share dependence on a common θ , they can only be estimated if they are repeated. Large amounts of data will then be needed.

6.1.2 Continuous-time models

We now turn to stationary continuous-time Markov models with finite state space $\mathcal{S}$. The basic assumption is that over small intervals $[t, t + \delta t)$, transitions between states have probabilities

$$\Pr(X_{t+\delta t} = s \mid X_t = r) = \begin{cases} \gamma_{rs}\delta t + o(\delta t), & s \neq r, \\ 1 + \gamma_{rr}\delta t + o(\delta t), & s = r, \end{cases} \quad (6.8)$$

$o(\delta t)$ is small enough that $o(\delta t)/\delta t \rightarrow 0$ as $\delta t \rightarrow 0$.

where γ_{rs} is interpreted as the rate at which transitions $r \rightarrow s$ occur. The transition probabilities do not depend on t , so X_t is time homogeneous. Note that $\sum_s \gamma_{rs} = 0$, for each r , because the probabilities in (6.8) sum to one.

Let $p(t)$ denote the $S \times 1$ vector whose r th element is $p_r(t) = \Pr(X_t = r)$; note that $1_S^T p(t) = 1$ for all t . Then

$$p_s(t + \delta t) = \sum_{r=1}^S \Pr(X_{t+\delta t} = s \mid X_t = r) p_r(t) \doteq p_s(t) + \sum_{r=1}^S \gamma_{rs} p_r(t) \delta t + o(\delta t),$$

implying that

$$\frac{dp_s(t)}{dt} = \lim_{\delta t \rightarrow 0} \frac{p_s(t + \delta t) - p_s(t)}{\delta t} = \sum_{r=1}^S \gamma_{rs} p_r(t), \quad s = 1, \dots, S,$$

written in matrix form as

$$\begin{pmatrix} \frac{dp_1(t)}{dt} & \cdots & \frac{dp_S(t)}{dt} \end{pmatrix} = (p_1(t) \quad \cdots \quad p_S(t)) \begin{pmatrix} \gamma_{11} & \cdots & \gamma_{1S} \\ \vdots & \ddots & \vdots \\ \gamma_{S1} & \cdots & \gamma_{SS} \end{pmatrix}.$$

In terms of the *infinitesimal generator* of the chain, the matrix G whose (r, s) element is γ_{rs} , we write

$$\frac{dp(t)^T}{dt} = p(t)^T G, \quad (6.9)$$

to which the formal solution is

$$p(t)^T = p(0)^T \exp(tG),$$

where $p(0)$ is the probability vector for the states of X_0 , and the matrix exponential $\exp(tG)$ is interpreted as $\sum_{m=0}^{\infty} (tG)^m / m!$, with $G^0 = I_S$. If the initial state was $X_0 = r$, $p(0)$ consists of zeros except for its r th component, implying that $\Pr(X_t = s \mid X_0 = r) = p_{rs}(t)$ is the (r, s) element of $\exp(tG)$.

Any stationary distribution π for X_t must be time-independent, so the right-hand side of (6.9) will be zero when $p(0) = \pi$. Hence π^T will be a left eigenvector of G with eigenvalue zero.

The chain is reversible if and only if there is a distribution π satisfying the detailed balance condition $\pi_r \gamma_{rs} = \pi_s \gamma_{sr}$.

If G is diagonalizable the eigendecomposition (6.3) is again useful. For if $G = E^{-1} L E$ then $G^m = E^{-1} L^m E$, so

$$\exp(tG) = E^{-1} \text{diag}\{\exp(tl_1), \dots, \exp(tl_S)\} E.$$

Hence the s th row of E and column of E^{-1} , e_s^T and e'_s , are left and right eigenvectors of $\exp(tG)$ with eigenvalue $\exp(tl_s)$. The fact that $\sum_s \gamma_{rs} = 0$ for each r implies that $G 1_S = 0$, so $e'_1 = 1_S$ is a right eigenvector of G with eigenvalue $l_1 = 0$, while $e_1 = \pi$,

as we saw above. The remaining eigenvalues of G all have strictly negative real parts. Hence

$$\begin{aligned}\exp(tG) &= (e'_1 \quad \cdots \quad e'_S) \begin{pmatrix} \exp(tl_1) & & 0 \\ & \ddots & \\ 0 & & \exp(tl_S) \end{pmatrix} \begin{pmatrix} e_1^T \\ \vdots \\ e_S^T \end{pmatrix} \\ &= \sum_{r=1}^S \exp(tl_r) e'_r e_r^T \\ &\rightarrow e'_1 e_1^T = 1_S \pi^T \quad \text{as } t \rightarrow \infty : \end{aligned}$$

starting from any X_0 , the (r, s) element of $\exp(tG)$, $\Pr(X_t = s \mid X_0 = r) \rightarrow \pi_s$. This transition probability may be written as a linear combination of exponentials, $c_{rs,1}e^{t l_1} + \cdots + c_{rs,S}e^{t l_S}$, where $c_{rs,v}$ is the (r, s) element of $e'_v e_v^T$, that is, the product of the r th element of e'_v and the s th element of e_v .

Fully observed trajectory

If X_t had been fully observed during $[0, t_0]$, say, we would know exactly when and between which states transitions occurred. To write down the likelihood we would need probabilities for events such as $X_u = r$, $0 \leq u < t$, followed by transition from r to s at time t , so $X_t = s$. To obtain this we divide $[0, t)$ into m intervals of length δt and apply the Markov property to see that

$$\Pr(X_{\delta t} = X_{2\delta t} = \cdots = X_{(m-1)\delta t} = r, X_{m\delta t} = s \mid X_0 = r)$$

equals

$$\Pr(X_{m\delta t} = s \mid X_{(m-1)\delta t} = r) \prod_{i=1}^{m-1} \Pr(X_{i\delta t} = r \mid X_{(i-1)\delta t} = r),$$

and this itself is

$$\{1 + \gamma_{rr}\delta t + o(\delta t)\}^{m-1} \{\gamma_{rs}\delta t + o(\delta t)\} = \left(1 + \frac{\gamma_{rr}t}{m}\right)^{m-1} \gamma_{rs}\delta t + o(\delta t).$$

On dividing by δt and letting $m \rightarrow \infty$, then recalling that $\gamma_{rr} = -\sum_{v \neq r} \gamma_{rv}$, we see that the density corresponding to observing $X_u = r$, $0 \leq u < t$, followed by transition to $X_t = s$, is

$$\gamma_{rs} \exp(t\gamma_{rr}) = \gamma_{rs} \exp\left(-t \sum_{v \neq r} \gamma_{rv}\right).$$

This has the simple interpretation that the first transition out of r occurs at $T = \min\{t : X_t \neq r\} = \min_{v \neq r} \{T_{rv}\}$, where the T_{rv} are independent exponential variables with parameters γ_{rv} , that is, with means γ_{rv}^{-1} . This suggests an algorithm for simulating data from such a process (Exercise 6.1.11).

The probability of a trajectory fully observed for the period $[0, t_0]$ and with transitions at $t_1 < \cdots < t_k$ is calculated by using the Markov property to express

$$\Pr(X_t = s_0, 0 \leq t < t_1, X_t = s_1, t_1 \leq t < t_2, \dots, X_t = s_k, t_k \leq t \leq t_0)$$

as

$$\begin{aligned} & \Pr(X_0 = s_0) \Pr(X_t = s_0, 0 < t < t_1, X_{t_1} = s_1 \mid X_0 = s_0) \\ & \times \prod_{j=1}^{k-1} \Pr(X_t = s_j, t_j < t < t_{j+1}, X_{t_{j+1}} = s_{j+1} \mid X_{t_j} = s_j) \\ & \times \Pr(X_t = s_k, t_k < t \leq t_0 \mid X_{t_k} = s_k). \end{aligned}$$

Thus the likelihood for the γ_{rs} based on such data is

$$\Pr(X_0 = s_0) \times \gamma_{s_0 s_1} e^{t_1 \gamma_{s_0 s_0}} \times \prod_{j=1}^{k-1} \gamma_{s_j s_{j+1}} e^{(t_{j+1} - t_j) \gamma_{s_j s_j}} \times e^{(t_0 - t_k) \gamma_{s_k s_k}}. \quad (6.10)$$

The initial probability $\Pr(X_0 = s_0)$ might be replaced by the s_0 th element of the stationary distribution of X_t , or dropped from the likelihood. In either case (6.10) may be maximized with respect to the γ_{rs} , $s \neq r$, if enough transitions have occurred — in general, no inferences can be made about transitions from r to s if none have been observed.

Partially observed trajectory

In practice trajectories may not be fully observed. One possibility is that the states $s_0, s_1, \dots, s_k$ of X_t at times $0 < t_1 < \dots < t_k$ are known, as are the numbers and types of transitions between the s_j , but that the times of these intervening transitions are unknown. A less informative possibility is that nothing is known about transitions, so that only the s_j and t_j are known. The likelihood is then (6.1) with $\Pr(X_{t_j} = s_j \mid X_{t_{j-1}} = s_{j-1})$ equal to the (s_{j-1}, s_j) element of $\exp\{(t_j - t_{j-1})G\}$, that is, $p_{s_{j-1}s_j}(t_j - t_{j-1})$, and $\Pr(X_0 = s_0)$ chosen according to context.

Example 6.9 (Two-state Markov chain) The simplest case has $S = 2$ states with transition intensities given by

$$G = \begin{pmatrix} -\gamma_{12} & \gamma_{12} \\ \gamma_{21} & -\gamma_{21} \end{pmatrix}, \quad \gamma_{12}, \gamma_{21} > 0.$$

Its eigendecomposition is

$$G = \begin{pmatrix} 1 & \gamma_{12} \\ 1 & -\gamma_{21} \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & -(\gamma_{21} + \gamma_{12}) \end{pmatrix} \frac{1}{\gamma_{12} + \gamma_{21}} \begin{pmatrix} \gamma_{21} & \gamma_{12} \\ 1 & -1 \end{pmatrix},$$

so the limiting distribution is $\pi^\top = (\gamma_{12} + \gamma_{21})^{-1}(\gamma_{21}, \gamma_{12})$, and

$$\exp(tG) = \frac{1}{\gamma_{12} + \gamma_{21}} \begin{pmatrix} \gamma_{21} + \gamma_{12}e^{l_2 t} & \gamma_{12}(1 - e^{l_2 t}) \\ \gamma_{21}(1 - e^{l_2 t}) & \gamma_{12} + \gamma_{21}e^{l_2 t} \end{pmatrix},$$

where $l_2 = -(\gamma_{12} + \gamma_{21}) < 0$ except in the trivial case $\gamma_{12} = \gamma_{21} = 0$, when the chain stays forever in its initial state.

The holding time in state r is exponential with parameter γ_{rs} , so the likelihood based on a trajectory fully observed on the interval $[0, t_0]$ with transitions $1 \rightarrow 2 \rightarrow 1 \rightarrow 2$ at $t_1 < t_2 < t_3$ is

$$\frac{\gamma_{21}}{\gamma_{12} + \gamma_{21}} \times \gamma_{12} e^{-t_1 \gamma_{12}} \times \gamma_{21} e^{-(t_2 - t_1) \gamma_{21}} \times \gamma_{12} e^{-(t_3 - t_2) \gamma_{12}} \times e^{-(t_0 - t_3) \gamma_{21}},$$

the first and last terms being the stationary probability $\Pr(X_0 = 1)$ and the probability that no transition occurs in $(t_3, t_0]$. Apart from the first term, the log likelihood is $n_{12} \log \gamma_{12} - \gamma_{12} t'_1 + n_{21} \log \gamma_{21} - \gamma_{21} t'_2$, where n_{rs} is the number of $r \rightarrow s$ transitions and t'_r the total time spent in state r .

Each row of $\exp(tG)$ tends to π^T as $t \rightarrow \infty$. One effect of this is that if the process is observed so intermittantly that $X_0, X_{t_1}, \dots$ are essentially independent, the transition probabilities $p_{rs}(t_j - t_{j-1})$ will almost equal elements of π^T , because $\exp\{l_2(t_j - t_{j-1})\} \doteq 0$. If so, then although $\gamma_{21}/(\gamma_{12} + \gamma_{21})$ will be estimable — it will be roughly the proportion of occasions that $X_t = 1$ — the individual rates γ_{12} and γ_{21} will not. The implication for design of studies involving such models is that X_t must be observed often enough that its successive values are correlated; otherwise only the stationary distribution is estimable. If several transitions occur every week, data obtained at monthly intervals will be essentially uninformative. ■

Example 6.10 (Breast cancer data) A model for these data has

$$G = \begin{pmatrix} -\gamma_{12} - \gamma_{13} & \gamma_{12} & \gamma_{13} \\ \gamma_{21} & -\gamma_{21} - \gamma_{23} & \gamma_{23} \\ 0 & 0 & 0 \end{pmatrix};$$

of course $\gamma_{31} = \gamma_{32} = 0$ because death is absorbing. A simpler model sets $\gamma_{13} = 0$, so a woman with the disease cannot die without first being unable to walk. Appropriate asymptotics take the number of women, rather than the number of observations on each, large; below we suppose that large-sample approximations are applicable with just 37 women. In practice it would be wise to check this by simulation.

The overall likelihood L is the product of independent contributions of form (6.1), one for each woman. Appreciable information might be lost by ignoring the terms $\Pr(X_0 = s_0)$, which comprise 37 of the 135 terms of L . Owing to the absorbing state, we cannot replace $\Pr(X_0 = s_0)$ with its stationary value

$$\lim_{t \rightarrow \infty} \Pr(X_t = 1) = \lim_{t \rightarrow \infty} \Pr(X_t = 2) = 0,$$

and we use $\lim_{t \rightarrow \infty} \Pr(X_t = s_0 \mid X_t \neq 3)$ instead, because only living women entered the study. Now for $s = 1, 2$,

$$\Pr(X_t = s \mid X_0 = r) = c_{rs,1} e^{l_1 t} + c_{rs,2} e^{l_2 t} + c_{rs,3} e^{l_3 t},$$

where $l_3 < l_2 < l_1 = 0$, and as this probability has limit zero we must have $c_{rs,1} = 0$. As $t \rightarrow \infty$, therefore,

$$\begin{aligned} \Pr(X_t = s \mid X_t \neq 3, X_0 = r) &= \frac{c_{rs,2} e^{l_2 t} + c_{rs,3} e^{l_3 t}}{c_{r1,2} e^{l_2 t} + c_{r1,3} e^{l_3 t} + c_{r2,2} e^{l_2 t} + c_{r2,3} e^{l_3 t}} \\ &\rightarrow \frac{c_{rs,2}}{c_{r1,2} + c_{r2,2}} \\ &= \frac{e_{2,s}}{e_{2,1} + e_{2,2}}, \end{aligned}$$

independent of r , where $e_{2,v}$ is the v th element of e_2^T , the left eigenvector of G corresponding to l_2 .

The missing value complicates the likelihood contribution for woman 24, which is

$$\frac{e_{2,1}}{e_{2,1} + e_{2,2}} \times p_{12}(3) \times \{p_{21}(3)p_{13}(6) + p_{22}(3)p_{23}(6)\}.$$

The maximized log likelihoods for the three- and four-parameter models are -107.43 and -107.39 . As $\gamma_{13} = 0$ lies on the boundary of the parameter space, the asymptotic distribution of the likelihood ratio statistic is $\frac{1}{2} + \frac{1}{2}\chi_1^2$; see Example 4.39. Its value, $2\{-107.39 - (-107.43)\} = 0.08$, supports the simpler model, for which maximum likelihood estimates and standard errors are $\hat{\gamma}_{12} = 0.116$ (0.025), $\hat{\gamma}_{21} = 0.057$ (0.035) and $\hat{\gamma}_{23} = 0.238$ (0.043). The transition rate γ_{21} is poorly determined, and taking the 95% confidence interval based on its profile likelihood, (0.014, 0.170), is preferable to using its standard error. The estimated mean times spent in states 1 and 2 are $\hat{\gamma}_{12}^{-1} = 8.6$ and $(\hat{\gamma}_{21} + \hat{\gamma}_{23})^{-1} = 3.4$ months, with death then occurring with estimated probability $\hat{\gamma}_{23}/(\hat{\gamma}_{21} + \hat{\gamma}_{23}) = 0.81$. Confidence intervals for these quantities should be based on profile likelihoods.

The non-zero eigenvalues of $\hat{G}$ are -0.33 and -0.08 , and examination of the estimated transition matrices between the later follow-up times suggests that there is some information in the small number of later transitions.

A more thorough analysis would assess the effect of initial status, for example by seeing if the likelihood increases significantly when the three-parameter model is fitted separately to each of the two initial groups. Of particular concern is the stationarity assumption, which is hard to justify here. The data are too sparse, however, for much further modelling to be conclusive. ■

Inhomogeneous chains

If the transition rates $\gamma_{rs}(t)$ depend on time then the fundamental equation (6.9) becomes $dp(t)^T/dt = p(t)^T G(t)$. This is a system of first-order ordinary differential equations, whose solution may be written formally as $p(t)^T = p(0)^T \exp\{\int_0^t G(s) ds\}$. Typically this will not be available explicitly, and the transition probabilities must be obtained using packages for solving systems of ordinary differential equations, or by discretizing time and fitting suitable models to the resulting transition probabilities.

Exercises 6.1

- 1 Classify the states of Markov chains with transition matrices

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

- 2 Find the eigendecomposition of

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}$$

and show that $p_{11}(n) = a + 2^{-n} \{b \cos(n\pi/2) + c \sin(n\pi/2)\}$ for some constants a , b and c . Write down $p_{11}(n)$ for $n = 0, 1$ and 2 and hence find a , b and c .

- 3 In Example 6.5, sketch how $p_{11}(n)$ depends on n when $l_2 < 0$, $l_2 > 0$ and $l_2 = 0$. Find $E(T_{11})$ by first showing that

$$\Pr(T_{11} = k) = \begin{cases} 1 - p, & k = 1, \\ pq(1 - q)^{k-2}, & k = 2, 3, \dots \end{cases}$$

- 4 Say when

$$P = \begin{pmatrix} 1 - p & p & 0 \\ 0 & 1 - p & p \\ p & 0 & 1 - p \end{pmatrix}, \quad 0 \leq p \leq 1,$$

has an equilibrium distribution, and write it down. Show that P has eigenvalues $1, (2 - 3p \pm i3^{1/2}p)/2$, and use them to say when the chain is ergodic.

- 5 Let X_t be a stationary first-order Markov chain with state space $\{1, \dots, S\}$, $S > 2$, and let I_t indicate the event $X_t = 1$. Is $\{I_t\}$ a Markov chain?
- 6 Consider a sequence 0100...10 of variables I_j and let $\bar{I}_t = (2k + 1)^{-1} \sum_{j=-k}^k I_{t+j}$ be the average of the $2k + 1$ variables centred at t .
- (a) Verify the calculations in Example 6.5.
- (b) Let the stationary first-order chain $\{I_t\}$ have state space $\{0, 1\}$ and transition probability matrix P . In the notation of Example 6.5, show that

$$\text{cov}(I_t, I_{t+j}) = \Pr(I_t = I_{t+j} = 1) - \Pr(I_t = 1)\Pr(I_{t+j} = 1) = pq l_2^j / (p + q)^2,$$

and deduce that with $m = 2k + 1$,

$$\text{var}(\bar{I}_t) = \frac{2}{m^2} \sum_{j=0}^{m-1} (m - j) \text{cov}(I_0, I_j) - \frac{\text{var}(I_0)}{m}.$$

It may be useful to know that for large n , $\sum_{j=0}^n j p^j \doteq p/(1 - p)^2$.

Give an expression for $\text{var}(\bar{I}_t)$, and show that it is roughly $(2 - p - q)/(p + q)$ times the corresponding expression for independent I_j .

- 7 Check the log likelihood for the six-parameter model given at the end of Example 6.8, obtain the maximum likelihood estimates and the fitted counts, and calculate Pearson's statistic. Give its degrees of freedom and assess the fit of the model.
- 8 A run of length m of a stationary Markov chain occurs when there is a sequence of form $X_t \neq s, X_{t+1} = \dots = X_{t+m} = s, X_{t+m+1} \neq s$. Show that this has probability $p_{ss}^{m-1}(1 - p_{ss})$ for $m = 1, 2, \dots$: the geometric density with mean $(1 - p_{ss})^{-1}$. Show that in a first-order chain the lengths of separate runs are independent. Is this true in higher-order chains? Can you construct a non-trivial 3×3 transition matrix for which it is impossible to use runs to falsify the independence model, whatever the length of the chain?
- 9 Recall that T_{rs} denotes the first-passage time from state r to state s . For the three-parameter model in Example 6.10, show that $E(T_{23}) = (g_{12} + g_{23})^{-1}(1 + g_{12})E(T_{13})$ and find the corresponding equation for $E(T_{13})$. Hence give expressions for $E(T_{13})$ and $E(T_{23})$ and show that their maximum likelihood estimates are 17 and 8.4 months respectively. What additional information do you need to compute standard errors for these estimates?
- 10 Modify the argument from the preceding question to find the moment-generating functions of T_{13} and T_{23} in terms of γ_{12} , γ_{21} , and γ_{23} . Hence check your formulae for $E(T_{13})$ and $E(T_{23})$.

- 11 Let $X_1, \dots, X_n$ be independent exponential variables with rates λ_j . Show that $Y = \min(X_1, \dots, X_n)$ is also exponential, with rate $\lambda_1 + \dots + \lambda_n$, and that $\Pr(Y = X_j) = \lambda_j / (\lambda_1 + \dots + \lambda_n)$. Hence write down an algorithm to simulate data from a continuous-time Markov chain with finite state space, using exponential and multinomial random number generators.
- 12 Observations $s_0, \dots, s_k$ on a discrete-time Markov chain with one-step transition matrix P are obtained at times $0 < t_1 < \dots < t_k$, where not all the $t_j - t_{j-1}$ equal unity. Write down the likelihood in terms of elements $p_{rs}(n)$ of P^n , $n = 1, 2, \dots$. Give explicitly the likelihood when the states 12311 of a three-state chain with stationary distribution π are observed at times 0, 1, 3, 4, 6. Explain how you would calculate the likelihood L for the data in Table 6.3, with three-month transition probability matrix

$$P = \begin{pmatrix} 1 - p_{12} & p_{12} & 0 \\ p_{21} & 1 - p_{21} - p_{23} & p_{23} \\ 0 & 0 & 1 \end{pmatrix}.$$

What value has L under this model? How could P be made more plausible?

Look carefully at the data.

- 13 Check the eigendecomposition of G in Example 6.9. Calculate the stationary distribution when $\gamma_{12} = 0$. Is this a surprise?

6.2 Markov Random Fields

6.2.1 Basic notions

The previous section described simple models for random variables indexed by a scalar, often time, so the variables can be visualized at points along an axis. Many applications require variables associated to points in space or in space-time, however, and then more general indexing sets are needed. Think, for example, of the colours of pixels in an image, the fertility of parts of a field or the occurrence of cancer cases at points on a map. This section outlines how our earlier ideas extend to some more complex settings. There is a close connection to notions of statistical physics, from which some of the terminology is derived.

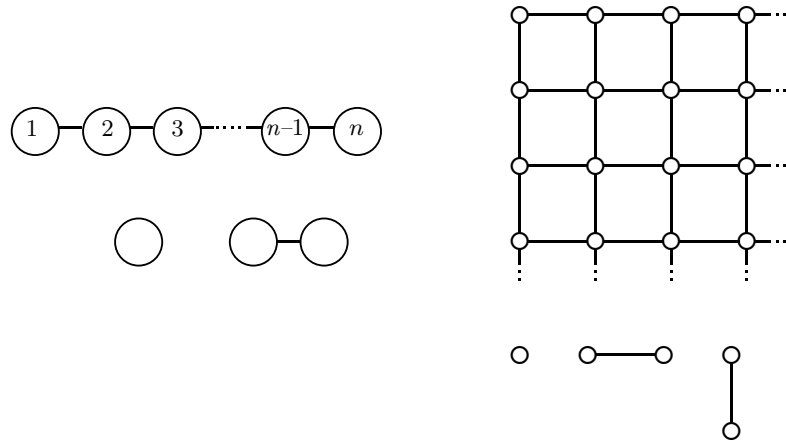
Our earlier discussion owed its relative simplicity to the Markov property — that the ‘future’ is independent of the ‘past’, conditional on the ‘present’ — whose importance suggests that we should seek its analogy here. The notions of ‘past’, ‘present’, and ‘future’ have no obvious spatial counterparts, but another formulation does generalize in a natural way. A sequence $Y_1, \dots, Y_n$ satisfies the Markov property if

$$\Pr(Y_{j+1} = y_{j+1} \mid Y_1 = y_1, \dots, Y_j = y_j) = \Pr(Y_{j+1} = y_{j+1} \mid Y_j = y_j)$$

for $j = 1, \dots, n-1$ and all y . This is equivalent to having each Y_j depend on the remaining variables $Y_{-j} = (Y_1, \dots, Y_{j-1}, Y_{j+1}, \dots, Y_n)$ only through the adjacent variables Y_{j-1} and Y_{j+1} (Exercise 6.2.1). To prepare for our generalization, let $\mathcal{N}_j$ denote the set of *neighbours* of j , given by $\mathcal{N}_j = \{j-1, j+1\}$ for $j = 2, \dots, n-1$, with $\mathcal{N}_1 = \{2\}$ and $\mathcal{N}_n = \{n-1\}$; hence $Y_{\mathcal{N}_j} = (Y_{j-1}, Y_{j+1})$ for $j \neq 1, n$, while $Y_{\mathcal{N}_1} = Y_2$ and $Y_{\mathcal{N}_n} = Y_{n-1}$. Then the Markov property for variables along an axis is equivalent to

$$\Pr(Y_j = y_j \mid Y_{-j} = y_{-j}) = \Pr(Y_j = y_j \mid Y_{\mathcal{N}_j} = y_{\mathcal{N}_j}), \quad (6.11)$$

Figure 6.4 Markov random fields. Left: neighbourhood structure for first-order Markov chain and its cliques and their subsets. Right: first-order neighbourhood structure, cliques and their subsets for rectangular grid of sites.



for all values of j and y . Thus Y_j depends on the other variables only through the neighbouring variables $Y_{\mathcal{N}_j}$. The probability densities on the left of (6.11) are known to statisticians as *full conditional densities*, while those on the right are called *local characteristics* in statistical physics.

For more complicated settings, let $\mathcal{J} = \{1, \dots, n\}$ be a finite set of *sites*, each with a random variable Y_j attached. In many applications each Y_j takes the same finite number k of values, and then $Y_1, \dots, Y_n$ may have at most k^n possible configurations; though finite, this number may be very large indeed. For any subset $\mathcal{A} \subset \mathcal{J}$, let $Y_{\mathcal{A}}$ denote the corresponding subset of $Y \equiv Y_{\mathcal{J}}$, and let $Y_{-\mathcal{A}}$ indicate $Y_{\mathcal{J}-\mathcal{A}}$, with $Y_j = Y_{\{j\}}$ and Y_{-j} defined as above. We impose a topology on $\mathcal{J}$ by defining a *neighbourhood system* $\mathcal{N} = \{\mathcal{N}_j, j \in \mathcal{J}\}$. The neighbours of j are the elements of $\mathcal{N}_j \subset \mathcal{J}$, the neighbourhoods $\mathcal{N}_j$ having the properties that

- $j \notin \mathcal{N}_j$ and
- $i \in \mathcal{N}_j$ if and only if $j \in \mathcal{N}_i$.

We visualize this as a graph $(\mathcal{J}, \mathcal{N})$ whose nodes correspond to sites, with two nodes joined by an edge if the sites are neighbours. We denote the union of $\{j\}$ and its neighbourhood by $\tilde{\mathcal{N}}_j = \mathcal{N}_j \cup \{j\}$. A subset $C \subset \mathcal{J}$ is *complete* if there are edges between all its nodes, and a maximal complete subset is a *clique* of $(\mathcal{J}, \mathcal{N})$; every pair of distinct elements of C are then neighbours, but C cannot be enlarged and retain this property. Let $\mathcal{C}$ denote the set of cliques and their subsets; in particular, $\mathcal{C}$ contains all singletons $\{j\}$ and the empty set $\emptyset$.

Example 6.11 (Markov chain) For the graph on the left of Figure 6.4, each interior variable has just two neighbours, and the end variables have just one. Hence $\mathcal{C} = \{\emptyset, \{1\}, \dots, \{n\}, \{1, 2\}, \dots, \{n-1, n\}\}$; the cliques are the $n-1$ adjacent pairs. ■

Example 6.12 (Pixillated image) Let $\mathcal{J}$ be an $m \times m$ rectangular array of sites, with neighbourhood structure shown on the right of Figure 6.4. Here $n = m^2$. Interior

Some authors do not insist that cliques be maximal.

sites have four neighbours, while boundary sites have two or three neighbours. The cliques are horizontal or vertical pairs of adjacent sites.

This neighbourhood system is said to be first-order. It is easy to envisage enlarging the neighbourhoods, for example by adding adjacent diagonal sites to give a second-order neighbourhood system. ■

Having defined a neighbourhood system analogous to that implicit in a Markov chain, the extension of the Markov property is clear: a probability distribution for Y is said to be a *Markov random field* with respect to $\mathcal{N}$ if Y_j is independent of $Y_{-\mathcal{N}_j}$ given $Y_{\mathcal{N}_j}$, or equivalently, if (6.11) holds: the conditional distribution of Y_j depends on the other variables only through those at the neighbouring sites.

Or sometimes a Markov field or a locally dependent Markov random field.

Although the local characteristics of Y are determined by its joint density, it is not true that any collection $\Pr(Y_1 | Y_{\mathcal{N}_1}), \dots, \Pr(Y_n | Y_{\mathcal{N}_n})$ of local characteristics yields a proper joint density. This is awkward, because in practice the local characteristics are much easier to deal with than the full joint density. Hence we ask which collections of $\Pr(Y_j | Y_{\mathcal{N}_j}) = \Pr(Y_j | Y_{-j})$ give well-defined joint distributions. It turns out that a *positivity condition* is needed, that for any $y_1, \dots, y_n$,

$$\Pr(Y_j = y_j) > 0 \text{ for } j = 1, \dots, n \text{ implies } \Pr(Y_1 = y_1, \dots, Y_n = y_n) > 0 : \quad (6.12)$$

if values of Y_j can occur singly they can occur together. In this case

$$\frac{\Pr(Y = y)}{\Pr(Y = y')} = \prod_{j=1}^n \frac{\Pr(y_j | y_1, \dots, y_{j-1}, y'_{j+1}, \dots, y'_n)}{\Pr(y'_j | y_1, \dots, y_{j-1}, y'_{j+1}, \dots, y'_n)} \quad (6.13)$$

for any two possible realizations y and y' of Y (Exercise 6.2.5). Hence (6.13) may be found for every possible y simply by taking a baseline y' and using the full conditional densities, the value of $\Pr(Y = y')$ being found by summing the ratios. Under the positivity condition, therefore, the full conditional densities determine a unique joint density for Y . This density must be unaffected by the labelling of the sites of $\mathcal{J}$, any change to which will leave (6.13) unaltered. This is a severe restriction, and we shall see at the end of this section that the joint density must have form

$$\Pr(Y = y) \propto \exp \{-\psi(y)\}, \quad (6.14)$$

where

$$\psi(y) = \sum_{C \in \mathcal{C}} \phi_C(y), \quad (6.15)$$

is a sum over all complete subsets C associated with the graph $(\mathcal{J}, \mathcal{N})$; this result, the *Hammersley–Clifford theorem*, is proved at the end of this section. Hence the only contributions to the joint density come from cliques of $(\mathcal{J}, \mathcal{N})$ and their subsets. Moreover the functions ϕ_C can be arbitrary, provided the total probability of (6.14) is finite. Many standard models have functions ϕ_C chosen so that (6.14) is an exponential family, but though convenient this is not essential. The sum in (6.15) could involve only cliques, as contributions from other complete subsets could be subsumed into those from the cliques. The collection of functions $\{\phi_C : C \in \mathcal{C}\}$ is called a *potential*.

The representation given by (6.14) and (6.15) is powerful because it enables systems whose global behaviour is very complex to be built from simple local components, namely the local characteristics determined by the ϕ_C . This is analogous to the notion that the transition probabilities of a Markov chain entirely determine its behaviour.

Example 6.13 (Markov chain) In Example 6.11 $\mathcal{C}$ contains the empty set, singletons, and pairs of adjacent sites, and hence

$$\psi(y) = a + \sum_{j=1}^n b_j(y_j) + \sum_{i \sim j} c_{ij}(y_i, y_j),$$

where the second sum is over all distinct pairs of neighbours, or equivalently all edges of the graph. The proportionality in (6.14) means that we can set $a = 0$, while setting $b_j \equiv b$ and $c_{ij}(\cdot, \cdot) = c(\cdot, \cdot)$ for all i and j gives a homogeneous field.

If the field is homogeneous and the Y_j take only values 0 and 1, we may write

$$\psi(y) = \sum_{j=1}^n b y_j + \sum_{j=1}^{n-1} (c_{10} y_j + c_{01} y_{j+1} + c_{11} y_j y_{j+1}),$$

and a little algebra gives

$$\Pr(Y_{j+1} = y_{j+1} \mid Y_1 = y_1, \dots, Y_j = y_j) = \frac{e^{(\beta + \gamma y_j) y_{j+1}}}{1 + e^{(\beta + \gamma y_j)}},$$

where β and γ are functions of b , c_{10} , c_{01} and c_{11} . As expected, this conditional probability depends on $y_1, \dots, y_j$ only through y_j and does not depend upon j directly. Hence it corresponds to a stationary first-order Markov chain with transition probabilities $\Pr(0 \mid 0) = (1 + e^\beta)^{-1}$ and $\Pr(0 \mid 1) = (1 + e^{\beta + \gamma})^{-1}$.

If the Y_j take values in the real line and we set

$$b(y_j) = \tau(y_j^2 - 2\mu y_j)/(2\sigma^2), \quad c(y_i, y_j) = (y_i - y_j)^2/(2\sigma^2),$$

then $\psi(y) = (y^T V y - 2\mu y^T V 1_n)/(2\sigma^2)$, where

$$V = \begin{pmatrix} \tau + 1 & -1 & 0 & \cdots & 0 & 0 \\ -1 & \tau + 2 & -1 & \cdots & 0 & 0 \\ 0 & -1 & \tau + 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \tau + 2 & -1 \\ 0 & 0 & 0 & \cdots & -1 & \tau + 1 \end{pmatrix},$$

and 1_n is an $n \times 1$ vector of ones. It follows that

$$\exp\{-\psi(y)\} \propto \exp\left\{-\frac{1}{2\sigma^2} (y - \mu 1_n)^T V (y - \mu 1_n)\right\},$$

which corresponds to the multivariate normal distribution with mean vector $\mu 1_n$ and covariance matrix $V/(2\sigma^2)$.

If $\tau = 0$, the rows of V sum to zero and the distribution is degenerate. Moreover (6.14) is integrable only if $\sigma^2 > 0$. This underlines the fact that although any choice of b_j and c_{ij} yields a proper joint density when each Y_j takes only a finite number

of values, restrictions may be needed to ensure this when any of the Y_j has infinite support.

Example 11.27 gives an application of this. ■

Example 6.14 (Ising model) Let $\mathcal{J}$ be an $m \times m$ grid of pixels, the j th of which can take values 0 and 1, corresponding to the colours white and black. As $n = m^2$, the sample space has size 2^{m^2} , about 10^{4932} even for a small image with $m = 128$. Under a first-order neighbourhood system the cliques are horizontal and vertical pairs of adjacent pixels; see Figure 6.4. Hence if b_j and c_{ij} are homogeneous, we can take

$$\psi(y) = \sum_j b(y_j) + \sum_{i \sim j} c(y_i, y_j)$$

the second sum being over all distinct cliques. The resulting probability distribution is the *Ising model* of statistical physics, which is important in investigations of ferromagnetism.

The conditional probability that $Y_j = 0$ given Y_{-j} is

$$\frac{\Pr(Y_j = 0, Y_{-j} = y_{-j})}{\Pr(Y_j = 0, Y_{-j} = y_{-j}) + \Pr(Y_j = 1, Y_{-j} = y_{-j})},$$

and on using (6.14) and cancelling all terms not involving y_j , we obtain

$$\frac{\exp \left\{ -b(0) - \sum_{i \in \mathcal{N}_j} c(y_i, 0) \right\}}{\exp \left\{ -b(0) - \sum_{i \in \mathcal{N}_j} c(y_i, 0) \right\} + \exp \left\{ -b(1) - \sum_{i \in \mathcal{N}_j} c(y_i, 1) \right\}};$$

thus the full conditional densities have form

$$\Pr(Y_j = 0 \mid Y_{-j}) = \frac{1}{1 + \exp \left\{ b(0) - b(1) + \sum_{i \in \mathcal{N}_j} c(y_i, 0) - c(y_i, 1) \right\}}.$$

Let n_1 denote $\sum_{i \in \mathcal{N}_j} I(Y_i = 1)$, the number of neighbours of site j that equal one, and define n_0 similarly; note that $n_0 = |\mathcal{N}_j| - n_1$. Now

$$\begin{aligned} \sum_{i \in \mathcal{N}_j} c(y_i, 0) - c(y_i, 1) &= n_0 c(0, 0) + n_1 c(1, 0) - n_0 c(0, 1) - n_1 c(1, 1) \\ &= n_0 \{c(0, 0) + c(1, 1) - c(0, 1) - c(1, 0)\} \\ &\quad + |\mathcal{N}_j| \{c(1, 0) - c(1, 1)\}, \end{aligned}$$

from which it follows that we can write

$$\Pr(Y_j = 0 \mid Y_{-j}) = \Pr(Y_j = 0 \mid Y_{\mathcal{N}_j}) = \frac{1}{1 + \exp(\beta + \gamma |\mathcal{N}_j| + \delta n_0)}. \quad (6.16)$$

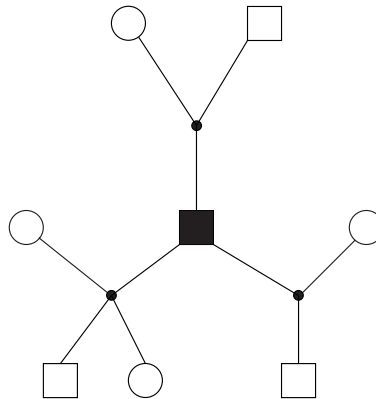
We interpret $\beta + |\mathcal{N}_j| \gamma$ as controlling the overall size of the probability and δ its dependence on the number of its white neighbours: $\gamma = 0$ means that the colour of cell j is independent of the colours around it, while (6.16) increases to one as $\gamma \rightarrow -\infty$.

Images with more colours may be dealt with by letting Y_j take $k > 2$ values, with an analogous argument giving the local characteristics. More complex neighbourhood

Ernst Ising (1900–1998) was one of the generation of German scientists whose careers were destroyed by the rise of Nazism. After a period of forced labour during the war he emigrated to the USA in 1949. The Ising model described in his 1924 PhD thesis was later used to account for the phase transition between the ferromagnetic and paramagnetic states.

$|\mathcal{A}|$ is the cardinality of the set $\mathcal{A}$.

Figure 6.5 A small genealogy. Females are shown as circles, males as squares, and marriages leading to offspring as dots. Thus the male shown by the solid square has two parents and three children by two marriages. This would be his neighbourhood in potentially a much larger pedigree.



structures will introduce more parameters into the model, while these ideas can be extended to fields that allow lines, textures and other features of real images. ■

Example 6.15 (Genetic pedigree) In the analysis of a genealogy, the sites typically correspond to individuals and Y_j to the genotype at a particular locus on the j th individual's DNA. Typically the genotype cannot be observed, but the phenotypes of some of the individuals are known. A simple example is in the ABO blood group system, where the observable phenotype blood group 'A' arises with genotypes AA and AO which are harder to observe; see Example 4.38. Two individuals in a pedigree are said to be *spouses* if they have mutual offspring in the pedigree, and each such pairing constitutes a *marriage*. A pedigree may be represented as a graph in which both individuals and marriages correspond to nodes, while the edges link each individual to his or her marriages and each marriage to the resulting offspring. See Figure 6.5.

The laws of genetic inheritance are Markovian. Genes are passed from parents to offspring in such a way that conditional on their parents' genotypes, individuals are independent of their earlier direct ancestors. It turns out that this dependence imposes a neighbourhood structure on the genotypes, with the neighbourhood for any individual defined to contain his parents, children and spouses. However distributions defined on this structure need not satisfy the positivity condition. A simple example is the ABO blood system: a person whose parents are both of type AB cannot be of type O. The fact that genetic models usually do not satisfy the positivity condition complicates statistical analysis of pedigree data. ■

Statistical inference for Markov random fields is generally based on the iterative simulation methods discussed in Section 11.3.3.

6.2.2 Directed acyclic graphs

Thus far we have supposed that all the Y_j have the same support and that the neighbourhood structure of the random field is known. The idea of expressing dependencies among variables as a graph is useful in more general settings, however, and it is then necessary to read off neighbourhoods from the joint distribution of $Y_1, \dots, Y_n$. Often

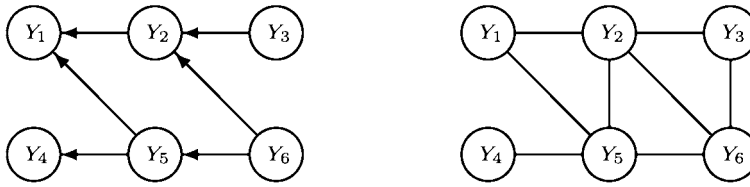


Figure 6.6 Directed acyclic and moral graphs. Left: directed acyclic graph representing (6.17). Right: moral graph, formed by moralizing the directed acyclic graph, that is, ‘marrying’ parents and dropping arrowheads.

the dependence structure is specified hierarchically, for example by stating the conditional distributions of Y_1 given Y_2 and of Y_2 given Y_3, Y_4 and so forth. The hierarchy may then be expressed using a directed graph, in which dependence of Y_1 on Y_2 is shown by an arrow from the *parent* Y_2 to the *child* Y_1 , and Y_1 is a *descendent* of Y_3 if there is a sequence of arrows from Y_3 to Y_1 . Such a graph is *directed* because each edge is an arrow, and *acyclic* if it is impossible to start from a node, traverse a path by following arrows, and return to the starting-point. The left of Figure 6.6 shows the directed acyclic graph for a model in which the joint density of $Y_1, \dots, Y_6$ factorizes as

$$f(y) = f(y_1 | y_2, y_5) f(y_2 | y_3, y_6) f(y_3) f(y_4 | y_5) f(y_5 | y_6) f(y_6). \quad (6.17)$$

For any directed acyclic graph we have

$$Y_j \perp \text{non-descendents of } Y_j \mid \text{parents of } Y_j, \quad \text{for all } j,$$

and (6.17) generalizes to

$$f(y) = \prod_{j \in \mathcal{J}} f(y_j \mid \text{parents of } y_j). \quad (6.18)$$

The density is then said to be *recursive* with respect to the directed acyclic graph. Acyclicity prevents a variable from introducing a degenerate density by being its own descendent.

A directed acyclic graph does not display all the neighbourhoods of the resulting Markov random field, but its *moral graph* does. This is obtained by *moralizing* the directed acyclic graph — ‘marrying’ or putting edges between any parents that share a child and then cutting off the arrowheads. In Figure 6.6, for example, the directed acyclic graph on the left shows us that Y_2 and Y_5 are parents of Y_1 , so they are joined in the moral graph on the right. This shows us that

$$\begin{aligned} \mathcal{N}_1 &= \{2, 5\}, \quad \mathcal{N}_2 = \{1, 3, 5, 6\}, \quad \mathcal{N}_3 = \{2, 6\}, \\ \mathcal{N}_4 &= \{5\}, \quad \mathcal{N}_5 = \{1, 2, 4, 6\}, \quad \mathcal{N}_6 = \{2, 3, 5\}. \end{aligned}$$

In general the full conditional density of y_j is

$$\begin{aligned} f(y_j \mid y_{-j}) &= \frac{f(y)}{\int f(y) dy_j} \\ &= \frac{\prod_{i \in \mathcal{J}} f(y_i \mid \text{parents of } y_i)}{\int \prod_{i \in \mathcal{J}} f(y_i \mid \text{parents of } y_i) dy_j} \\ &\propto f(y_j \mid \text{parents of } y_j) \prod_{i: y_j \text{ is child of } y_i} f(y_i \mid \text{parents of } y_i), \end{aligned}$$

$\perp$ means ‘is independent of’.

Also called a conditional independence graph.

A moral graph contains no unmarried parents.

because the integral only affects terms where y_j appears. In order for the denominator to be positive for any y_{-j} , the positivity condition must hold. If so, we see that $\mathcal{N}_j$ comprises the parents and children of Y_j , and any parents of Y_j 's children, precisely those variables joined to Y_j in the moral graph. Thus the distribution of Y satisfies (6.11), also called the *local Markov property*.

Consider a directed acyclic graph, let the *family* $\mathcal{F}_j$ consist of j and its parents, if any, and let $\mathcal{C}$ denote the cliques of the corresponding moral graph. Then as the families $\mathcal{F}_j$ yield cliques $C \in \mathcal{C}$, we may write

$$f(y) = \prod_j g(y_{\mathcal{F}_j}) = \prod_{C \in \mathcal{C}} h_C(y), \quad (6.19)$$

taking $g(y_{\mathcal{F}_j}) = f(y_j \mid \text{parents of } y_j)$. Thus we may write the joint density in terms of the cliques of an moral graph, analogous to (6.14) and (6.15). Let $\mathcal{A}$ and $\mathcal{B}$ be disjoint subsets of $\mathcal{J}$ that are *separated* by $\mathcal{D}$, that is, any path from an element of $\mathcal{A}$ to an element of $\mathcal{B}$ must pass through $\mathcal{D}$. Then under the positivity condition the distribution on the moral graph has the *global Markov property*, that $Y_{\mathcal{A}}$ and $Y_{\mathcal{B}}$ are independent conditional on $Y_{\mathcal{D}}$. To see this in the case where all the variables are discrete, suppose for now that $\mathcal{A} \cup \mathcal{B} \cup \mathcal{D} = \mathcal{J}$, and note that as no clique can contain elements of both $\mathcal{A}$ and $\mathcal{B}$, (6.19) implies that the joint density can be written as

$$f(y) = f(y_{\mathcal{A}}, y_{\mathcal{B}}, y_{\mathcal{D}}) = g_1(y_{\mathcal{A}}, y_{\mathcal{D}})g_2(y_{\mathcal{B}}, y_{\mathcal{D}}).$$

Thus

$$f(y_{\mathcal{A}}, y_{\mathcal{B}} \mid y_{\mathcal{D}}) = \frac{g_1(y_{\mathcal{A}}, y_{\mathcal{D}})g_2(y_{\mathcal{B}}, y_{\mathcal{D}})}{\sum_{y_{\mathcal{A}}} \sum_{y_{\mathcal{B}}} g_1(y_{\mathcal{A}}, y_{\mathcal{D}})g_2(y_{\mathcal{B}}, y_{\mathcal{D}})},$$

which factorizes in terms of $y_{\mathcal{A}}$ and $y_{\mathcal{B}}$, showing that any subset of $Y_{\mathcal{A}}$ is independent of any subset of $Y_{\mathcal{B}}$, conditional on $Y_{\mathcal{D}}$. The positivity condition ensures that the denominator here is positive for any $y_{\mathcal{D}}$. We now have only to note that if $\mathcal{A} \cup \mathcal{B} \cup \mathcal{D} \neq \mathcal{J}$, then $\mathcal{A}, \mathcal{B}$ can be enlarged to give sets $\mathcal{A}', \mathcal{B}'$ which together with $\mathcal{D}$ partition $\mathcal{J}$ such that $\mathcal{D}$ separates $\mathcal{A}', \mathcal{B}'$. Then $Y'_{\mathcal{A}} \perp Y'_{\mathcal{B}} \mid Y_{\mathcal{D}}$, implying that $Y_{\mathcal{A}} \perp Y_{\mathcal{B}} \mid Y_{\mathcal{D}}$, which is the global Markov property. The moral graph in Figure 6.6, for example, shows that

$$Y_1 \perp Y_3, Y_4 \mid Y_2, Y_5,$$

as can be verified from (6.17).

Markov properties of this sort are useful because they enable the computation of $f(y)$ or derived quantities to be broken into practicable steps. Sometimes the moral graph must be *triangulated* by adding edges to ensure that every cycle of length four or more contains an edge between two nodes that are not adjacent in the cycle itself. Triangulation can accelerate computation of $f(y)$ by making closed-form calculations possible for some model classes.

Example 6.16 (Belief network) Graphs may be used to represent supposed logical or causal relationships among variables and play an important role in probabilistic expert systems. Figure 6.7, for instance, shows a directed acyclic graph that represents

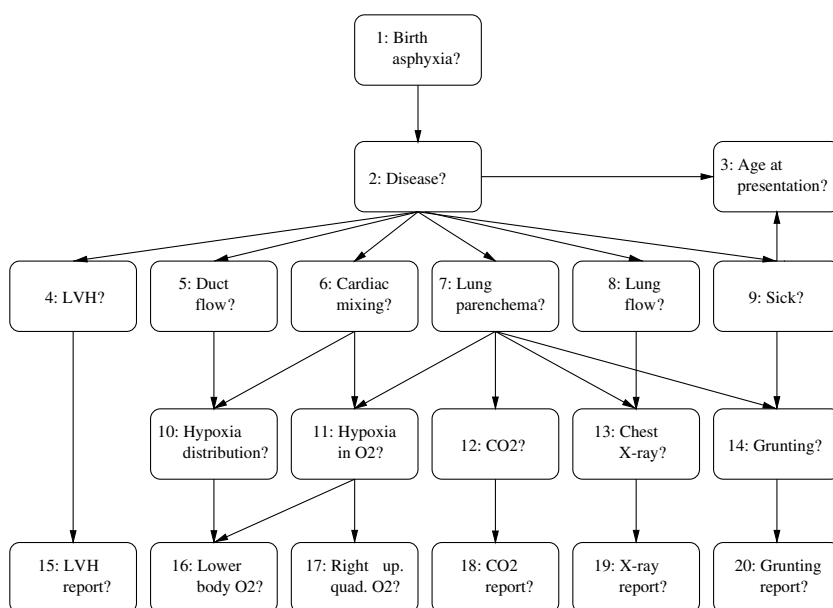


Figure 6.7 Directed acyclic graph representing the incidence and presentation of six possible diseases that would lead to a 'blue' baby (Spiegelhalter *et al.*, 1993). LVH means left ventricular hypertrophy.

the incidence and presentation of six diseases that would lead to a 'blue' baby. Early appropriate treatment is essential when such a child is born, and this expert system was developed to increase the accuracy of preliminary diagnoses. The graph shows, for example, that the level of oxygen in the lower body (node 16) is thought to be directly related to hypoxia distribution (node 10) and to its level when breathing oxygen (node 11). This last variable depends on the degree of mixing of blood in the heart (node 6) and the state of the blood vessels (parenchyma) in the lungs (node 7), and these two variables are directly influenced by which of the six possible levels the variable disease (node 2) has taken. Links such as those between nodes 6 and node 11 might be regarded as causal if poor cardiac mixing was known to contribute to hypoxia.

Each variable in such a network is typically treated as discrete, so the joint distribution of the variables is determined by a large number of multinomial distributions giving the terms on the right of (6.18). These are often obtained by eliciting opinions from experts and then updating these opinions, and perhaps the structure of the graph, as data become available. Table 6.6, for example, shows the expert view that left ventricular hypertrophy (LVH) would be present in 10% of cases of persistent foetal circulation, and that if present, it would be correctly reported in 90% of cases. The full distribution is given by specifying such tables for each of the 20 nodes of the graph, giving a sample space with more than one billion elements.

Now imagine that the LVH report for a baby is positive. In the light of this evidence the probabilities for the other variables will need updating, for example to ascribe new probabilities to the diseases or to determine which other diagnostic report will be most informative. Thus evidence must be propagated through the network to give the joint distribution of the other variables conditional on a positive LVH report. This involves

Table 6.6 Subjective expert assessments of conditional probability tables for links node 2 → node 4 and node 4 → node 15 in Figure 6.7 (Spiegelhalter *et al.*, 1993).

Node 2: Disease	Node 4: LVH	
	Yes	No
Persistent foetal circulation	0.10	0.90
Transposition of the great arteries	0.10	0.90
Tetralogy of Fallot	0.10	0.90
Pulmonary atresia with intact ventricular septum	0.90	0.10
Obstructed total anomalous pulmonary venous connection	0.05	0.95
Lung disease	0.10	0.90

Node 4: LVH	Node 15: LVH report	
	Yes	No
Yes	0.90	0.10
No	0.05	0.95

the cliques of the triangulated moral graph of Figure 6.7. Details are given in the references in the bibliographic notes. ■

Directed acyclic and their moral graphs play a useful role in the iterative simulation methods described in Section 11.3.3.

Hammersley–Clifford theorem

We now show that if the positivity condition (6.12) holds when all the Y_j take values in $\{0, \dots, L\}$, then the most general form that their joint density $f(y)$ can take is given by (6.14) and (6.15). Conversely these equations entail the Markov property (6.11) and positivity condition (6.12).

Let $\mathcal{Y} = \{0, \dots, L\}^n$ denote the sample space for $Y_1, \dots, Y_n$, and for any $y \in \mathcal{Y}$ let y_j^0 denote the vector $(y_1, \dots, y_{j-1}, 0, y_{j+1}, \dots, y_n)$. Under the positivity condition every element of $\mathcal{Y}$ occurs with positive probability, so we can define $\psi(y) = \log\{f(y)/f(0)\}$, where 0 represents a vector of n zeros. Now

$$\exp\{\psi(y) - \psi(y_j^0)\} = \frac{f(y)}{f(y_j^0)} = \frac{f(y_j | y_1, \dots, y_{j-1}, y_{j+1}, \dots, y_n)}{f(0 | y_1, \dots, y_{j-1}, y_{j+1}, \dots, y_n)} = \frac{f(y_j | y_{\mathcal{N}_j})}{f(0 | y_{\mathcal{N}_j})},$$

because the joint density satisfies the local Markov property, so knowing ψ will determine the full conditional densities and therefore the local characteristics of $f(y)$. Note that this implies that $\psi(y) - \psi(y_j^0)$ depends only on y_j and $y_{\mathcal{N}_j}$.

Now any function $\psi(y)$ has an expansion

$$\begin{aligned} \psi(y) &= \sum_{j=1}^n y_j a_j(y_j) + \sum_{1 \leq j < k \leq n} y_j y_k a_{jk}(y_j, y_k) \\ &\quad + \sum_{1 \leq j < k < l \leq n} y_j y_k y_l a_{jkl}(y_j, y_k, y_l) + \dots + y_1 \dots y_n a_{1\dots n}(y_1, \dots, y_n), \end{aligned}$$

This can be omitted at a first reading.

because we can set $y_j a_j(y_j) = \psi(0, \dots, 0, y_j, 0, \dots, 0) - \psi(0)$, with analogous formulae for the other a -functions. We must now show that for any subset C of $\{1, \dots, n\}$, the corresponding a -function may be non-null if and only if C is a clique of the graph. Consider y_1 without loss of generality, and recall that owing to (6.11), $\psi(y) - \psi(y_1^0)$ depends only on y_1 and $y_{\mathcal{N}_1}$. Now $\psi(y) - \psi(y_1^0)$ equals

$$y_1 \left\{ a_1(y_1) + \sum_{2 \leq j \leq n} y_j a_{1j}(y_1, y_j) + \sum_{2 \leq j < k \leq n} y_j y_k a_{1jk}(y_1, y_j, y_k) \right. \\ \left. + \dots + y_2 \dots y_n a_{1\dots n}(y_1, \dots, y_n) \right\},$$

and this must be free of y_l for any $l \notin \mathcal{N}_1$. On setting $y_j = 0$ for $j \neq 1, l$, we see that $a_{1l}(y_1, y_l) = 0$ for every possible y_l . Suitable choices of y show in like wise that every other a -function involving y_l must be identically zero. As the same is true for every other node, the Markov property (6.11) implies that the only non-zero functions $a_{j_1 \dots j_k}$ are those in which $j_1, \dots, j_k$ are all neighbours, that is, form a subset of a clique, and this entails (6.14) and (6.15).

For the converse, note that any set of a -functions gives a density $f(y)$ that satisfies the positivity condition (6.12). Now $\psi(y) - \psi(y_j^0)$ depends on x_l only if there is a non-null a -function containing both y_j and y_l , so the local characteristic $f(y_j | y_1, \dots, y_{j-1}, y_{j+1}, \dots, y_n)$ also depends only on those y_l that are neighbours of y_j . Thus (6.14) and (6.15) together imply (6.11) and (6.12).

Exercises 6.2

- 1 Show that the Markov property for a sequence $Y_1, \dots, Y_n$ is equivalent to (6.11).
- 2 Give the cliques for the second-order neighbourhood system in Example 6.12.
- 3 Give the cliques for a second-order Markov chain, and hence write down the form of the most general density for it, under the positivity condition.
- 4 Consider two binary random variables with local characteristics

$$\begin{aligned} \Pr(Y_1 = 1 | Y_2 = 0) &= \Pr(Y_1 = 0 | Y_2 = 1) = 1, \\ \Pr(Y_2 = 0 | Y_1 = 0) &= \Pr(Y_2 = 1 | Y_1 = 1) = 1. \end{aligned}$$

Show that these do not determine a joint density for (Y_1, Y_2) . Is the positivity condition satisfied?

- 5 Let the density of a random variable $Y = (Y_1, \dots, Y_n)$ satisfy (6.12), and let the sample space be the set $\mathcal{Y} = \{y : f(y) > 0\}$. If $(y_1, \dots, y_n)$ and $(x_1, \dots, x_n)$ are two elements of $\mathcal{Y}$, use the identity

$$f(y_1, \dots, y_n) = f(y_n | y_1, \dots, y_{n-1}) f(y_1, \dots, y_{n-1})$$

to show that

$$f(y_1, \dots, y_n) = \frac{f(y_n | y_1, \dots, y_{n-1})}{f(x_n | y_1, \dots, y_{n-1})} f(y_1, \dots, y_{n-1}, x_n),$$

and then that

$$f(y_1, \dots, y_{n-1}, x_n) = \frac{f(y_{n-1} | y_1, \dots, y_{n-2}, x_n)}{f(x_{n-1} | y_1, \dots, y_{n-2}, x_n)} f(y_1, \dots, y_{n-2}, x_{n-1}, x_n),$$

Hence establish (6.13).
(Besag, 1974)

- 6 Use induction on the number of variables to prove (6.18).
- 7 Let $\mathcal{G}_m$ be the graph obtained by moralizing a finite directed acyclic graph $\mathcal{G}$. Show that every family of $\mathcal{G}$ is a clique of $\mathcal{G}_m$ but that the converse is false (consider the moral graph for Figure 6.7).
- 8 A subset $\mathcal{A}$ of the nodes of a graph $\mathcal{G}$ is *ancestral* if $\mathcal{A}$ contains the parents and neighbours of a whenever $a \in \mathcal{A}$. Show that if the density of Y is recursive with respect to $\mathcal{G}$, the marginal density of $Y_{\mathcal{A}}$ is recursive with respect to subgraph induced on $\mathcal{A}$. Now consider the moral graph $\mathcal{G}'_m$ of the smallest ancestral set containing $\mathcal{A} \cup \mathcal{B} \cup \mathcal{D}$, and suppose that $\mathcal{D}$ separates $\mathcal{A}$ from $\mathcal{B}$ in $\mathcal{G}'_m$. Show that $Y_{\mathcal{A}} \perp Y_{\mathcal{B}} \mid Y_{\mathcal{D}}$.
- 9 Are the moral graphs for Figures 6.6 and 6.7 triangulated?
- 10 Write down the directed acyclic and moral graphs for X , Y , and I under the missing data models described in Section 5.5. Use them to give an equation-free explanation of the differences among the models and of their consequences.
- 11 Give the form of the a -functions of page 253 when $L = 1$ and $n = 3$, and hence verify the Hammersley–Clifford theorem when (i) none of 1, 2, and 3 are neighbours; (ii) $1 \sim 2$ only; (iii) $1 \sim 2 \sim 3$ only; and (iv) $1 \sim 2 \sim 3 \sim 1$, where $\sim$ means ‘is a neighbour of’. In each case give (6.15).
- 12 Suppose that the variables $U_1, \dots, U_n$ of a Markov random field have joint density

$$f(u; \theta) = c(\theta) \exp \left[-\frac{1}{2} \left\{ \theta_1 \sum_{j \sim j'} (u_j - u_{j'})^2 + \theta_2 \sum_{j=1}^n u_j^2 \right\} \right], \quad \theta_1 > 0, \theta_2 \geq 0,$$

where the first sum is over all pairs of neighbours, each pair being taken only once. Let $\mathcal{N}_j$ denote the neighbours of node j .

(a) Show that their joint distribution is normal with covariance matrix determined by $\theta_1 I_n + \theta_2 A$, where the j th diagonal element of the $n \times n$ adjacency matrix A is the number of neighbours of node j , and its off-diagonal elements are

$$a_{jj'} = \begin{cases} -1, & j \sim j' \\ 0, & \text{otherwise.} \end{cases}$$

(b) Show that the density is degenerate when $\theta_2 = 0$, but that otherwise $c(\theta) = (2\pi)^{-n/2} \prod_j (\theta_2 + \theta_1 a'_j)$, where the a'_j are the eigenvalues of A .

(c) Suppose that conditional on the values u of unseen variables U , $Y_1, \dots, Y_n$ are independent with $Y_j \mid U \sim N(u_j, \sigma^2)$. Show that the joint density of U given Y is well-defined even if $\theta_2 = 0$ provided that $\sigma^2 > 0$, and that the density of $U_j \mid Y, U_{-\mathcal{N}_j}$ involves on y_j and $u_{\mathcal{N}_j}$.

(d) Work out the details using the matrix V in Example 6.13.

6.3 Multivariate Normal Data

6.3.1 Multivariate dependence

When there is a single response variable, analysis is relatively simple, the crucial aspect being how the distribution of that variable depends on any covariates. Problems with just one response variable are common in practice and typically are readily interpretable, but cases arise where relations among two or more responses are to be modelled, and to these we now briefly turn. We motivate our discussion by an example.

Mechanics (C)	Vectors (C)	Algebra (O)	Analysis (O)	Statistics (O)
77	82	67	67	81
63	78	80	70	81
75	73	71	66	81
55	72	63	70	68
63	63	65	70	63
⋮	⋮	⋮	⋮	⋮
15	38	39	28	17
5	30	44	36	18
12	30	32	35	21
5	26	15	20	20
0	40	21	9	14

Table 6.7 Marks out of 100 in five mathematics examinations for the first and last five of 88 students (Mardia *et al.*, 1979, pp. 3–4). Some of the examinations were closed-book (C), and others were open-book (O).

Example 6.17 (Maths marks data) Table 6.7 gives marks out of 100 for the first and last five students out of 88 who took five mathematics examinations. As we would anticipate, the top students tend to do best in all the exams, and the worst dismally in all, so the marks for each student are related. This is shown by the modified *scatterplot matrix* in Figure 6.8, the below-diagonal part of which contains scatterplots of the marks for each examination against those for every other one, for all 88 students. Each column of marks has been centred at zero for comparison with the panels above the diagonal, which are discussed in Example 6.20. The original marks are shown by the light histograms along the diagonal. The average marks in mechanics and in vectors were about 40 and 50, but the panel in the second row and first column shows that one candidate got about 40 more marks than this in mechanics and about 30 more in vectors; this is the first person in Table 6.7. The below-diagonal panels show generally positive association between marks for different subjects, but its strength varies. For example, algebra is strongly associated with all the other subjects, whereas the first column suggests that while mechanics is related to vectors and algebra, its association with analysis and statistics is weaker.

We could rank students by their overall averages, but this would not be useful if the question of interest is how the marks for different exams relate to one another. We would then have five response variables, and we would need to model the forms of dependence that might arise. ■

Simpson’s paradox

The easiest way to deal with the multitude of potential dependencies in such situations is to ignore as many of them as possible. For example, if the response is bivariate, (Y_1, Y_2) , and there is a single explanatory variable, x , one might just model Y_1 as a function of x , regardless of Y_2 . Unfortunately this can be badly misleading, as Figure 6.9 illustrates. Its left panel shows how a continuous variable Y_1 depends on x for two values of the discrete variable Y_2 . For each value of y_2 , the mean of Y_1 , $E(Y_1 \mid Y_2 = y_2, x)$, increases with x , as shown by the positive slope of the regression of Y_1 on x . However, the right panel shows that when Y_2 is ignored the mean of Y_1

Figure 6.8 Modified scatterplot matrix for the full maths marks data. Below the diagonal are scatterplots (and sample correlation coefficient) for the centred pairs of marks; for example, the lower left panel shows results for statistics plotted against those for mechanics. Above the diagonal are scatterplots of residuals (and sample partial correlation coefficients): for example, the top right panel shows the dependence remaining between mechanics and statistics after adjusting for the other variables. The diagonal shows histograms of the variables (light) and of the residuals from regression on all other variables, centred at the variable mean (dark), with the marginal and partial standard deviations.

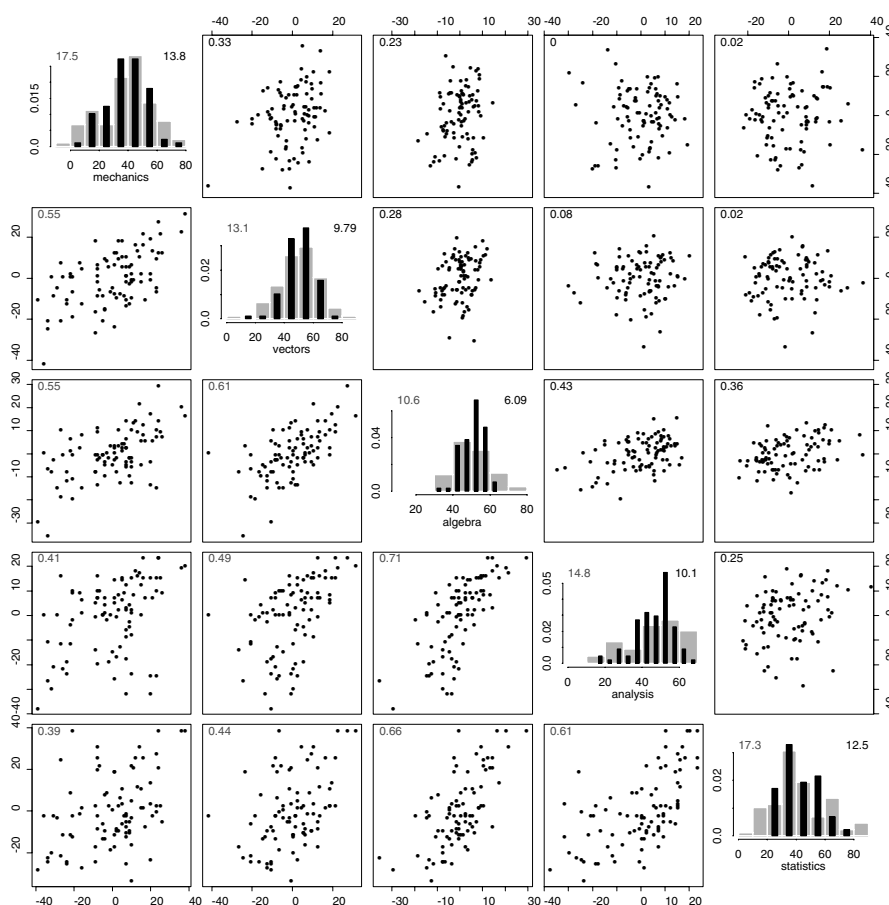
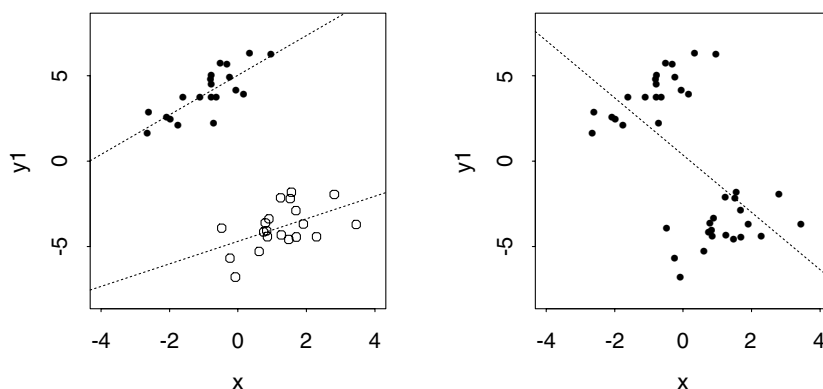


Figure 6.9 Artificial data illustrating Simpson's paradox. The left panel shows how Y_1 depends on x for each of two values of Y_2 , with observations with $y_2 = 0$ shown by blobs and those with $y_2 = 1$ shown as circles. The lines are from separate straight-line regression fits of y_1 on x for each value of y_2 and show positive association. The right panel shows the fit to the data ignoring Y_2 , for which the association is negative.



E. H. Simpson called attention to this effect in 1951, although it was known to G. U. Yule almost 50 years earlier.

decreases with x , that is, $E(Y_1 | x)$ has negative slope as a function of x . This effect — *Simpson's paradox* — is due to the fact that marginalization of the joint distribution of (Y_1, Y_2) over Y_2 has reversed the sign of the association between Y_1 and x . Here a plot at once reveals that it is a bad idea to fit a common line to both groups, but the

Age (years)	Smokers	Non-smokers
Overall	139/582 (24)	230/732 (31)
18–24	2/55 (4)	1/62 (2)
25–34	3/124 (2)	5/157 (3)
35–44	14/109 (13)	7/121 (6)
45–54	27/130 (21)	12/78 (15)
55–64	51/115 (44)	40/121 (33)
65–74	29/36 (81)	101/129 (78)
75+	13/13 (100)	64/64 (100)

Table 6.8 Twenty-year survival and smoking status for 1314 women (Appleton *et al.*, 1996). The smoker and non-smoker columns contain number dead/total (% dead).

paradox arises also in contexts where it is not obvious what to plot, and association may be strengthened or weakened as well as reversed.

Example 6.18 (Smoking and the Grim Reaper) Table 6.8 shows data on smoking and survival for 1314 women in Whickham, near Newcastle upon Tyne in the north of England. In 1972–1974 data were collected by surveying people on the town electoral register, and among the many variables collected were age and smoking habits at that time. Twenty years later a follow-up survey was conducted and it was determined if each woman in the original survey had yet died. Just 162 women had smoked before the first survey but were non-smokers at that time, and there were only 18 whose smoking habits were not recorded; these 180 women have been excluded.

The first line of the table shows a surprising apparent health bonus of smoking: the death rate for smokers, 24%, is lower than that for non-smokers, 31%. However a breakdown of the overall figures by age shows that the death rate for smokers is higher for every age category except 25–34 and over 75 years. On inspection the reason for this discrepancy is obvious: in the early 1970s there were many more non-smokers than smokers in the age range 65–74, and as old age makes death even more likely than does smoking, most of these did not survive until the follow-up survey. To learn about the effect of smoking on health, we should compare women of the same age. Thus the appropriate analysis involves comparisons within age groups, not after merging the data across them. ■

These cautionary examples show that it is unwise to collapse data by ignoring variables without first examining the full dependence structure to ensure that it is safe to do so. Data with several responses are common in the social sciences, psychology, epidemiology, and public health, where outcomes may depend on numerous variables that are related in a complicated way. The modelling of such data has been extensively studied, and we only scratch its surface here, confining our discussion to the most basic models for continuous data. Discrete data such as those in Table 6.8 arise in many applications; see Chapter 10.

6.3.2 Multivariate normal distribution

Let $Y_1, \dots, Y_n$ be a random sample from the $N_p(\mu, \Omega)$ density. That is, $Y_j = (Y_{1j}, \dots, Y_{pj})^T$ has the multivariate normal distribution with mean vector $\mu = (\mu_1, \dots, \mu_p)^T$ and $p \times p$ nonsingular covariance matrix Ω , whose (r, s) element is ω_{rs} . As each of the Y_j has density (3.20), the log likelihood is

$$\begin{aligned}\ell(\mu, \Omega) &\equiv -\frac{1}{2} \left\{ n \log |\Omega| + \sum_{j=1}^n (Y_j - \mu)^T \Omega^{-1} (Y_j - \mu) \right\} \\ &= -\frac{1}{2} \left\{ n \log |\Omega| + \sum_{j=1}^n (Y_j - \bar{Y} + \bar{Y} - \mu)^T \Omega^{-1} (Y_j - \bar{Y} + \bar{Y} - \mu) \right\} \\ &= -\frac{1}{2} \left\{ n \log |\Omega| + n(\bar{Y} - \mu)^T \Omega^{-1} (\bar{Y} - \mu) + \sum_{j=1}^n (Y_j - \bar{Y})^T \Omega^{-1} (Y_j - \bar{Y}) \right\} \\ &= -\frac{1}{2} \{ n \log |\Omega| + n(\bar{Y} - \mu)^T \Omega^{-1} (\bar{Y} - \mu) + (n-1) \text{tr}(\Omega^{-1} S) \} \quad (6.20)\end{aligned}$$

where the $p \times 1$ vector and $p \times p$ matrix

$$\bar{Y} = n^{-1} \sum Y_j, \quad S = (n-1)^{-1} \sum (Y_j - \bar{Y})(Y_j - \bar{Y})^T,$$

are the sample average and covariance matrix. These are minimal sufficient statistics for μ and Ω . The second and third equalities before (6.20) rest on the identities

$$\begin{aligned}\sum (Y_j - \bar{Y}) \Omega^{-1} (\bar{Y} - \mu) &= 0, \\ \sum (Y_j - \bar{Y})^T \Omega^{-1} (Y_j - \bar{Y}) &= \sum \text{tr}\{(Y_j - \bar{Y})^T \Omega^{-1} (Y_j - \bar{Y})\} \\ &= \sum \text{tr}\{\Omega^{-1} (Y_j - \bar{Y})(Y_j - \bar{Y})^T\} \\ &= \text{tr}\left\{ \Omega^{-1} \sum (Y_j - \bar{Y})(Y_j - \bar{Y})^T \right\}.\end{aligned}$$

As Ω is a positive definite matrix, so is $\Delta = \Omega^{-1}$, and it follows that for any Ω , the maximum of (6.20) with respect to μ is at $\hat{\mu} = \bar{Y}$, the maximum likelihood estimator. In order to maximize $\ell(\hat{\mu}, \Omega)$, we make the 1–1 transformation from Ω to Δ , in terms of which $\ell(\hat{\mu}, \Delta) = \frac{1}{2} \{ n \log |\Delta| - (n-1) \text{tr}(\Delta S) \}$. Differentiation with respect to Δ shows that the maximum likelihood estimate of $\Delta^{-1} = \Omega$ is $\hat{\Omega} = n^{-1}(n-1)S$; with probability one this has rank p if $n > p$. Otherwise its rank is $n-1$.

As in the scalar case S is unbiased. We let $\hat{\omega}_{rs}$ denote the (r, s) element of S ; this is the sample covariance between the r th and s th components of Y . The sample variances lie on the diagonal of S . The sample correlations are $\hat{\omega}_{rs}/(\hat{\omega}_{rr}\hat{\omega}_{ss})^{1/2}$, the (r, s) elements of $D^{-1/2}SD^{-1/2}$, where D is the diagonal matrix $\text{diag}(\hat{\omega}_{1,1}, \dots, \hat{\omega}_{p,p})$ (Exercise 6.3.2).

Example 6.19 (Maths marks data) Table 6.9 shows the averages, variances, and correlations for the maths marks data. The best results are on vectors and algebra, and the worst on mechanics and statistics. The numbers below the diagonal show positive correlations among the variables, with the strongest those between algebra and the other subjects. The most variable marks are for mechanics and statistics, with

Exercise 6.3.1 gives the details.

	Mechanics	Vectors	Algebra	Analysis	Statistics
Mechanics	17.5/13.8	0.33	0.23	−0.00	0.03
Vectors	0.55	13.2/9.8	0.28	0.08	0.02
Algebra	0.55	0.61	10.6/6.1	0.43	0.36
Analysis	0.41	0.49	0.71	14.8/10.1	0.25
Statistics	0.39	0.44	0.66	0.61	17.3/12.5
Average	39.0	50.6	50.6	46.7	42.3

Table 6.9 Summary statistics for maths marks data. The sample correlations between variables are below the diagonal, and the sample partial correlations are above the diagonal. The diagonal contains sample standard deviation/ sample partial standard deviation.

sample standard deviations $\widehat{\omega}_{rr}^{1/2}$ of 17.5 and 17.3 respectively, while that for algebra is smallest, at 10.6. Although the averages for mechanics and statistics are smallest, there is a wider spread of results for these subjects. The values above the diagonal are discussed in Example 6.20. ■

Extensions of the arguments for univariate data show that

$$\bar{Y} \sim N_p(\mu, n^{-1}\Omega), \quad \text{independent of} \quad (n-1)S \sim W_p(n-1, \Omega), \quad (6.21)$$

where $W_p(v, \Omega)$ denotes the p -dimensional Wishart distribution with $p \times p$ parameter matrix Ω and v degrees of freedom. In fact, if $Z_1, \dots, Z_v$ is a random sample from the $N_p(0, \Omega)$ distribution, then $Z_1 Z_1^\top + \dots + Z_v Z_v^\top \sim W_p(v, \Omega)$; when $p = 1$ and $\Omega = 1$, the Wishart distribution reduces to the chi-squared.

The multivariate extension of the t statistic is Hotelling's T^2 statistic,

$$T^2 = n(\bar{Y} - \mu)^\top S^{-1}(\bar{Y} - \mu) \sim \frac{p(n-1)}{n-p} F_{p, n-p},$$

which can be used to test hypotheses and form confidence regions for elements of μ .

6.3.3 Graphical Gaussian models

The structure of the multivariate normal density means that variables depend on each other in a particularly simple way. Before getting into details, we need some notation. Let $\mathcal{S}$ be a subset of the integers $\{1, \dots, p\}$, of cardinality $|\mathcal{S}|$, and let $Y_{\mathcal{S}}$ and $Y_{-\mathcal{S}}$ be the sets of variables $\{Y_s, s \in \mathcal{S}\}$ and $\{Y_s, s \notin \mathcal{S}\}$. If $\mathcal{S} = \{r\}$, we write $Y_{\mathcal{S}} = Y_r$ and $Y_{-\mathcal{S}} = Y_{-r}$. For two such subsets $\mathcal{A}$ and $\mathcal{B}$, let $\Omega_{\mathcal{A}, \mathcal{B}}$ be the $|\mathcal{A}| \times |\mathcal{B}|$ matrix with elements $\omega_{ab} = \text{cov}(Y_a, Y_b)$, and let $\Omega_{\mathcal{A}|\mathcal{B}} = \text{cov}(Y_{\mathcal{A}} | Y_{\mathcal{B}})$ be the $|\mathcal{A}| \times |\mathcal{A}|$ conditional covariance matrix of $Y_{\mathcal{A}}$ given the value of $Y_{\mathcal{B}}$; we write its elements as $\omega_{a_1, a_2|\mathcal{B}}$.

Equation (3.21) establishes that the conditional distribution of $Y_{\mathcal{S}}$ given $Y_{-\mathcal{S}} = y_{-\mathcal{S}}$ is normal with mean vector and covariance matrix

$$\mu_{\mathcal{S}} + \Omega_{\mathcal{S}, -\mathcal{S}} \Omega_{-\mathcal{S}, -\mathcal{S}}^{-1} (y_{-\mathcal{S}} - \mu_{-\mathcal{S}}), \quad \Omega_{\mathcal{S}, \mathcal{S}} - \Omega_{\mathcal{S}, -\mathcal{S}} \Omega_{-\mathcal{S}, -\mathcal{S}}^{-1} \Omega_{-\mathcal{S}, \mathcal{S}}. \quad (6.22)$$

Thus the conditional mean depends linearly on the values of the known variables, and the conditional variance is independent of them. If $\mathcal{S} = \{r\}$ and the conditional variance of Y_r , $\omega_{rr|-r}$, is much smaller than the unconditional variance ω_{rr} , then

knowing Y_{-r} is highly informative about the distribution of Y_r . Thus it will be useful to compare estimates of these variances. It is also useful to learn how knowledge of the other variables affects the covariance of Y_r and Y_s . Their 2×2 conditional covariance matrix is given by (6.22), with $\mathcal{S} = \{r, s\}$, and their *partial correlation*,

$$\rho_{rs|-S} = \frac{\omega_{rs|-S}}{(\omega_{rr|-S} \omega_{ss|-S})^{1/2}},$$

represents the correlation between Y_r and Y_s conditional on the remaining variables. The quantities on the right are sometimes called the *partial variances* and *partial covariance*. On page 264 we show that the partial correlation equals minus one times the (r, s) element of the correlation matrix constructed from Ω^{-1} . Thus partial variances, correlations and covariances of Y are readily computed from Ω , and we can use the transformation property of maximum likelihood estimators to estimate $\rho_{rs|-S}$ and so forth by the same functions of $\hat{\Omega}$.

Example 6.20 (Maths marks data) The second diagonal elements in Table 6.9 give the sample partial standard deviations $\hat{\omega}_{rr|-r}^{1/2}$ for each subject. According to the normal model, our best guess of a student's mark in algebra without knowledge of his other marks would be 50.6, with standard deviation 10.6: a 95% confidence interval is $51 \pm 1.96 \times 11 = (29, 73)$, which is virtually useless. If we knew $\bar{y}$ and $\hat{\Omega}$ and his marks y_{-r} for the other four subjects, however, we could replace the components of μ and Ω in (6.22) with $\mathcal{S} = \{r\}$ by estimates, giving estimated score $\bar{y}_r + \hat{\Omega}_{r,-r} \hat{\Omega}_{-r,-r}^{-1} (y_{-r} - \bar{y}_{-r})$. The estimated conditional standard deviation $\hat{\omega}_{rr|-r}^{1/2} = 6.1$ is appreciably smaller than the unconditional value.

The above-diagonal part of Table 6.9 shows the sample partial correlations. A good mark at algebra is correlated positively with each of the other variables, given the remainder. Given the other variables, however, mechanics seems to be unrelated to analysis or statistics, and likewise for vectors: the upper right corner of the matrix is essentially zero. Thus the subjects split into three groups: vectors and mechanics; analysis and statistics; and algebra. Variables in the first two pairs are partially correlated with each other and with algebra, which itself is partially correlated with all four other variables.

This information is displayed more fully in the above-diagonal panels of Figure 6.8. Set $\mathcal{S} = \{r, s\}$, and let y denote the $n \times p$ data matrix whose j th row is $y_j^\top$, y_r the r th column of y , and y_{-S} the $n \times (p - 2)$ array comprising all columns of y but the r th and s th. Then the vertical axes show the $n \times 1$ vectors of sample values

$$y_{r|-S} = y_r - \bar{y}_r - \hat{\Omega}_{r,-S} \hat{\Omega}_{-S,-S}^{-1} (y_{-S} - \bar{y}_{-S})$$

of the scalar random variable

$$Y_{r|-S} = Y_r - \mu_r - \Omega_{r,-S} \Omega_{-S,-S}^{-1} (Y_{-S} - \mu_{-S}),$$

while the horizontal axes show the $y_{s|-S}$'s. The quantities $Y_{r|-S}$ are normal with means zero and variances $\Omega_{r,-S} \Omega_{-S,-S}^{-1} \Omega_{-S,r}$, and partial correlation

$$\text{corr}(Y_r, Y_s | Y_{-S}) = \text{corr}(Y_{r|-S}, Y_{s|-S}) = \rho_{rs|-S},$$

while the correlation coefficient between the sample versions is the corresponding sample quantity $\widehat{\rho}_{rs|S}$. Thus the scatterplot in the first row and third column shows the association between mechanics on the vertical axis and algebra on the horizontal axis after adjusting for dependence on the other variables. The partial correlation of 0.23 shows that some positive correlation remains after allowing for the other variables. Summary in terms of partial correlations seems reasonable, as none of the panels shows much nonlinearity, but there is a possible outlier in the lower left corner of panels (1, 2) and (2, 1). This is a person whose marks $y_{81}^T = (3, 9, 51, 47, 40)$ are dire for applied mathematics but not for pure mathematics or statistics. Dropping him makes little change to the correlations or partial correlations.

The diagonal of the scatterplot matrix compares histograms of the raw marks y_r and the marks $y_{r|-r} + \bar{y}_r$ after adjusting for all the other variables, with the sample standard deviations of these vectors. ■

Conditional independence graphs

As their third and higher-order joint cumulants are identically zero (Section 3.2.3), dependence among normal variables is expressed through their correlations, calculated from Ω , or equivalently their partial correlations, calculated from Ω^{-1} . Consider the graph with p nodes corresponding to the variables $Y_1, \dots, Y_p$. Now Y_r and Y_s are independent conditional on all the other variables if and only if their partial correlation is zero, and we encode this by the absence of an edge between the corresponding nodes. Thus two nodes are neighbours — joined by an edge — if and only if the corresponding partial correlation is non-zero and hence if and only if the corresponding element of Δ is non-zero. This yields a conditional independence graph for $Y_1, \dots, Y_p$ (Section 6.2.2).

If the density of $Y_1, \dots, Y_p$ is non-degenerate, then the global Markov property holds. To see this, let $\mathcal{A}, \mathcal{B}$, and $\mathcal{D}$ be any disjoint nonempty subsets of $\mathcal{J} = \{1, \dots, p\}$ such that $\mathcal{D}$ separates $\mathcal{A}$ from $\mathcal{B}$ and $\mathcal{A} \cup \mathcal{B} \cup \mathcal{D} = \mathcal{J}$. As there are no edges between $\mathcal{A}$ and $\mathcal{B}$, the density of Y has exponent

$$-\frac{1}{2}(y - \mu)^T \Omega^{-1}(y - \mu) = -\frac{1}{2}(y - \mu)^T \begin{pmatrix} \Delta_{AA} & \Delta_{AD} & 0 \\ \Delta_{DA} & \Delta_{DD} & \Delta_{DB} \\ 0 & \Delta_{BD} & \Delta_{BB} \end{pmatrix} (y - \mu),$$

with quadratic term in y_A and y_B identically zero. Hence

$$f(y) = f(y_A, y_B, y_D) = g_1(y_A, y_D)g_2(y_B, y_D),$$

for some positive functions g_1 and g_2 , implying that Y_A and Y_B are conditionally independent given y_D ; of course this property is inherited by any subsets of Y_A and Y_B . As any disjoint subsets of $\mathcal{J}$ separated by $\mathcal{D}$ can be augmented to give sets $\mathcal{A}, \mathcal{B}$ which are separated by $\mathcal{D}$ and which together with $\mathcal{D}$ partition $\mathcal{J}$, the global Markov property holds.

In graphical terms it is natural to restrict the degree of dependence among components of Y by deleting edges from its graph, and this means setting elements of Ω^{-1} to zero. Suppose that the inverse covariance matrix resulting from

such deletions is $\Omega_0^{-1} = \Delta_0$, for which the profile log likelihood is $\ell(\hat{\mu}, \Delta_0) \equiv \frac{1}{2}\{n \log |\Delta_0| - (n-1)\text{tr}(\Delta_0 \hat{\Omega}^{-1})\}$. For an idea of the difficulties involved in maximizing this with respect to the non-zero elements of Δ_0 , we consider the simplest non-trivial case, with $p = 3$ variables and $\delta_{32} = 0$, implying that Y_2 and Y_3 are independent given Y_1 . In this case the log likelihood may be written down and differentiated directly, giving five simultaneous equations to be solved for the non-zero components of $\hat{\Delta}_0$. We lay these equations out as

$$\frac{1}{|\hat{\Delta}_0|} \begin{pmatrix} \hat{\delta}_{22}\hat{\delta}_{33} & \hat{\delta}_{11}\hat{\delta}_{33} - \hat{\delta}_{31}^2 & \\ -\hat{\delta}_{21}\hat{\delta}_{33} & ? & \hat{\delta}_{11}\hat{\delta}_{22} - \hat{\delta}_{21}^2 \\ -\hat{\delta}_{31}\hat{\delta}_{22} & & \end{pmatrix} = \frac{n-1}{n} \begin{pmatrix} \hat{\omega}_{11} & & \\ \hat{\omega}_{21} & \hat{\omega}_{22} & \\ \hat{\omega}_{31} & ? & \hat{\omega}_{33} \end{pmatrix},$$

where there is a missing equation $? = ?$ corresponding to δ_{32} , which does not appear in the likelihood. The structure of these equations shows that in general we must solve a system of polynomial equations of degree p , and the properties of the graph of Δ_0 play a crucial role in determining the character of the solution. Here it turns out that if the missing equation is replaced by $\hat{\delta}_{21}\hat{\delta}_{31}/|\hat{\Delta}_0| = (n-1)\hat{\omega}_{21}\hat{\omega}_{31}/(n\hat{\omega}_{11})$ and the matrices are completed by symmetry, the $\hat{\delta}_{rs}$ can be found explicitly in terms of the $\hat{\omega}_{rs}$.

Comparisons between two nested graphical models may be based on likelihood ratio statistics, though large-sample asymptotics can be unreliable. Exact comparison of the full model with the one with a single edge missing may be based on the corresponding partial correlation coefficient (Exercise 6.3.6).

Example 6.21 (Maths marks data) The above-diagonal part of Table 6.9 suggests a graphical model in which the upper right 2×2 corner of Δ is set equal to zero. The likelihood ratio statistic for comparison of this model with the full model is 0.90, which is not large relative to the χ_4^2 distribution. This suggests strongly that the simpler model fits as well as the full one, an impression confirmed by comparing the original and fitted partial correlations,

0.33	0.23	-0.00	0.03	0.33	0.24	0.00	0.00
	0.28	0.08	0.02		0.33	0.00	0.00
		0.43	0.36			0.45	0.37
			0.25				0.26

Figure 6.10 shows the graphs for these two models. In the full model every variable is joined to every other, and there is no simple interpretation. The reduced model has a butterfly-like graph whose interpretation is that given the result for algebra, results for mechanics and vectors are independent of those for analysis and statistics. Thus a result for mechanics can be predicted from those for algebra and vectors alone, while prediction for algebra requires all four other results. ■

The graphs described above have the drawback of taking no account of the logical status of the variables. For example, it may be known that Y_1 influences Y_2 but not vice versa, but this is not reflected in an undirected graph. In applications, therefore, it is useful to have different types of edges, with directed edges representing supposed causal effects and undirected edges linking variables that are to be put on an equal

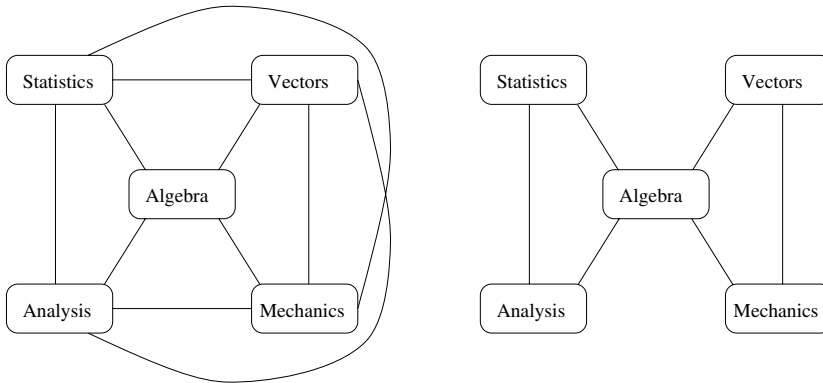


Figure 6.10 Graphs for the full model (left) and a reduced model (right) for the maths marks data. The interpretation of the reduced model is that given the result for algebra, results for vectors and mechanics are independent of those for analysis and statistics.

footing. This important topic is beyond the scope of this book; see the bibliographic notes.

Calculation of partial correlation

Let $\mathcal{S} = \{r, s\}$, where without loss of generality $r < s$. Then the conditional variance matrix for Y_r and Y_s given $Y_{-\mathcal{S}}$ is $\Omega_{\mathcal{S},\mathcal{S}} - \Omega_{\mathcal{S},-\mathcal{S}}\Omega_{-\mathcal{S},-\mathcal{S}}^{-1}\Omega_{-\mathcal{S},\mathcal{S}}$, and hence their partial correlation is

$$\rho_{rs|-\mathcal{S}} = \frac{\omega_{rs} - \Omega_{r,-\mathcal{S}}\Omega_{-\mathcal{S},-\mathcal{S}}^{-1}\Omega_{-\mathcal{S},s}}{\left\{(\omega_{rr} - \Omega_{r,-\mathcal{S}}\Omega_{-\mathcal{S},-\mathcal{S}}^{-1}\Omega_{-\mathcal{S},r})(\omega_{ss} - \Omega_{s,-\mathcal{S}}\Omega_{-\mathcal{S},-\mathcal{S}}^{-1}\Omega_{-\mathcal{S},s})\right\}^{1/2}}.$$

The (r, s) element of Ω^{-1} is $(-1)^{r+s}\Omega_{rs}/|\Omega|$, where Ω_{rs} is the (r, s) minor of Ω . Thus the (r, s) element of the ‘correlationized’ version of Ω^{-1} is $(-1)^{r+s}\Omega_{rs}/(\Omega_{rr}\Omega_{ss})^{1/2}$. To show how this is related to $\rho_{rs|-\mathcal{S}}$, we use the formula

$$\begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix} = |A_{11} - A_{12}A_{22}^{-1}A_{21}| \cdot |A_{22}| \quad (6.23)$$

for the determinant of a partitioned matrix for which A_{22}^{-1} exists. On making the row and column interchanges that bring ω_{ss} to the $(1, 1)$ position of $\Omega_{-r,-r}$, we see that

$$\Omega_{rr} = (-1)^{2(s-1)} \begin{vmatrix} \omega_{ss} & \Omega_{s,-\mathcal{S}} \\ \Omega_{-\mathcal{S},s} & \Omega_{-\mathcal{S},-\mathcal{S}} \end{vmatrix} = (\omega_{ss} - \Omega_{s,-\mathcal{S}}\Omega_{-\mathcal{S},-\mathcal{S}}^{-1}\Omega_{-\mathcal{S},s})|\Omega_{-\mathcal{S},-\mathcal{S}}|,$$

with a similar expression for Ω_{ss} , while Ω_{rs} equals

$$(-1)^{r+(s-1)} \begin{vmatrix} \omega_{sr} & \Omega_{s,-\mathcal{S}} \\ \Omega_{-\mathcal{S},r} & \Omega_{-\mathcal{S},-\mathcal{S}} \end{vmatrix} = (-1)^{r+s-1}(\omega_{rs} - \Omega_{s,-\mathcal{S}}\Omega_{-\mathcal{S},-\mathcal{S}}^{-1}\Omega_{-\mathcal{S},r})|\Omega_{-\mathcal{S},-\mathcal{S}}|,$$

as $\omega_{rs} = \omega_{sr}$ by symmetry of Ω . On substituting the expressions for Ω_{rr} , Ω_{ss} , and Ω_{rs} into $(-1)^{r+s}\Omega_{rs}/(\Omega_{rr}\Omega_{ss})^{1/2}$, we see that the (r, s) element of the ‘correlationized’ version of Ω^{-1} equals $-\rho_{rs|-\mathcal{S}}$, as was to be proved.

This may be skipped on a first reading.

Exercises 6.3

- 1 If A is a $p \times p$ matrix, all of whose elements are distinct and if A^{ij} denotes the cofactor of the (i, j) element a_{ij} of A , then $\partial|A|/\partial a_{ij} = A^{ij}$, whereas if A is symmetric, then

$$\frac{\partial|A|}{\partial a_{ij}} = \begin{cases} A^{ii}, & i = j, \\ 2A^{ij}, & i \neq j. \end{cases}$$

If A and B have dimensions $p \times q$ and $q \times p$, then

$$\frac{\partial \text{tr}(AB)}{\partial A} = \begin{cases} B^T, & \text{all elements of } A \text{ distinct,} \\ B + B^T - \text{diag}(B), & A \text{ symmetric.} \end{cases}$$

Use these identities to verify that $n^{-1}(n-1)S$ solves the likelihood equations for Ω for the multivariate normal model on page 259. Check that this maximizes the likelihood when $p = 2$.

- 2 Show that the (r, s) element of $\hat{\Omega}$ is $\hat{\omega}_{rs} = (n-1)^{-1} \sum_j (y_{rj} - \bar{y}_r)(y_{sj} - \bar{y}_s)$, where $\bar{y}_r$ is the r th element of the $p \times 1$ vector $\bar{y}$, and that although $\hat{\omega}_{rs}$ is not the maximum likelihood estimate of ω_{rs} , the maximum likelihood estimate of the correlation between Y_r and Y_s equals $\hat{\omega}_{rs}/(\hat{\omega}_{rr}\hat{\omega}_{ss})^{1/2}$.
- 3 Let Ω be the variance matrix of a p -dimensional normal variable Y . Use Cramer's rule to show that the r th diagonal element of Ω^{-1} is $\text{var}(Y_r | Y_{-r})$.
- 4 Let $Y^T = (Y_1, \dots, Y_3)$ be a multivariate normal variable with

$$\Omega = \begin{pmatrix} 1 & m^{-1/2} & \frac{1}{2} \\ m^{-1/2} & \frac{2}{m} & m^{-1/2} \\ \frac{1}{2} & m^{-1/2} & 1 \end{pmatrix}.$$

Find Ω^{-1} and hence write down the moral graph for Y .

If $m \rightarrow \infty$, show that the distribution of Y becomes degenerate while that of (Y_1, Y_3) given Y_2 remains unchanged. Is the graph an adequate summary of the joint limiting distribution? Is the Markov property stable in the limit?

- 5 Suppose that $W_1, \dots, W_n$ may be written $W_j = \mu + \sigma Z_j + \tau X$, where $Z_1, \dots, Z_n$ and X are independent standard normal variables. Obtain the correlation matrix Ω of $Y^T = (X, W_1, \dots, W_n)$, write down the moral graph for Y , and hence obtain Ω^{-1} .
- 6 Let $y_1, \dots, y_n$ be a $N_p(\mu, \Omega)$ random sample and let $\Delta = \Omega^{-1}$ have elements δ_{rs} . Show that apart from constants, the value of (6.20) maximized over both μ and Ω is $-\frac{1}{2}n \log |\hat{\Omega}|$, and deduce that the likelihood ratio statistic for comparison of the full model and a sub-model obtained by constraining elements of Ω (or Δ) may be written $n \log |\hat{\Omega}^{-1} \hat{\Omega}_0|$, in an obvious notation.
- (a) Show that the likelihood ratio statistic for testing if all the components of Y are independent is a function of the determinant of the sample correlation matrix.
- (b) Use (6.23) to show that the likelihood ratio statistic to test if $\delta_{12} = 0$ may be written $-n \log(1 - \hat{\rho}_{1,2|-\mathcal{S}}^2)$, where $\mathcal{S} = \{1, 2\}$, and check for what values of the partial correlation $\hat{\rho}_{12|-\mathcal{S}}$ this is large.
- 7 In the discussion on page 263, verify that if $\delta_{32} = 0$, then the likelihood equations are equivalent to

$$\hat{\Delta}_0^{-1} = \frac{n-1}{n} \begin{pmatrix} \hat{\omega}_{11} & & \\ \hat{\omega}_{21} & \hat{\omega}_{22} & \\ \hat{\omega}_{31} & \hat{\omega}_{21}\hat{\omega}_{31}/\hat{\omega}_{11} & \hat{\omega}_{33} \end{pmatrix},$$

and hence find $\hat{\Delta}_0$ in terms of the $\hat{\omega}_{rs}$.

Find also the maximum likelihood estimate of $\hat{\Delta}_0$ when $\delta_{31} = \delta_{32} = 0$ and when $\delta_{31} = \delta_{32} = \delta_{21} = 0$.

Give the graphs corresponding to each of these models.

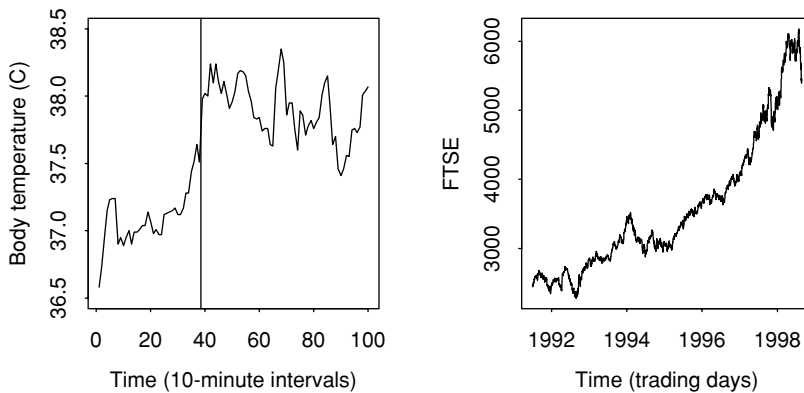


Figure 6.11 Example time series. Left: body temperatures ($^{\circ}\text{C}$) of a female Canadian beaver measured at 10-minute intervals (Reynolds, 1994). The vertical line marks where she left her lodge. Right: FTSE closing prices, 1991–1998.

6.4 Time Series

A time series consists of data recorded in time order. Examples are monthly inflation rate, weekly demand for electricity, daily maximum temperature, number of packets of information sent per second over a communication network, and so forth. The measurements may be instantaneous, such as the daily closing prices of some stock, or may be an average, such as annual temperature averaged over the surface of the globe. Typically such data show variation on several scales. Data on internet traffic, for example, show strong diurnal variation as well as long-term upward trend. Time series are ubiquitous and their analysis is well-developed, with many techniques specific to particular areas of application. In many cases the goal of time series modelling is the forecasting of future values, while in others the intention is to control the underlying process. Here we simply introduce a few basic notions in the most common situation, where the observations are continuous and arise at regular intervals. Irregular and discrete time series also occur — see Example 6.2 — but their modelling is less well explored.

Example 6.22 (Beaver body temperature data) The left panel of Figure 6.11 shows 100 consecutive telemetric measurements on the body temperature of a female Canadian beaver, *Castor canadensis*, taken at 10-minute intervals. The animal remains in its lodge for the first 38 recordings and then moves outside, at which point there is a sustained temperature rise. This is likely to be of main interest in such an application, with the dependence structure of the series regarded as secondary. The dependence must be accounted for, however, if confidence intervals for the rise are to be reliable. ■

Example 6.23 (FTSE data) The right panel of Figure 6.11 shows the closing prices of the Financial Times Stock Exchange index of London closing prices from 1991–1998. Prices are available only for days on which the exchange was open so there are many fewer than 365 observations per year. The dominant feature is the strong upward trend. Here interest would typically focus on short-term forecasting, though

portfolio managers will also wish to understand the relationship between this and other markets. In either case the dependence structure is of crucial importance. ■

Stationarity and autocorrelation

Statistical inference cannot proceed without some assumption of stochastic regularity, and in time series this is provided by the notion of *stationarity*.

$\{Y_t\}$ is also called covariance stationary, weakly stationary, or stationary in the wide sense.

Consider data $y_1, \dots, y_n$, supposed to be a realization of the random variables $Y_1, \dots, Y_n$, themselves forming a contiguous stretch of a stochastic process $\{Y_t\} = \{\dots, Y_{-1}, Y_0, Y_1, \dots\}$. Then $\{Y_t\}$ is said to be *second-order stationary* if its first and second moments are finite and time-independent, so that the mean $E(Y_s) = \mu$ is constant and the covariances $\text{cov}(Y_s, Y_{s+t}) = \gamma_t$ do not depend on s . Finiteness of $\gamma_0 = \text{var}(Y_t)$ guarantees that $|\mu|, |\gamma_t| < \infty$ for all t . The first and second moments of a second-order stationary series do not depend on the point at which they are calculated. Neither panel of Figure 6.11 looks stationary, though it is plausible that the temperature data to the right of the vertical line are.

A series is said to be *strictly stationary* if the joint distribution of any finite subset Y_A does not depend on the origin; thus the distributions of Y_{s+A} and of Y_A are the same for any s . This is a stronger condition than second-order stationarity, because it constrains the entire distribution of the series. In particular it implies that the joint cumulants of Y_{s+A} are independent of s , if they exist. Evidently strict stationarity yields more powerful theoretical results, but as it is impossible to verify from data, they are less useful in practice. The definitions coincide if $\{Y_t\}$ has a multivariate normal distribution, as this is determined by its first and second moments. The term stationary used without qualification in this section means second-order stationary.

The second-order structure of a stationary process is summarized in its *autocorrelation function* $\rho_t = \text{corr}(Y_0, Y_t)$, $t = \pm 1, \pm 2, \dots$, where $\rho_t = \gamma_t / \gamma_0$; $\gamma_0 = \text{var}(Y_0)$ is the marginal variance of the process $\{Y_t\}$. Note that

$$\rho_{-t} = \text{corr}(Y_s, Y_{s-t}) = \text{corr}(Y_{s+t}, Y_s) = \rho_t$$

by stationarity. A related function is the *partial autocorrelation function* $\rho'_t = \text{corr}(Y_0, Y_t \mid Y_1, \dots, Y_{t-1})$, which summarizes any correlation between observations t lags apart after conditioning on the intervening data; see Section 6.3.3.

A *white noise process* $\{\varepsilon_t\}$ is an uncorrelated sample from some distribution with mean zero and variance σ^2 ; evidently it has $\rho_t = \rho'_t \equiv 0$. We shall use the term *normal white noise* when $\varepsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$.

Plots of estimated ρ_t and ρ'_t against positive values of t are called the *correlogram* and *partial correlogram*. Under mild conditions their ordinates are asymptotic independent $N(0, n^{-1})$ variables for a white noise series of length n , from which significance can be assessed; see Figure 6.12.

Example 6.24 (Autoregressive process) About the simplest time series model is the autoregressive process of order one, or AR(1) model

$$Y_t - \mu = \alpha(Y_{t-1} - \mu) + \varepsilon_t, \quad t = \dots, -1, 0, 1, \dots, \quad (6.24)$$

where the innovation series $\{\varepsilon_t\}$ is normal white noise and ε_t is independent of $\dots, Y_{t-2}, Y_{t-1}$. Taking variances in (6.24) yields $\gamma_0 = \alpha^2 \gamma_0 + \sigma^2$. Hence $\gamma_0 = \sigma^2 / (1 - \alpha^2)$, so a necessary condition for stationarity is $|\alpha| < 1$. This condition is also sufficient, and if it is satisfied then $E(Y_t) = \mu$ and $\rho_t = \alpha^{-|t|}$ (Exercise 6.4.1).

This is a Markov process, because Y_t depends on the previous observations only through Y_{t-1} , and hence the only non-zero partial autocorrelation is $\rho'_1 = \alpha$. If the ε_t are normal, then Y_t is a linear combination of normal variables and so $Y_1, \dots, Y_n$ are jointly normal with mean vector $\mu \mathbf{1}_n$ and covariance matrix

$$\Omega = \frac{\sigma^2}{1 - \alpha^2} \begin{pmatrix} 1 & \alpha & \alpha^2 & \dots & \alpha^{n-1} \\ \alpha & 1 & \alpha & \dots & \alpha^{n-2} \\ \alpha^2 & \alpha & 1 & \dots & \alpha^{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \alpha^{n-1} & \alpha^{n-2} & \alpha^{n-3} & \dots & 1 \end{pmatrix}.$$

One can verify directly that Ω^{-1} is the tridiagonal matrix (Example 6.13)

$$\sigma^{-2} \begin{pmatrix} 1 & -\alpha & 0 & \dots & 0 & 0 \\ -\alpha & 1 + \alpha^2 & -\alpha & \dots & 0 & 0 \\ 0 & -\alpha & 1 + \alpha^2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 + \alpha^2 & -\alpha \\ 0 & 0 & 0 & \dots & -\alpha & 1 \end{pmatrix}.$$

The autoregressive process of order p or AR(p) model satisfies

$$Y_t - \mu = \sum_{j=1}^p \alpha_j (Y_{t-j} - \mu) + \varepsilon_t, \quad t = \dots, -1, 0, 1, \dots,$$

and is therefore a Markov process of order p . Constraints on $\alpha_1, \dots, \alpha_p$ are needed for this process to be stationary, but if they are satisfied, there is a sharp cut-off in the partial autocorrelations: $\rho'_t = 0$ when $t > p$. This should be reflected in the partial correlogram of AR(p) data. The constraints are discussed after Example 6.26. ■

Example 6.25 (Beaver body temperature data) Figure 6.12 shows the correlogram and partial correlogram for the apparently stationary observations 39–100 of the beaver temperature data. The correlogram shows positive correlations at lags 1–3. Any further evidence of structure must be treated very cautiously, as the values around lag 15 are not very significant, and as each panel of the figure shows 20 correlations estimated from only 62 observations. The partial correlogram is suggestive of an AR(1) model with $\alpha \doteq 0.75$, consistent with the geometric decrease in the correlogram at short lags.

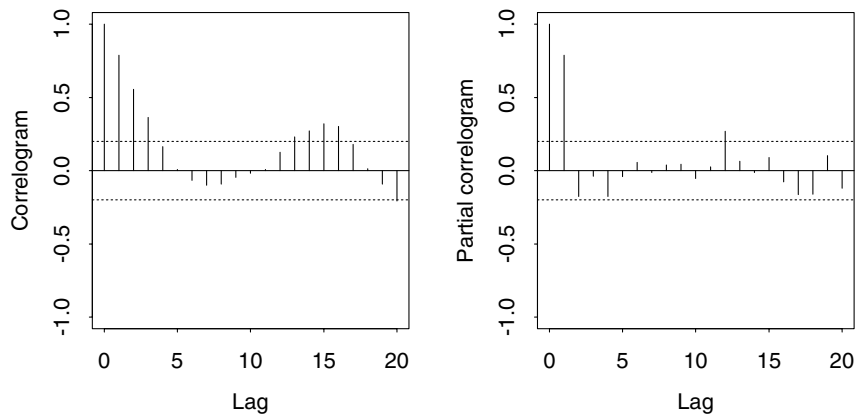
The change in level evident in Figure 6.11 suggests that we take

$$Y_t = \begin{cases} \beta_0 + \eta_t, & t = 1, \dots, 38, \\ \beta_0 + \beta_1 + \eta_t, & t = 39, \dots, 100, \end{cases} \quad (6.25)$$

while the partial correlogram suggests that the η_t follow (6.24) with $\mu = 0$. This yields a Markov model with parameters $(\beta_0, \beta_1, \alpha, \sigma^2)$. If we assume normal white

Figure 6.12

Correlogram and partial correlogram for observations 39–100 of the beaver body temperature data. The dotted horizontal lines at $\pm 2n^{-1/2}$ show 95% confidence bounds for the correlation coefficients, if the data are white noise. Strong systematic departures from these are suggestive of structure in the data.



noise and initial $N\{\beta_0, \sigma^2/(1 - \alpha^2)\}$ distribution for y_1 then the log likelihood is readily obtained from (4.8); see Exercise 6.4.3. The log likelihood can be maximized numerically and standard errors obtained from the inverse observed information matrix, giving $\hat{\beta}_0 = 37.19$ (0.119), $\hat{\beta}_1 = 0.61$ (0.138), $\hat{\alpha} = 0.87$ (0.068), and $\hat{\sigma}^2 = 0.015$ (0.002). Body temperature rises by about 0.6°C when the beaver is active, and successive measurements are quite highly correlated. Treating the data as independent gives standard error 0.044 for $\hat{\beta}_1$, so the autocorrelation greatly increases the uncertainty for β_1 .

Residuals can be constructed by estimating the scaled innovations ε_t/σ . In the inactive period we define residuals $r_t = \{y_t - \hat{\beta}_0 - \hat{\alpha}(y_{t-1} - \hat{\beta}_0)\}/\hat{\sigma}$, with a similar expression in the active period. Then the correlogram, partial correlogram, and probability plots of $r_2, \dots, r_{100}$ help assess model adequacy. Judged by these criteria, the model seems to fit well, though (6.25) does not account for the gradual rise in body temperature before the beaver left the lodge. ■

Example 6.26 (Moving average process) A moving average process of order q or MA(q) model satisfies the equation

$$Y_t - \mu = \sum_{j=1}^q \beta_j \varepsilon_{t-j} + \varepsilon_t, \quad t = \dots, -1, 0, 1, \dots$$

where $\{\varepsilon_t\}$ is white noise. Here $E(Y_t) = \mu$ and $\text{var}(Y_t) = \sigma^2(1 + \beta_1^2 + \dots + \beta_q^2)$ for all t , and it is easy to check that this process is stationary and that $\rho_t = 0$ for $t > q$ (Exercise 6.4.2). Thus the correlogram of such data should show a sharp cut-off after lag q . ■

Stationary autoregressive and moving average processes are linear processes, as the current observation Y_t may be expressed as an infinite moving average of the innovations,

$$Y_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j}, \quad t = \dots, -1, 0, 1, \dots, \quad \text{with} \quad \sum_{j=0}^{\infty} |c_j| < \infty. \quad (6.26)$$

This expresses the current Y_t in terms of past innovations, provides useful models in many applications, and leads to simple computations. For example, $\text{var}(Y_t) = \sum c_j^2 < \infty$ and $\gamma_t = \sum c_j c_{j+t}$.

Evidently an MA(q) model with zero mean has a representation (6.26). To see when this is true for an AR(p) model, it is useful to introduce the *backshift operator* B such that $BY_t = Y_{t-1}$ and $B^d Y_t = Y_{t-d}$, with $B^0 = I$ the identity operator. Then an AR(p) process is expressible as $a(B)Y_t = \varepsilon_t$, where the polynomial $a(z) = 1 - \sum_{j=1}^p \alpha_j z^j$ corresponds to the autoregression, and we can formally write $Y_t = a(B)^{-1} \varepsilon_t = \sum_{i=0}^{\infty} c_i \varepsilon_{t-i}$, say, which is stationary if and only if $\sum c_i^2 < \infty$. Now $a(z) = \prod_{j=1}^p (1 - a_j z)$, where a_j^{-1} are the possibly complex roots of $a(z)$, and provided that no two of the a_j are equal, $a(z)^{-1}$ may be written using partial fractions as $\sum_{j=1}^p b_j / (1 - a_j z)$ for some b_j . If we take z sufficiently small then $a(z)^{-1}$ can be expressed as a sum of geometric series with coefficients $c_i = \sum_{j=1}^p b_j a_j^i$, giving the infinite moving average (6.26). For this to be stationary we must have $\sum c_i^2 < \infty$, which occurs if and only if $|a_j| < 1$ for each j , or equivalently all the roots of $a(z)$ lie outside the unit disk in the complex plane. Thus properties of the polynomial $a(z)$ are intimately related to those of the process $\{Y_t\}$.

Example 6.27 (ARMA process) The autoregressive process is formed as a linear combination of previous observations, while a moving average process is based on a weighted combination of the innovations at previous steps. An obvious generalization is to combine the two, giving the *autoregressive moving average* process or ARMA(p, q) model

$$Y_t - \mu = \sum_{j=1}^p \alpha_j (Y_{t-j} - \mu) + \sum_{i=1}^q \beta_i \varepsilon_{t-i} + \varepsilon_t, \quad t = \dots, -1, 0, 1, \dots$$

As in the preceding examples, the Y_t will have a joint normal distribution if the process is stationary and the ε_t represent normal white noise. Let $\mu = 0$ for simplicity.

In terms of the backshift operator we have $a(B)Y_t = b(B)\varepsilon_t$, where the polynomials $a(z) = 1 - \sum_{j=1}^p \alpha_j z^j$ and $b(z) = 1 + \sum_{i=1}^q \beta_i z^i$ represent the autoregressive and moving average components. Thus $Y_t = a(B)^{-1} b(B) \varepsilon_t = \sum_{j=-\infty}^{\infty} c_j \varepsilon_{t-j}$, where the coefficients c_j are those of the infinite series $a(z)^{-1} b(z)$. Once again, properties of these polynomials determine those of $\{Y_t\}$.

The class of ARMA processes is typically regarded as a useful ‘black box’ for fitting and forecasting, though fitted models sometimes have a substantive interpretation. For instance, the values of AIC when (6.25) is fitted to the beaver data and the η_t follow an ARMA(p, q) process with (p, q) equal to (1, 1), (0, 1), (1, 2), and (2, 0) are -128.34 , -90.06 , -126.54 , and -128.78 , compared with -127.55 for the AR(1) model, which therefore seems a good compromise between quality of fit and simplicity of interpretation, the latter following from its Markov structure. It is considerably harder to explain the ARMA(1,2) model in simple terms, despite its slightly better fit. ■

Trend removal

In practice data are rarely stationary, and trends or periodic changes must be removed before fitting standard models. One simple approach to removing polynomial trends is differencing. Suppose that $Y_t = \gamma_0 + \gamma_1 t + \varepsilon_t$, so there is linear trend with possibly correlated noise superimposed. Then

$$X_t = Y_t - Y_{t-1} = (\gamma_0 + \gamma_1 t + \varepsilon_t) - \{\gamma_0 + \gamma_1(t-1) + \varepsilon_{t-1}\} = \gamma_1 + \eta_t,$$

say, where $\eta_t = \varepsilon_t - \varepsilon_{t-1}$. Thus differencing removes linear trend but complicates the error structure: if $\{\varepsilon_t\}$ had been white noise, then the differenced process $\{\eta_t\}$ follows an MA(1) model with $\beta_1 = -1$. It is straightforward to show that d -fold differencing will remove a polynomial trend of order d (Exercise 6.4.4). Over-differencing does little harm: if there had been no trend originally present then $\{X_t\}$ merely has a more complicated error structure than had $\{Y_t\}$. Differencing can also be used to remove seasonal components.

If an ARMA(p, q) model fits the d -fold difference of $\{Y_t\}$, then we have $a(B)(I - B)^d Y_t = b(B)\varepsilon_t$, and this is known as an *integrated autoregressive-moving average* or ARIMA(p, d, q) process. This generalizes the class of ARMA models to allow non-stationarity.

Example 6.28 (FTSE data) Trends such as that in the right panel of Figure 6.11 are generally removed by differencing the log closing prices, and the upper panel of Figure 6.13 shows $y_t = 100 \log(x_t/x_{t-1})$, where x_t is the original series. Thus y_t is proportional to the differences of the log x_t and represents daily percentage returns to investors. Differencing has removed the trend, but it is not clear that the y_t are stationary — their variability seems to increase from time to time. Such changes in volatility cannot be mimicked by linear processes and much effort has been expended in modelling them. Probability plots show that the y_t are somewhat asymmetric with heavier tails than the normal distribution, so the marginal distribution of $\{Y_t\}$ is non-normal.

The partial correlogram of y_t shows small but significant autocorrelation at lag one, suggestive of slight autoregressive behaviour. Its value, $\hat{\rho}_1 = 0.09$, is too small to be of much use in predicting movements of y_t . This makes sense: high correlation could be exploited by everyone for gain, but there must be both winners and losers when shares are traded. The partial correlogram of the $(y_t - \bar{y})^2$ shows generally positive autocorrelations to about lag 20.

The y_t have average 0.043 with standard error 0.018, so if the data were independent there would be evidence that $E(Y_t) > 0$, corresponding to an average daily increase of about 0.043% in the FTSE over 1991–1998. ■

Other approaches to trend removal can involve local smoothing by methods like those to be described in Section 10.7; very roughly the idea is to use weighted averages of the data to estimate changes in the process mean. Such averaging can be applied on different scales, for example giving separate estimates of systematic decadal, annual, and monthly variation. Robust versions of these smoothers exist and are often preferable in practice.

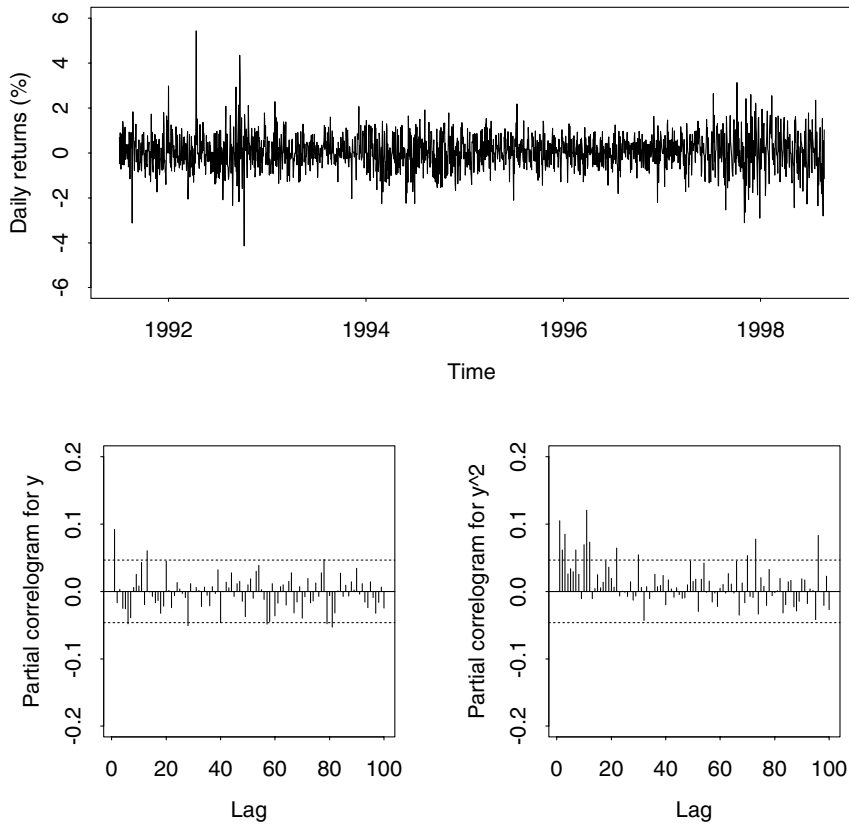


Figure 6.13 Daily returns (%) from the FTSE, 1991–1998. The lower panels show the partial correlograms of the y_t and their squares. The 95% confidence bands shown by the dotted horizontal lines are much narrower than in Figure 6.12 because there are many more data.

Volatility models

A key feature of financial time series such as that in the top panel of Figure 6.13 is their changing volatility, which leads to periods of high variability interspersed with quieter periods. A standard model for this in the financial context is the linear *autoregressive conditional heteroscedastic model of order one* or linear ARCH(1) process, which sets

$$Y_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \beta_0 + \beta_1 Y_{t-1}^2, \quad t = \dots, -1, 0, 1, \dots, \quad (6.27)$$

where $\{\varepsilon_t\}$ is normal white noise with unit variance with ε_t independent of Y_{t-1} , $\beta_0 > 0$ and $\beta_1 \geq 0$. The current variance σ_t^2 is increased if the previous observation was far from zero, giving bursts of high volatility when this occurs. A necessary condition for stationarity is $E(Y_t^2) = E(\sigma_t^2)E(\varepsilon_t^2) < \infty$, implying that $\gamma_0 = \beta_0 + \beta_1 \gamma_0$ or equivalently that $\beta_1 < 1$. In this case $\{Y_t\}$ is zero-mean white noise, but as we can write $Y_t^2 = \sigma_t^2 + (Y_t^2 - \sigma_t^2) = \beta_0 + \beta_1 Y_{t-1}^2 + \eta_t$, where $\eta_t = \sigma_t^2(\varepsilon_t^2 - 1)$ has mean zero, we see that $\{Y_t^2\}$ follows an autoregressive process, albeit with non-constant variance. In order for the process $\{Y_t^2\}$ to be stationary $E(Y_t^4)$ must be finite, and this occurs when $\beta_1^2 < 1/3$. Then Y_t has fatter tails than the normal distribution.

Thus ARCH models mimic two important features of financial time series: volatility clustering and fat-tailed marginal distributions.

The assumption of normal innovations can be replaced by other distributions, a popular choice being to set $\nu\varepsilon_t/(\nu-2) \stackrel{\text{iid}}{\sim} t_\nu$; the scaling ensures that $\text{var}(\varepsilon_t) = 1$. ARCH models can be extended to allow dependence on $Y_{t-2}^2, \dots$ and on $\sigma_{t-1}^2, \dots$, a particularly widely-used case being the generalized ARCH or GARCH(1,1) process in which $\sigma_t^2 = \beta_0 + \beta_1 Y_{t-1}^2 + \delta \sigma_{t-1}^2$.

Example 6.29 (FTSE data) Example 6.28 suggests that an unadorned ARCH model is unlikely to fit these data because it cannot account for the non-zero mean and non-zero correlations. Inspired by (6.27), we therefore let $Y_t - \mu = \alpha(Y_{t-1} - \mu) + \sigma_t \varepsilon_t$ with $\sigma_t^2 = \beta_0 + \beta_1(Y_{t-1} - \mu)^2$. This combines autoregressive structure for the means of the Y_t with ARCH structure for their variance. The result is a Markov process, and with normal ε_t the log likelihood contribution from the conditional density $f(y_t | y_{t-1})$ is

$$-\frac{1}{2} \log\{\beta_0 + \beta_1(y_{t-1} - \mu)^2\} - \frac{\{y_t - \mu - \alpha(y_{t-1} - \mu)\}^2}{2\{\beta_0 + \beta_1(y_{t-1} - \mu)^2\}}.$$

The overall log likelihood is a sum of such terms for $t = 2, \dots, n$ plus $\log f(y_1)$, but the series is so long that this initial term, which involves knowing the stationary density of Y_t , can safely be ignored.

The log likelihood is readily maximized numerically, but a correlogram suggests that structure remains in the squares of the residuals

$$r_t = \frac{y_t - \hat{\mu} - \hat{\alpha}(y_{t-1} - \hat{\mu})}{\{\hat{\beta}_0 + \hat{\beta}_1(y_{t-1} - \hat{\mu})^2\}^{1/2}},$$

so this model is not adequate. As an alternative, we retain the AR mean structure but use GARCH structure $\sigma_t^2 = \beta_0 + \beta_1(Y_{t-1} - \mu)^2 + \delta \sigma_{t-1}^2$ for the variances. A crude way to fit this is to estimate σ_m^2 by the variance of $y_1, \dots, y_m$, and then to compute $\sigma_t^2 = \beta_0 + \beta_1(y_{t-1} - \mu)^2 + \delta \sigma_{t-1}^2$ for $t = m+1, \dots, n$. The likelihood based on $f(y_{m+1}, \dots, y_n | y_1, \dots, y_m)$ is then readily obtained and may be maximized. Here n is large so little information is lost by conditioning on $y_1, \dots, y_m$. With $m = 30$ the maximized log likelihood is -2100.27 , and both the residuals and their squares look like white noise, so the structure of the model seems correct. However a normal probability plot of the residuals suggests that slightly heavier-tailed innovations may be needed. We therefore let the ε_t have t_ν distributions, scaled so that $\text{var}(\varepsilon_t) = 1$. The resulting log likelihood is -2075.64 , an appreciable improvement. The maximum likelihood estimates and standard errors are $\hat{\mu} = 0.051$ (0.018), $\hat{\alpha} = 0.070$ (0.024), $\hat{\beta}_0 = 0.006$ (0.004), $\hat{\beta}_1 = 0.036$ (0.011), $\hat{\delta} = 0.955$ (0.016) and $\hat{\nu} = 9.7$ (1.86). Thus μ and α seem necessary for successful modelling. Over the period of these data the return on investment was on average $100\hat{\mu} \doteq 5\%$ every 100 trading days, but little would be gained from using the estimated correlation $\hat{\alpha} = 0.07$ between Y_t and Y_{t+1} for short-term prediction. The value of $\hat{\delta}$ shows the strong dependence of σ_t^2 on σ_{t-1}^2 that leads to volatility persistence. A condition for

stationarity of a GARCH process $\{Y_t\}$ is that $\beta_1 + \delta < 1$, and this is satisfied by the estimates. The value of $\hat{\nu}$ indicates innovations somewhat heavier than normal, in agreement with the residual plot. Overall the model seems to fit surprisingly well. ■

Time series is a large and important topic, whose surface has barely been scratched above. The bibliographic notes give some points of entry to the literature.

Exercises 6.4

- 1 Consider (6.24) for $t = 1, \dots, n$, and suppose that Y_0 has a known distribution with finite variance, independent of $\varepsilon_1, \dots, \varepsilon_n$. Deduce that

$$Y_n - \mu = \sum_{j=1}^n \alpha^{n-j} \varepsilon_j + \alpha^n (Y_0 - \mu)$$

and establish that a limiting distribution for Y_n as $n \rightarrow \infty$ exists only when $\lim_{n \rightarrow \infty} \sum_{j=1}^n \alpha^{2j} < \infty$.

Hence show that a condition for stationarity is $|\alpha| < 1$, in which case the limiting distribution for Y_n is normal with mean μ and variance $\sigma^2/(1 - \alpha^2)$. Show also that if Y_0 has this distribution, so too do all the Y_j . Show that the covariance matrix Ω of $Y_1, \dots, Y_n$ is then that given in Example 6.24, and write down the corresponding moral graph.

- 2 Consider the MA(1) process; see Example 6.26. Show that its covariances are

$$\text{cov}(Y_t, Y_{t+s}) = \begin{cases} \sigma^2(1 + \beta_1^2), & s = 0, \\ \sigma^2\beta_1, & s = 1, \\ 0 & \text{otherwise,} \end{cases}$$

find the autocorrelation function and use the matrices in Example 6.24 to deduce that there is no cut-off in the partial autocorrelations.

Generalize this to the MA(q) model.

- 3 Give an expression for the log likelihood in Example 6.25.
- 4 Suppose that $Y_t = \sum_{j=0}^k \xi_j t^j + \varepsilon_t$, where $\{\varepsilon_t\}$ is a stationary process. Show by induction that d -fold differencing yields a series that is stationary for any $d \geq k$.
Let $Y_t = s(t) + \varepsilon_t$, where $s(t) = s(t + kp)$, for a fixed integer p and all integers t and k . Show that $(I - B^p)Y_t$ is stationary, and discuss the implications for removal of seasonality from a monthly time series.
- 5 Give a formula for the residual r_t when $\sigma_t^2 = \beta_0 + \beta_1(Y_{t-1} - \mu)^2 + \delta\sigma_{t-1}^2$ in Example 6.29.

6.5 Point Processes

Data that can be summarized by points in a continuum arise in many applications. Examples are the epicentres of earthquakes, the locations of cases of leukaemia, and the times at which emails are sent. The ‘point’ may be merely a convenient representation of something small compared to its surroundings, and other information may be available, such as the strength of the earthquake, but here we assume that summary as a point is sensible and ignore other aspects.

6.5.1 Poisson process

The Poisson process in the line is the simplest point process and the basis for many more complex models. Suppose that we observe points in a time interval $[0, t_0]$.

Let $N(w, w + t)$ denote how many fall into the subinterval $(w, w + t]$; we write $N(t) = N(0, t)$, $t > 0$, and $N(\mathcal{A})$ for the number of points in the set $\mathcal{A}$. Let $\lambda(t)$ be a well-behaved non-negative function whose integral is finite on $[0, t_0]$, and suppose that

$o(\delta t)$ is small enough that $o(\delta t)/\delta t \rightarrow 0$ as $\delta t \rightarrow 0$.

- events in disjoint subsets of $[0, t_0]$ are independent, that is, $N(\mathcal{A}_1)$ is independent of $N(\mathcal{A}_2)$ whenever $\mathcal{A}_1 \cap \mathcal{A}_2 = \emptyset$;
- $\Pr\{N(t, t + \delta t) = 0\} = 1 - \lambda(t)\delta t + o(\delta t)$ for small δt ; and
- $\Pr\{N(t, t + \delta t) = 1\} = \lambda(t)\delta t + o(\delta t)$ for small δt .

The last two properties imply that $\Pr\{N(t, t + \delta t) > 1\} = o(\delta t)$, so the process is *orderly*: multiple occurrences at the same t may not occur. The *intensity* $\lambda(t)$ is interpreted as the rate at which points occur in a small interval at t , so more points fall where $\lambda(t)$ is relatively high. Finiteness of $\int_0^{t_0} \lambda(u) du$ ensures that $N(t_0) < \infty$ with probability one, as we shall see below.

We find the probability that there are no points in the interval $(w, w + t]$ by dividing it into k subintervals of length $\delta t = t/k$, and then letting $\delta t \rightarrow 0$. Then the properties above imply that

$$\begin{aligned} \Pr\{N(w, w + t) = 0\} &= \prod_{i=0}^{k-1} \Pr\{N\{w + i\delta t, w + (i + 1)\delta t\} = 0\} \\ &\doteq \prod_{i=0}^{k-1} \{1 - \lambda(w + i\delta t)\delta t + o(\delta t)\} \\ &= \exp \left[\sum_{i=0}^{k-1} \log \{1 - \lambda(w + i\delta t)\delta t + o(\delta t)\} \right] \\ &= \exp \left\{ - \sum_{i=0}^{k-1} \lambda(w + i\delta t)\delta t + o(k\delta t) \right\} \\ &\rightarrow \exp \left\{ - \int_w^{w+t} \lambda(u) du \right\}, \end{aligned} \quad (6.28)$$

where the limit follows because as $\delta t \rightarrow 0$ with t fixed, $o(k\delta t) = t o(\delta t)/\delta t \rightarrow 0$. As the length of the random time T from w to the next point exceeds t if and only if $N(w, w + t) = 0$, T has probability density function

$$f_T(t) = - \frac{d\Pr\{N(w, w + t) = 0\}}{dt} = \lambda(w + t) \exp \left\{ - \int_w^{w+t} \lambda(u) du \right\}, \quad t > 0,$$

and hazard function $f_T(t)/\Pr(T \geq t) = \lambda(w + t)$.

Now suppose that points in $(0, t_0]$ have been observed at times $t_1, \dots, t_n$, where $0 < t_1 < \dots < t_n < t_0$. As events in non-overlapping sets are independent, the joint probability density of the data is

$$\lambda(t_1)e^{-\int_0^{t_1} \lambda(u) du} \times \lambda(t_2)e^{-\int_{t_1}^{t_2} \lambda(u) du} \times \dots \times \lambda(t_n)e^{-\int_{t_{n-1}}^{t_n} \lambda(u) du} \times e^{-\int_{t_n}^{t_0} \lambda(u) du},$$

where the final term is the probability of no events in $(t_n, t_0]$. This joint density reduces to

$$\exp \left\{ - \int_0^{t_0} \lambda(u) du \right\} \prod_{j=1}^n \lambda(t_j), \quad 0 < t_1 < \cdots < t_n < t_0. \quad (6.29)$$

Given a parametric form for $\lambda(t)$, (6.29) gives the likelihood on which inferences may be based. In practice the integral is usually unavailable in closed form and a numerical approximation must be used.

The probability of n events occurring in the interval $[0, t_0]$ is obtained by integrating (6.29) with respect to $t_1, \dots, t_n$ and is (Exercise 6.5.2)

$$\Pr \{N(t_0) = n\} = \frac{\Lambda(t_0)^n}{n!} \exp \{-\Lambda(t_0)\}, \quad n = 0, 1, \dots, \quad (6.30)$$

where we have written $\Lambda(t_0) = \int_0^{t_0} \lambda(u) du$. Thus $N(t_0)$ is a Poisson variable with mean $\Lambda(t_0)$. As events in disjoint subsets are independent and sums of independent Poisson variables are Poisson (Example 2.35), we see that in a Poisson process, the number of events in a subset $\mathcal{A}$ is a Poisson variable whose mean $\Lambda(\mathcal{A}) = \int_{\mathcal{A}} \lambda(u) du$ is the integral of the rate function λ over $\mathcal{A}$. Moreover these counts are independent for disjoint subsets.

Division of (6.29) by (6.30) gives the probability that points arise at $t_1, \dots, t_n$ conditional on there being n points, namely

$$n! \prod_{j=1}^n \frac{\lambda(t_j)}{\Lambda(t_0)}, \quad 0 < t_1 < \cdots < t_n < t_0.$$

This is the joint density of the order statistics of a random sample of size n with density $\lambda(t)/\Lambda(t_0)$ on the interval $[0, t_0]$; see (2.25). As we shall see, this result is useful in model-checking.

Example 6.30 (Exponential trend) Let $\lambda(t) = \exp(\beta_0 + \beta_1 t)$, so $\Lambda(t_0) = e^{\beta_0}(e^{\beta_1 t_0} - 1)/\beta_1$. When $\beta_1 = 0$ this yields a constant intensity. The log likelihood corresponding to (6.29) equals

$$\ell(\beta_0, \beta_1) = n\beta_0 + \beta_1 \sum_{j=1}^n t_j - e^{\beta_0}(e^{\beta_1 t_0} - 1)/\beta_1$$

and is of exponential family form.

The ratio $\lambda(t)/\Lambda(t_0)$ equals $\beta_1 e^{\beta_1 t}/(e^{\beta_1 t_0} - 1)$, corresponding to an exponential tilt of the uniform density on $[0, t_0]$, so when $\beta_1 > 0$ events tend to pile up toward the right end of the interval, and conversely. ■

There is an intimate connection between two ways to think about such data, in terms of the counts in subsets of the region of observation and in terms of the spacings between points. Although the second approach is natural in one dimension, the count representation is generally simpler in several dimensions. To see how it extends, let $\mathcal{S}$ be a subset of $\mathbb{R}^d$ and suppose that an integrable non-negative function $\lambda(t)$ is defined such that $\Lambda(\mathcal{S}) = \int_{\mathcal{S}} \lambda(u) du$ is finite. Then under conditions that extend those for

the univariate case, the numbers of events in disjoint subsets $\mathcal{A}_1, \dots, \mathcal{A}_m$ of $\mathcal{S}$ have independent Poisson distributions with means $\Lambda(\mathcal{A}_1), \dots, \Lambda(\mathcal{A}_m)$. The probability density for points observed at $\{t_1, \dots, t_n\} \subset \mathcal{S}$ is

$$\prod_{j=1}^n \lambda(t_j) \times \exp \{-\Lambda(\mathcal{S})\}, \quad (6.31)$$

from which a likelihood can again be constructed. Such models play an important role in event history and survival data, as described in Sections 5.4 and 10.8. In terms of Figure 5.8, the idea is to treat failures as events of an inhomogeneous Poisson process in the region of the plane bounded by the line $x = y$, the horizontal axis, and the vertical line marking the end of the trial; see Section 10.8.2. Another application, to statistics of extremes, will be described shortly.

Homogeneous Poisson process

$|\mathcal{A}|$ is the length
(Lebesgue measure) of the
set $\mathcal{A}$.

The simplest situation is when the intensity function $\lambda(t)$ is a constant λ . Then $\Lambda(\mathcal{A}) = \lambda|\mathcal{A}|$ and $\Lambda(t_0) = \lambda t_0$. The number of points in $[0, t_0]$ is then Poisson with mean λt_0 , and intervals between them are independent exponential variables with density $\lambda e^{-\lambda y}$. The log likelihood from (6.29) is

$$\ell(\lambda) \equiv n \log \lambda - \lambda t_0,$$

from which the maximum likelihood estimate $\hat{\lambda} = n/t_0$ and information quantities may be derived; see Example 4.19.

When $\lambda(t)$ is constant, the density $\lambda(t)/\Lambda(t_0) = t_0^{-1}$ is uniform on the interval $[0, t_0]$, and hence the n points $u_j = t_j/t_0$ are distributed as order statistics of a random sample from the uniform distribution on $[0, 1]$; see Section 2.3. A graphical check of this is to plot the empirical distribution function of the u_j , $\hat{F}(u)$. Departures from the uniform distribution $F(u) = u$, $0 \leq u \leq 1$ suggest that the intensity is not constant. Formal tests of fit using this are discussed in Section 7.3.1.

Data often exhibit clustering relative to a Poisson process. If so, there will tend to be an excess of short intervals between points, relative to the exponential distribution. Under the Poisson process model the *spacings* $y_1 = t_1 - 0$, $y_2 = t_2 - t_1, \dots, y_{n+1} = t_0 - t_n$ form a (non-independent) sample from the exponential distribution with mean λ^{-1} , so a plot of ordered spacings against exponential order statistics should be a straight line, departures from which will suggest model failure.

Example 6.31 (Danish fire data) Figure 6.14 shows data on the times and amounts of major insurance claims due to fire in Denmark from 1980–1990. The upper left panel shows the original 2492 claims; the original amounts have been rescaled. The data are dominated by a few large claims, shown in more detail in the upper right panel, which gives the logarithms of the 254 claims that exceed 5 units. This is a two-dimensional point process of times and log amounts, which reduces to the one-dimensional data shown as a rug at the foot of the panel if the amounts are ignored.

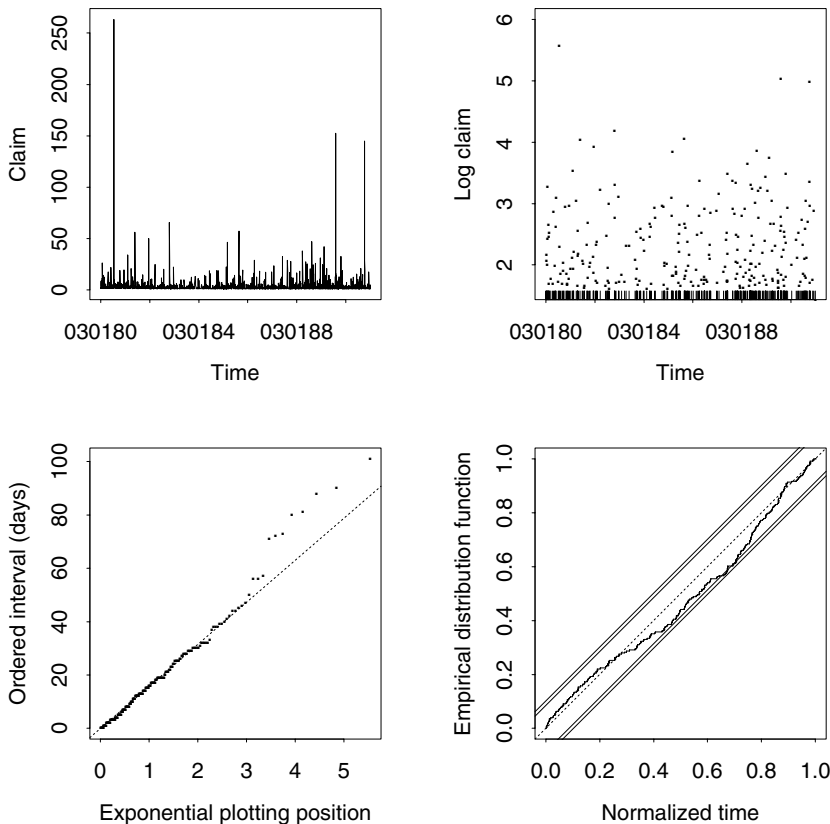


Figure 6.14 Data on major insurance claims due to fires in Denmark, 1980–1990 (Embrechts *et al.*, 1997, pp. 298–303). The upper left panel shows the original data and the upper right panel the logs of the 254 losses exceeding five units, with the rug below showing their times. The lower right panel shows the empirical distribution of the 254 $u_j = t_j/t_0$, and the lower left panel an exponential probability plot of spacings between these t_j . In each case the dotted line shows the expected pattern under a homogeneous Poisson process. The lower right panel suggests that the rate of the process may be non-uniform, with an excess of early points followed by a deficiency. The solid diagonal lines in the lower right panel show significance for a Kolmogorov–Smirnov statistic at levels 0.05 and 0.01 and are explained in Example 7.23. The lower left panel suggests that the spacings are close to exponentially distributed.

We consider only the times of these 254 largest claims. The lower right panel shows the empirical distribution function of the corresponding $u_j = t_j/t_0$, with $t_1, \dots, t_n$ the rug in the panel above. Relative to the uniform distribution there is a slight excess of claims up to about 1983, followed by a deficiency from 1984 to 1990. Example 7.23 gives further discussion of the fit.

The exponential probability plot of the spacings in the lower left panel of the figure suggests that the times between claims are fairly close to exponential, though perhaps with a slightly longer tail. The value of $\hat{\lambda}$ is roughly $254/(11 \times 365) = 0.063 \text{ days}^{-1}$. Thus the rate of arrival of claims per day is about 0.06, corresponding to a mean time between claims of $\hat{\lambda}^{-1} = 15.8$ days; this has standard error 1.0 calculated from the observed information.

We return to these data in Examples 6.34 and 7.23. ■

6.5.2 Statistics of extremes

An important application of Poisson processes is to rare events — high sea levels, low temperatures, record times to run a mile, large insurance claims, and so forth. To see how, we make a detour and consider properties of the maximum of a random

The upper support point is the smallest x_0 such that $\lim_{x \nearrow x_0} F(x) = 1$; possibly $x_0 = +\infty$.

sample $X_1, \dots, X_m$ from a continuous distribution function $F(x)$ with upper support point x_0 . As $m \rightarrow \infty$, independence of the X_i implies that for any fixed $x < x_0$,

$$\begin{aligned}\Pr\{\max(X_1, \dots, X_m) \leq x\} &= \Pr(X_i \leq x, i = 1, \dots, m) \\ &= \Pr(X_1 \leq x) \times \dots \times \Pr(X_m \leq x) \\ &= F(x)^m \rightarrow 0,\end{aligned}$$

so in order to obtain a non-degenerate limiting distribution for the maximum, we must rescale the X_i . We consider $Y_m = a_m^{-1}(\max_i X_i - b_m)$ for sequences of constants $\{a_m\} > 0$ and $\{b_m\}$, and ask under what conditions $Y_m \xrightarrow{D} Y$ as $m \rightarrow \infty$ for some non-degenerate random variable Y . As $m \rightarrow \infty$,

$$\begin{aligned}\Pr(Y_m \leq y) &= \Pr[a_m^{-1}\{\max(X_1, \dots, X_m) - b_m\} \leq y] \\ &= F(b_m + a_m y)^m \\ &= \left[1 - \frac{m\{1 - F(b_m + a_m y)\}}{m}\right]^m\end{aligned}\quad (6.32)$$

can be shown to possess a limit if and only if $\lim_{m \rightarrow \infty} m\{1 - F(b_m + a_m y)\}$ exists. As $m\{1 - F(b_m + a_m y)\}$ is the number of the $X_1, \dots, X_m$ expected to exceed $b_m + a_m y$, suitable sequences $\{a_m\}$ and $\{b_m\}$ exist for most, but not all, continuous distributions. If they do exist, a remarkable result is that the only possible non-trivial limit is of form

$$\lim_{m \rightarrow \infty} m\{1 - F(b_m + a_m y)\} = \left(1 + \xi \frac{y - \eta}{\tau}\right)_+^{-1/\xi}, \quad (6.33)$$

with the right-hand side taken to be $\exp\{-(y - \eta)/\tau\}$ if $\xi = 0$. The parameters τ and η control the scale and location of the limit, and account for the effect of minor changes to $\{a_m\}$ and $\{b_m\}$ — for example, replacing a_m by $\frac{1}{2}a_m$ would rescale any limit, but would not affect its existence or its shape.

On putting together (6.32) and (6.33), we see that if a limiting distribution for the maximum exists, it must be the *generalized extreme-value distribution*

$$H(y; \eta, \tau, \xi) = \exp\left\{-\left(1 + \xi \frac{y - \eta}{\tau}\right)_+^{-1/\xi}\right\}, \quad -\infty < \xi, \eta < \infty, \tau > 0, \quad (6.34)$$

where the range of y is such that $1 + \xi(y - \eta)/\tau > 0$. The parameter ξ controls the shape of the density, which has a heavy right tail and finite lower support point if $\xi > 0$, and a finite upper support point if $\xi < 0$. The *Gumbel distribution*

$$H(y; \eta, \tau, 0) = \exp[-\exp\{-(y - \eta)/\tau\}], \quad -\infty < y < \infty,$$

arises as $\xi \rightarrow 0$; see Problem 6.11. Expression (6.34) gives the only possible limiting distribution for maxima. Minima are dealt with by noting that any limit distribution for $\min_i(X_i) = -\max_i(-X_i)$ must have form $1 - H(-y; \eta, \tau, \xi)$.

Emil Julius Gumbel (1891–1966) was born and studied in Munich. His radical pacifist views and Jewish background caused conflict with his university colleagues and authorities in Heidelberg, and led to his exile in France in 1932 and later in the USA. He highlighted the importance of statistical extremes, on which he wrote an important book (Gumbel, 1958), and through his consulting strongly influenced hydrologists, meteorologists, and engineers.

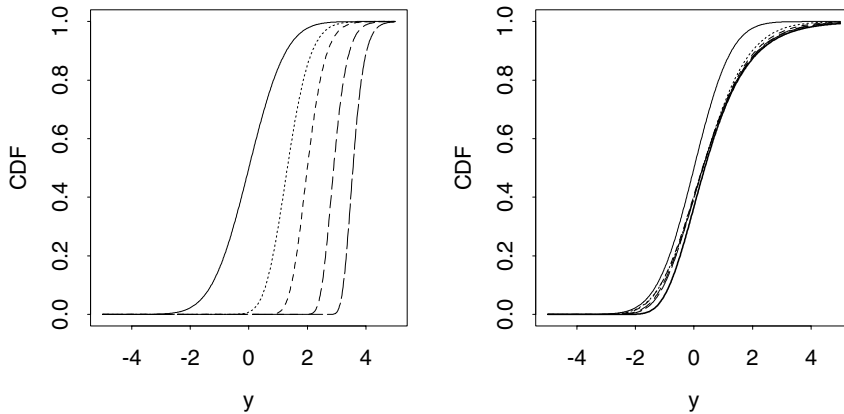


Figure 6.15
Convergence for sample maxima. Left panel: distributions of maxima of $m = 1, 7, 30, 365, 3650$ standard normal variables (from left to right). Right panel: distributions of renormalized maxima of $m = 1, 7, 30, 365, 3650$ standard normal variables. The distributions on the right converge to the Gumbel distribution (heavy).

Example 6.32 (Normal distribution) For the standard normal distribution, integration by parts gives $1 - F(x) = \int_x^\infty \phi(x) dx \doteq \phi(x)/x$ as $x \rightarrow \infty$. Hence $m \{1 - F(b_m + a_m y)\}$ approximately equals

$$\exp \left\{ -\frac{1}{2}(b_m + a_m y)^2 - \log(b_m + a_m y) + \log m - \frac{1}{2} \log 2\pi \right\}, \quad (6.35)$$

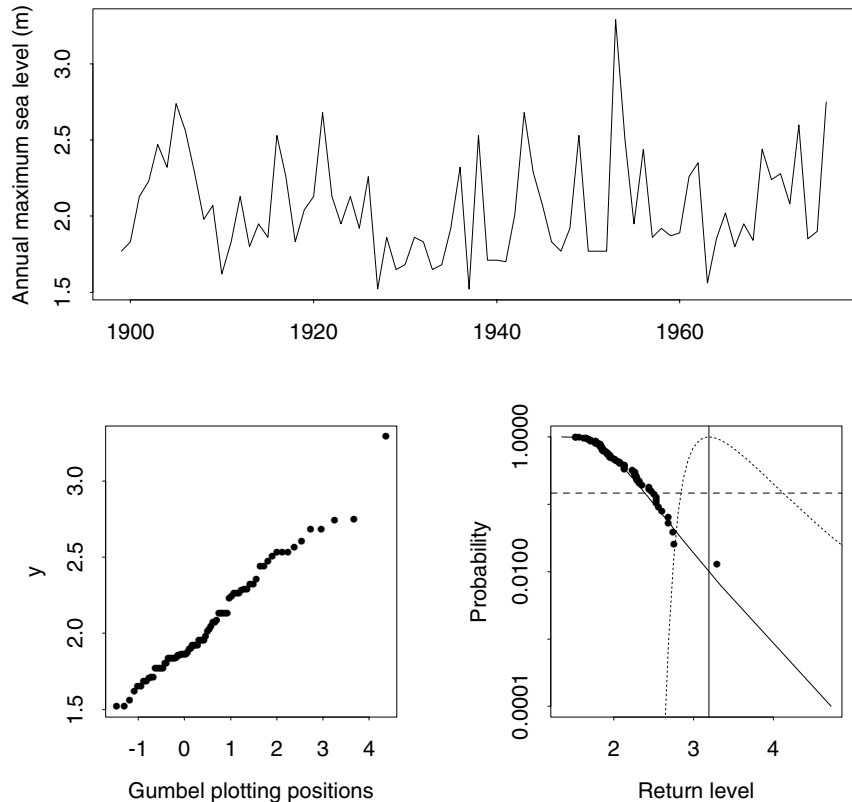
and some tedious algebra shows that with $a_m = (2 \log m)^{-1/2}$ and $b_m = a_m^{-1} - \frac{1}{2} a_m (\log \log m + \log 4\pi)$, (6.35) converges to $\exp(-y)$ as $m \rightarrow \infty$. However the convergence is very slow. With $y = 4$ the probabilities $\Phi(b_m + a_m y)^m$ are 0.9907, 0.9871, 0.9859, 0.9855 for $m = 30, 365, 1825, 3650$, while the target Gumbel probability is 0.9819. These values of m are chosen to correspond to random sampling of a normal distribution daily for periods of one month, and one, five, and ten years. Even with this amount of daily data the limiting probability is not attained, because the right tail of the normal distribution is so light compared to that of the Gumbel distribution that enormous samples are needed for the limit to work well.

Figure 6.15 shows the convergence graphically. The left panel shows the distributions of maxima of m standard normal variables, with $m = 1, 7, 30, 365$, and 3650 , corresponding to maxima over a day, a week, a month, a year and ten years of daily normal data. The distribution becomes increasingly concentrated as m increases, and does not converge to a useful limit. The right panel shows how the distribution of $a_m^{-1} \{\max(X_1, \dots, X_m) - b_m\}$ does converge to a limiting Gumbel distribution, given by the heavy solid line. As mentioned above, the convergence is rather slow. Fortunately the generalized extreme-value distribution usually gives a better approximation for sample maxima than this example might suggest. ■

The upshot is that the generalized extreme-value distribution provides the natural model to fit to sample maxima or minima. For example, if a series of annual maximum sea levels $y_1, \dots, y_n$ is available, we suppose that they are a random sample from (6.34) and fit it by maximum likelihood. Often the parameter of interest is the p quantile of the distribution, that is $y_p = \eta + \tau \{(-\log p)^{-\xi} - 1\}/\xi$, which is known in this context as the $(1 - p)^{-1}$ -year *return level*: it is the level exceeded once on average

$1/(1 - p)$ is known as the return period.

Figure 6.16 Annual maximum sea levels (m) at Yarmouth, 1899–1976. Lower left: Gumbel probability plot of the data. Lower right: fitted (solid) and empirical exceedance probabilities (points), with inference tools for 100-year return level $y_{0.99}$. The vertical line shows the value of $\hat{y}_{0.99}$, while its profile likelihood and 95% confidence interval are shown by the dotted and dashed lines. Note the strong asymmetry of the confidence interval.



every $(1 - p)^{-1}$ years. This would be important if the data were being analyzed in order to suggest how high coastal defenses should be built. Of course quantities such as the expected insurance loss should flooding occur are also of interest.

Maximum likelihood estimation is regular if $\xi > -1/2$, as seems common in applications. When $\xi \leq -1/2$, the likelihood derivatives do not have their usual properties and Example 4.43 is relevant, as the upper support point of the density can be estimated with rate faster than the usual $n^{-1/2}$.

The return level is estimated by replacing η , τ , and ξ by their maximum likelihood estimates. Its standard error may be obtained using the delta method (page 122), though the profile log likelihood for y_p gives a more reliable confidence set. In practice n is often substantially smaller than $(1 - p)^{-1}$ and the return level is estimated well outside the range of the data. Then it is important to consider whether there are enough data underlying the $y_1, \dots, y_n$ for the generalized extreme-value model to give a good approximate distribution for the maxima, and to check whether n is large enough for large-sample likelihood theory to be a good basis for inference. The crucial aspect is however the extent to which extrapolation to high quantiles of the distribution is sensible based on limited data, and this bears careful consideration.

Example 6.33 (Yarmouth sea level data) The upper panel of Figure 6.16 shows a time series of annual maximum sea levels at Yarmouth on the east coast of England for

1899–1976. As is typical with such data, the largest value is considerably greater than the rest; it arose in 1953 when there was widespread flooding. The correlogram and partial correlogram show no serial dependence, so we treat the values as independent.

The lower left panel of the figure shows a probability plot of the data against Gumbel plotting positions. Upward curvature would here suggest that $\xi > 0$, and downward curvature that $\xi < 0$. In fact the plot is close to straight, indicating that $\xi \doteq 0$. The large value from 1953 does not appear outlying, because of the heavy right tail of the density.

The maximum likelihood estimates and standard errors are $\hat{\eta} = 1.90$ (0.034), $\hat{\tau} = 0.26$ (0.025), and $\hat{\xi} = 0.04$ (0.096); the latter give no evidence against the Gumbel model, in agreement with the probability plot. The location and scale parameters are well determined compared to ξ .

The lower right panel of Figure 6.16 compares the estimated survivor function $\Pr(Y > y)$ with its empirical counterpart, obtained by plotting $1 - j/(n + 1)$ against $y_{(j)}$. The vertical line indicates the estimated 100-year return level, $\hat{y}_{0.99}$, while the broken lines show the profile likelihood for $y_{0.99}$ and the corresponding 95% confidence interval. This is highly asymmetric, so this interval is much preferable to using normal approximation. In practice 1000- or even 10,000-year return levels may be needed, and then of course the statistical uncertainty is very large indeed. ■

Point process approximation

If more extensive data are available it is potentially wasteful to use only the annual maxima, and we now show how a Poisson process model can overcome this. Let $X_1, \dots, X_{\lceil mt_0 \rceil}$ be a random sample from $F(x)$ and consider the pattern of points $(i/m, a_m^{-1}(X_i - b_m))$, $i = 1, \dots, \lceil mt_0 \rceil$ that fall into the subset $\mathcal{S} = [0, t_0] \times [u, \infty)$ of the plane. The event $a_m^{-1}(X_i - b_m) > y$ occurs if and only if $X_i > b_m + a_m y$, so the number of points that fall into $\mathcal{A} = [t_1, t_2] \times [y, \infty)$ may be expressed as the sum of indicator random variables

$$N_m(\mathcal{A}) = \sum_{i=\lceil mt_1 \rceil}^{\lceil mt_2 \rceil} I(X_i > b_m + a_m y), \quad 0 \leq t_1 < t_2 \leq t_0, y \geq u.$$

The X_i are independent and identically distributed, so $N_m(\mathcal{A})$ is binomial with denominator $\lceil mt_2 \rceil - \lceil mt_1 \rceil + 1$ and probability $1 - F(b_m + a_m y)$ that satisfies (6.33). Hence the Poisson limit for the binomial distribution (Problem 2.3) gives

$$\lim_{m \rightarrow \infty} \Pr\{N_m(\mathcal{A}) = n\} = \frac{\Lambda(\mathcal{A})^n}{n!} \exp\{-\Lambda(\mathcal{A})\}, \quad n = 0, 1, \dots,$$

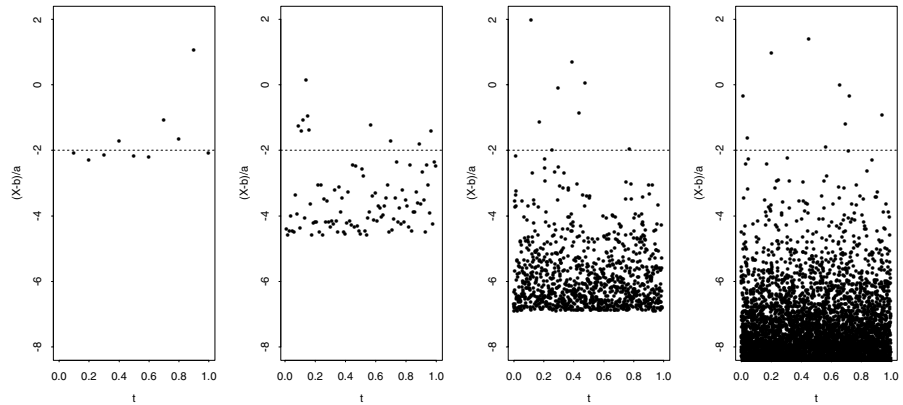
where $\Lambda(\mathcal{A})$ equals

$$\Lambda\{[t_1, t_2] \times [y, \infty)\} = (t_2 - t_1) \left(1 + \xi \frac{y - \eta}{\tau}\right)_+^{-1/\xi}, \quad 0 \leq t_1 < t_2 \leq t_0, y \geq u, \quad (6.36)$$

with the second term on the right replaced by $\exp\{-(y - \eta)/\tau\}$ if $\xi = 0$. That is, $N_m(\mathcal{A}) \xrightarrow{D} N(\mathcal{A})$, where $N(\mathcal{A})$ is Poisson with mean $\Lambda(\mathcal{A})$.

$\lceil a \rceil$ and $\lfloor a \rfloor$ are respectively the smallest integer larger than a and the largest integer smaller than a .

Figure 6.17 Poisson process limit for rare events. The panels show the values of $a_m^{-1}(X_i - b_m)$ plotted against i/m for random samples of size $m = 10, 100, 1000$ and $10,000$ from the exponential distribution. The pattern of points above the threshold at $u = -2$ tends to a bivariate Poisson process with intensity given by (6.36).



More sophisticated techniques reveal that as $m \rightarrow \infty$, the limiting joint distributions of counts $N_m(\mathcal{A}_1), N_m(\mathcal{A}_2), \dots$ in any collection of disjoint subsets $\mathcal{A}_1, \mathcal{A}_2, \dots$ of $\mathcal{S}$ is that of independent Poisson variables with means $\Lambda(\mathcal{A}_1), \Lambda(\mathcal{A}_2), \dots$. Hence as $m \rightarrow \infty$, the limiting positions of random values X_i , suitably rescaled, have the joint distribution of points of a Poisson process N in $\mathcal{S}$ with intensity (6.36), with arbitrary u . Figure 6.17 illustrates this for exponential samples.

To see the connection to extremes, suppose we have daily data for t_0 years and that $t_2 - t_1 = 1$ year. Then if we apply the Poisson limit to these data with $\mathcal{A} = [t_1, t_2] \times [y, \infty)$, effectively assuming that the limit has set in when $m = 365$ days, and let $Y^1 \geq \dots \geq Y^r$ denote the r largest values for that year, we see that in an obvious shorthand notation,

$$\begin{aligned} \Pr(Y^1 \leq y) &= \Pr\{N(\mathcal{A}) = 0\} \\ &= \exp \left\{ - \left(1 + \xi \frac{y - \eta}{\tau} \right)_+^{-1/\xi} \right\}, \\ \Pr(Y^r \leq y^r, \dots, Y^1 \leq y^1) &= \Pr\{N(y^r, y^{r-1}) = 1, \dots, N(y^2, y^1) = 1\}. \end{aligned}$$

The first of these identities recovers (6.34), while the joint density of $Y^1 \geq \dots \geq Y^r$ at $y^1 \geq \dots \geq y^r$ is obtained either by differentiating the second identity or from (6.31), with $\mathcal{S}$ replaced by $[t_1, t_2] \times [y^r, \infty)$. Both routes show that the limiting joint density of the r largest values is

$$\prod_{i=1}^r \tau^{-1} \left(1 + \xi \frac{y^i - \eta}{\tau} \right)_+^{-1/\xi - 1} \times \exp \left\{ - \left(1 + \xi \frac{y^r - \eta}{\tau} \right)_+^{-1/\xi} \right\}. \quad (6.37)$$

Independence of counts in disjoint subsets implies that data for different years may be treated as independent, so an overall likelihood based on the r largest values for each year is simply the product of such terms for all t_0 years.

In many ways a more satisfactory approach to inference starts by noticing that (6.36) has form $\Lambda_1\{[t_1, t_2]\}\Lambda_2\{[y, \infty)\}$, implying that the points result from two independent Poisson processes, one giving the random ‘times’ T at which $X_i > u$, and the other giving the rescaled sizes $a_m^{-1}(X_i - b_m)$ of these X_i . The times of

exceedances fall according to a homogeneous Poisson process of intensity $\lambda_1(t) = \{1 + \xi(u - \eta)/\tau\}_+^{-1/\xi} \equiv \lambda$, say, while their sizes follow an inhomogeneous Poisson process whose intensity is

$$\lambda_2(y) = -\frac{d\Lambda_2\{[y, \infty)\}}{dy} = \tau^{-1} \left(1 + \xi \frac{y - \eta}{\tau}\right)_+^{-1/\xi - 1}, \quad y > u.$$

This implies that the number of exceedances over level u has a Poisson distribution with mean λt_0 , and conditional on n_u exceedances, their sizes $W_j = X_j - u$ are a random sample of size n_u from the *generalized Pareto distribution* (Problem 6.15)

$$G(w) = \begin{cases} 1 - (1 + \xi w/\sigma)_+^{-1/\xi}, & \xi \neq 0, \\ 1 - \exp(-w/\sigma), & \xi = 0. \end{cases} \quad (6.38)$$

The log likelihood (6.31) may be written as

$$\ell(\lambda, \sigma, \xi) \equiv n_u \log \lambda - t_0 \lambda - n_u \log \sigma - \left(\frac{1}{\xi} + 1\right) \sum_{j=1}^{n_u} \log \left(1 + \xi \frac{w_j}{\sigma}\right). \quad (6.39)$$

We apply this discussion by taking a threshold u over which the Poisson approximation seems to hold; then the exceedance times should be a homogeneous Poisson process, and their sizes should follow (6.38), as typically assessed by a probability plot. If the fit is satisfactory, estimates and standard errors are obtained by our usual likelihood methods. As with the generalized extreme-value distribution, estimation of σ and ξ is not regular if $\xi \leq -1/2$, and Example 4.43 is again relevant.

We now briefly discuss the choice of u . If it is chosen so that the number of exceedances is small, then the Poisson process approximation to the extremes may be good, but the parameter estimators will have large variance. The variance can be reduced by lowering u , but at the cost of bias because the Poisson approximation for extremes cannot be expected to give good inferences when applied to the bulk of the data. Formal procedures for choosing u attempt to trade off these two aspects, but in practice graphical approaches are more common. These rest on the *threshold stability* property of a random variable W following (6.38), that is,

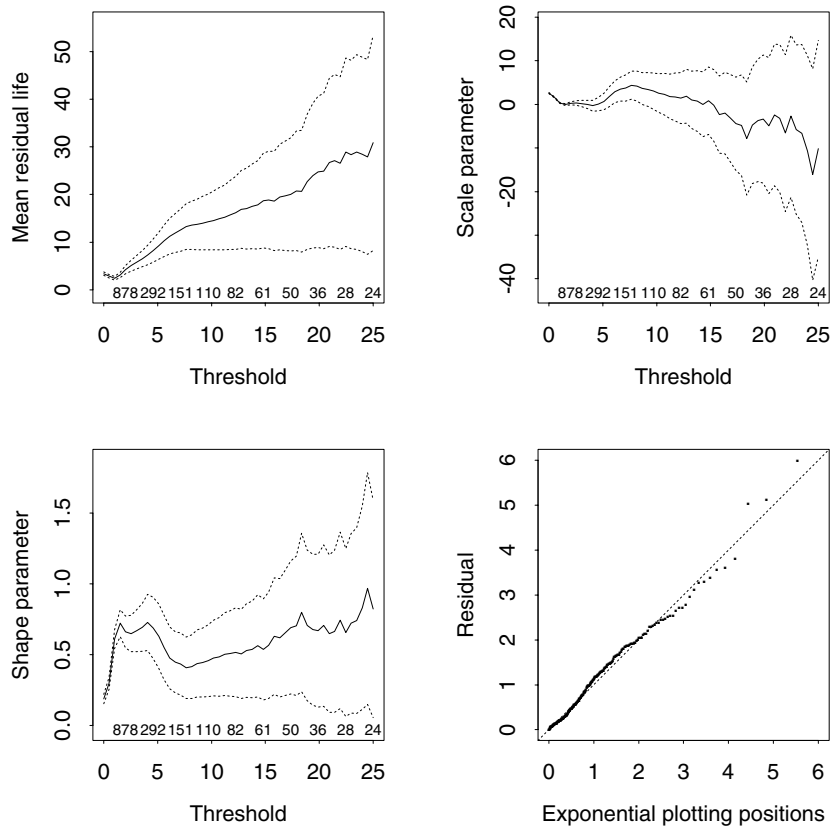
$$\Pr(W > w \mid W > u) = \{1 + \xi(w - u)/\sigma_u\}_+^{-1/\xi}, \quad w \geq u \geq 0,$$

where $\sigma_u = \sigma + \xi u$. The operation of thresholding by considering only the tail of W above u yields another random variable $W_u = W - u$, say, following (6.38) but transforms the parameters as $(\sigma, \xi) \mapsto (\sigma + \xi u, \xi)$. When $\xi = 0$ this is the lack-of-memory property of the exponential distribution.

One graphical approach uses the fact that $E(W) = \sigma/(1 - \xi)$ provided $\xi < 1$, so $E(W_u \mid W > u) = (\sigma + \xi u)/(1 - \xi)$, for $u \geq 0$. Thus if the generalized Pareto approximation is adequate for the upper tail of a random sample $X_1, \dots, X_n$, a graph against u of the empirical version of this conditional mean, given by

$$n_u^{-1} \sum_{j=1}^n (X_j - u) I(X_j > u), \quad \text{where} \quad n_u = \sum_{j=1}^n I(X_j > u), \quad (6.40)$$

Figure 6.18 Analysis of Danish fire data. Upper left: mean residual life plot, with 95% confidence band (dots) and number of exceedances n_u at the foot of the panel. Upper right and lower left: plots of $\hat{\sigma}_u - \hat{\xi}_u u$ and $\hat{\xi}_u$ against threshold u , with 95% confidence bands. Lower right: exponential probability plot of residuals $\hat{\xi}^{-1} \log(1 + \hat{\xi} w_j / \hat{\sigma})$.



should be a straight line of gradient $\xi/(1 - \xi)$. The idea is to take the threshold to be the smallest u above which this *mean residual life plot* appears linear.

Another approach to choosing u uses the fact that if $\hat{\xi}_u$ and $\hat{\sigma}_u$ are maximum likelihood estimators based on the n_u positive exceedances $X_j - u$ over u , and if the generalized Pareto approximation holds, then $\hat{\xi}_u$ and $\hat{\sigma}_u - \hat{\xi}_u u$ should estimate ξ and σ for all u . Thus graphs of $\hat{\xi}_u$ and $\hat{\sigma}_u - \hat{\xi}_u u$ against u should be constant above a certain point, and this is the minimum threshold for which it is reasonable to apply the approximation. Interpretation of such graphs is aided by adding confidence intervals.

Example 6.34 (Danish fire data) In Example 6.31 we saw that exceedance times for the data in the upper right panel of Figure 6.14 seem to follow a homogeneous Poisson process with rate about 0.06 days^{-1} . For threshold modelling we first choose the threshold u . Figure 6.18 shows the mean residual life plot and values of $\hat{\sigma}_u - \hat{\xi}_u u$ and $\hat{\xi}_u$ plotted against u . The mean residual life plot is roughly linear from $u = 7$ onwards, and its positive slope suggests that $\xi > 0$. The other two plots do not tend to constants, but in each case the confidence intervals are wide enough to contain a constant above about $u = 5$. For illustration we take $u = 5$, let $w_j = y_j - u$ denote the 254 claims that exceed $u = 5$ units, and fit the generalized Pareto distribution

(6.38) to the w_j . The maximum likelihood estimates are $\hat{\sigma} = 3.809$ and $\hat{\xi} = 0.632$, with standard errors 0.464 and 0.111 from observed information. The value of $\hat{\xi}$ corresponds to a very heavy upper tail for $W = Y - u$.

The form of (6.38) shows that $\xi^{-1} \log(1 + \xi W/\sigma)$ has a standard exponential distribution, so the fit of the model for exceedances can be assessed by an exponential probability plot of the residuals $\hat{\xi}^{-1} \log(1 + \hat{\xi} w_j/\hat{\sigma})$, shown in the left panel of Figure 6.18. The distribution fits fairly well but not perfectly.

Estimates and confidence regions for quantities of interest such as return levels are found in ways analogous to Example 6.33. In practice it is important to vary the threshold to see if the conclusions depend strongly on u . ■

In applications the underlying variables are typically neither identically distributed nor independent. For concreteness, consider using daily temperature data to model the occurrence of hot days at a site in England. These will occur in the summer months, so one way to proceed is to retain only the data for June, July, and August, to suppose that over this period the temperature distribution is roughly constant, and then to hope that about 90 rather than 365 days of data will suffice for the point process paradigm to be applicable. However, even if the summer data are roughly stationary, they will display short-term correlation owing to clustering of hot days. Some detailed mathematics establishes that if extremes far apart are asymptotically independent and the data are stationary — so that in particular all the X_i have the same marginal distribution — then the Poisson process representation with intensity (6.36) still applies, but now to the largest value in a cluster. Clusters then occur at the times of a homogeneous Poisson process, but the cluster size is random and its distribution depends on the local dependence of the X_i . This leads to the practical issues of identifying clusters from data, and of modelling their properties, which are topics of current research.

6.5.3 More general models

In a Poisson process events in disjoint intervals are independent. In practice point process data can show complex dependencies, so this property must be weakened for realistic modelling. This weakening can be done in many ways and below we merely sketch a few possibilities. We continue to suppose that the process is orderly, so events cannot coincide.

Let $\mathcal{H}_t$ denote the entire history of the process up to time t , that is, the positions of all the points in $(-\infty, t]$, and define the *complete intensity function* to be

$$\lambda_{\mathcal{H}}(t) = \lim_{\delta t \rightarrow 0} (\delta t)^{-1} \Pr\{N(t, t + \delta t) > 0 \mid \mathcal{H}_t\};$$

this is the intensity of arrival of points just after t , given the history to t . It is akin to the hazard function of Section 5.4, but here potentially dependent on the entire history of the process. The requirement of orderliness is that

$$\Pr\{N(t, t + \delta t) > 1 \mid \mathcal{H}_t\} = o(\delta t)$$

for all t and all possible $\mathcal{H}_t$. The complete intensity must be uniquely defined and well-behaved for any possible $\mathcal{H}_t$ and must moreover determine the probabilistic structure of the process. We shall take this for granted here, though a careful mathematical argument is needed in a formal discussion.

Now consider the probability of no event in $(w, w + t]$ conditional on $\mathcal{H}_w$. We divide $(w, w + t]$ into disjoint subintervals $\mathcal{I}_i = (w + i\delta t, w + (i + 1)\delta t]$, $i = 0, \dots, k - 1$, where $\delta t = t/k$, and note that

$$\begin{aligned} \Pr\{N(w, w + t) = 0 \mid \mathcal{H}_w\} &\doteq \prod_{i=0}^{k-1} \Pr\{N(\mathcal{I}_i) = 0 \mid \mathcal{H}_{w+i\delta t}\} \\ &= \prod_{i=0}^{k-1} \{1 - \lambda_{\mathcal{H}}(w + i\delta t)\delta t + o(\delta t)\}, \end{aligned}$$

where $\mathcal{H}_{w+i\delta t}$ represents $\mathcal{H}_w$ followed by no events up to time $w + i\delta t$. The argument leading to (6.28) applies with $\lambda(u)$ replaced by $\lambda_{\mathcal{H}}(u)$, so

$$\Pr\{N(w, w + t) = 0 \mid \mathcal{H}_w\} = \exp\left\{-\int_w^{w+t} \lambda_{\mathcal{H}}(u) du\right\},$$

and the probability density that the first point subsequent to w is at t , given $\mathcal{H}_w$, is $-d\Pr\{N(w, w + t) = 0 \mid \mathcal{H}_w\}/dt$. At least in principle, this enables the likelihood for points in an interval $(0, t_0]$, conditional on $\mathcal{H}_0$, to be written down by extending our arguments for the Poisson process, giving

$$\prod_{j=1}^n \lambda_{\mathcal{H}}(t_j) \exp\left\{-\int_0^{t_0} \lambda_{\mathcal{H}}(u) du\right\} \quad (6.41)$$

as the likelihood based on events at $t_1, \dots, t_n$ when the process is observed over $(0, t_0]$. In practice it is often hard to specify a tractable but realistic form for $\lambda_{\mathcal{H}}(t)$.

A useful implication is that if events are observed at times $0 < T_1 < \dots < T_n < t_0$ and we write $\Lambda_{\mathcal{H}}(t) = \int_0^t \lambda_{\mathcal{H}}(u) du$, then the transformed times $\Lambda_{\mathcal{H}}(T_1), \dots, \Lambda_{\mathcal{H}}(T_n)$ form a Poisson process of unit rate on $(0, \Lambda_{\mathcal{H}}(t_0)]$, the transformation $\Lambda_{\mathcal{H}}$ being random. Thus our earlier tools may be used to check the adequacy of an estimated $\hat{\Lambda}_{\mathcal{H}}$.

Example 6.35 (Poisson process) The complete intensity function for a Poisson process may depend on t , but not on the history of the process. Thus $\lambda_{\mathcal{H}}(t) = \lambda(t)$, which is a constant λ for a homogeneous process. ■

Example 6.36 (Renewal process) The inter-event intervals in a homogeneous Poisson process are independent exponential variables. The renewal process generalizes this to possibly non-exponential intervals and is a standard model in reliability studies, where failing components in a system may be immediately replaced by apparently identical ones, thereby renewing the system. If system failure is identified with failure of the component and the process is stationary then the complete intensity function depends only on the time since the last event. Thus if previous events have taken place at times t_i , the complete intensity at time t depends only on $v = \min(t - t_i)$

and has form $\lambda(v)$. This is the hazard function corresponding to the density of interval lengths, f . Statistical analysis for such a process is straightforward. Time series tools such as the correlogram and partial correlogram can be used to find serial dependence among successive intervals between events, though it may be clear from the context that these are independent. If independent and stationary, they can be treated as a random sample from f and inference performed in the usual way. ■

Example 6.37 (Birth process) In a birth process the intensity at time t depends on the number of previous events. Assuming that the number n of events up to t is finite, then $\lambda_{\mathcal{H}}(t) = \beta_0 + \beta_1 n$, where $\beta_0 > 0$, $\beta_1 \geq 0$. The complete intensity function is a step function which jumps β_1 at each event; if $\beta_1 = 0$ the process is a homogeneous Poisson process. ■

Before giving a numerical example, we briefly describe two functions useful for model checking and exploratory analysis of stationary processes.

The *variance-time curve* is defined as $V(t) = \text{var}\{N(t)\}$, for $t > 0$. A homogeneous Poisson process of intensity λ has $V(t) = \lambda t$, comparisons with which may be informative. Estimation of $V(t)$ is described in Problem 6.12.

The *conditional intensity function* is defined as

$$m_f(t) = \lim_{\delta s, \delta t \rightarrow 0} (\delta t)^{-1} \Pr\{N(t, t + \delta t) > 0 \mid N(-\delta s, 0) > 0\}, \quad t > 0,$$

which gives the intensity of events at t conditionally on there being an event at the origin. Evidently $m_f(t) = \lambda$ for a homogeneous Poisson process. An event at time t need not be the first event after that at the origin.

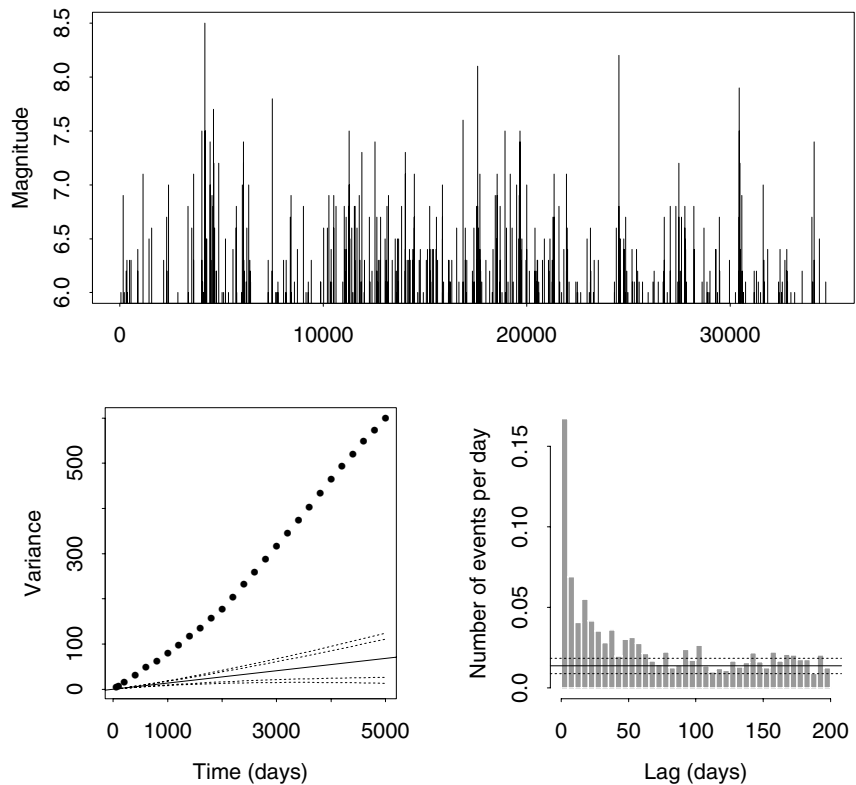
Example 6.38 (Japanese earthquake data) Figure 6.19 shows the times and magnitudes of earthquakes with epicentre less than 100km deep in an offshore region west of the main Japanese island of Honshū and south of the northern island of Hokkaidō. The figure shows all 483 earthquakes of magnitude 6 or more on the Richter scale in the period 1885–1980, about 5 tremors per year, in one of the most seismically active areas of Japan. A cumulative plot of the times rises fairly evenly and suggests that the data may be regarded as stationary; we shall assume this below. We take days as the units, giving $t_0 = 35,175$.

This is a *marked point process*, as in addition to the event times there is a *mark* — the magnitude — attached to each event. If we let the times be $0 < t_1 < \dots < t_n < t_0$ and the associated magnitudes $m_1, \dots, m_n$, their joint density may be written

$$\prod_{j=1}^n f(m_j \mid m_{(j-1)}, t_{(j)}) \prod_{j=1}^n f(t_j \mid m_{(j-1)}, t_{(j-1)}), \quad (6.42)$$

where $t_{(j-1)}$ and $m_{(j-1)}$ represent $t_1, \dots, t_{j-1}$ and $m_1, \dots, m_{j-1}$. Here we concentrate on inference for the times using the second term, leaving the magnitudes to Examples 10.7 and 10.31. The lower panels of Figure 6.19 show the estimated variance-time curve and conditional intensity function for the times, which are clearly far from Poisson. The variance-time curve grows more quickly than for a Poisson process, indicating clustering of events, and this is confirmed by the

Figure 6.19 Japanese earthquake data (Ogata, 1988). The upper panel shows the times and magnitudes (Richter scale) of 483 shallow earthquakes. Lower left: estimated variance-time curve for earthquake times, with theoretical line for a Poisson process (solid) and two-sided 95% and 99% pointwise confidence limits (dots). Lower right: estimated conditional intensity, with baseline for Poisson process (solid) and two-sided 95% pointwise confidence limits (dots).



conditional intensity: for about 2–3 months after each shock the probability of another is increased.

One possible model for such data is a *self-exciting process* in which

$$\lambda_{\mathcal{H}}(t) = \mu + \sum_{j:t_j < t} w(t - t_j),$$

where μ is a positive constant and $w(u)$ is non-negative for $u > 0$ and otherwise zero. Here the intensity at any time is affected by the occurrence of previous events; often $w(u)$ is monotonic decreasing, so recent events affect the current intensity more than distant ones. This may be interpreted as asserting that events occur in clusters, whose centres occur as a Poisson process of rate μ . Subsidiary events are then spawned by the increase in intensity that occurs due to the superposition of the $w(t - t_j)$ for previous events. Seismological considerations suggest letting this function depend on m_j also, taking

$$w(t - t_j; m_j) = \frac{\kappa e^{\beta(m_j - 6)}}{(t - t_j + \gamma)^\rho}, \quad t > t_j,$$

where $\rho, \gamma, \kappa, \beta, \mu > 0$, with $\beta \doteq 2$. Under this formulation the increase in intensity depends not only on the time since an event but also on its magnitude.

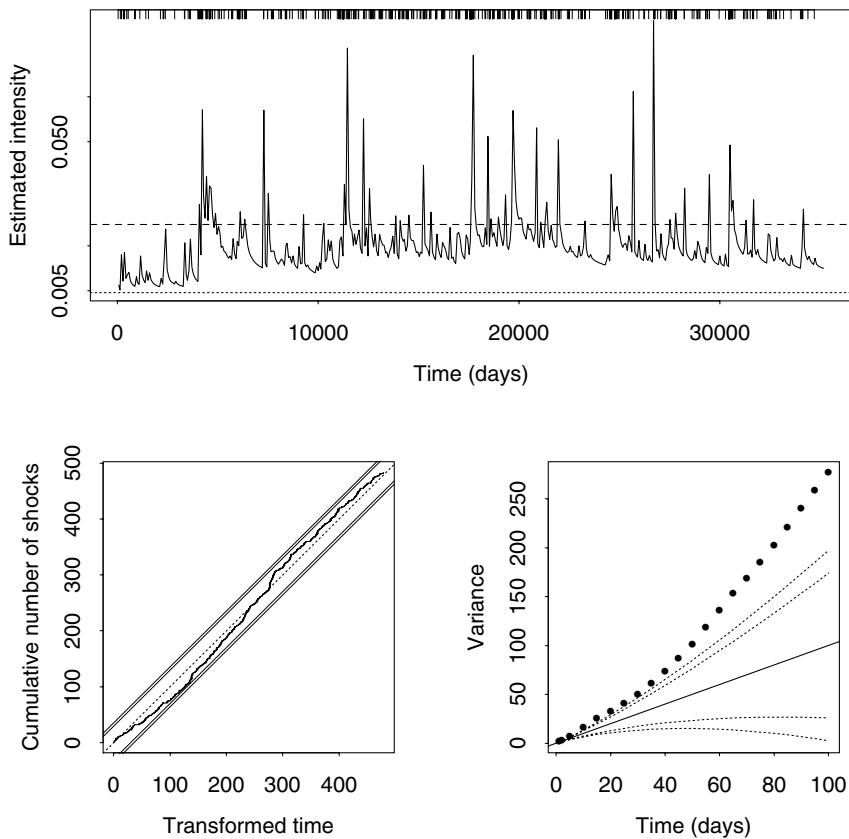


Figure 6.20 Japanese earthquake data fit. The upper panel shows the estimated intensity $\hat{\lambda}_H(t)$ events/day with $\hat{\mu}$ (dots) and the mean intensity (dashes). The tick marks at the top of panel show the event times. Lower left: estimated cumulative number of events $\hat{\Lambda}_H(t_j)$ (solid) and two-sided 95% and 99% overall confidence limits (solid diagonal), based on the Kolmogorov–Smirnov statistic; the dotted line shows perfect fit of the model. Lower right: variance-time function for transformed process $\hat{\Lambda}_H(t_j)$ (blobs), with baseline for Poisson process (solid) and two-sided 95% and 99% pointwise confidence limits (dots)

The log likelihood (6.41) corresponding to the second term of (6.42) with the self-exciting model is readily obtained. Its maximized value is -2232.01 , but this changes only to -2232.25 on fixing $\rho = 1$. With this restriction the estimates and standard errors are $\hat{\mu} = 0.0049$ (0.0007) events/day, $\hat{\kappa} = 0.020$ (0.003) events/day, $\hat{\gamma} = 0.054$ (0.024) days, and $\hat{\beta} = 1.61$ (0.14). These imply that after an earthquake of size $m_j = 6$, $\lambda_H(t)$ jumps by $\hat{\kappa}/\hat{\gamma} \doteq 0.37$ events/day, while a shock of size $m_j = 8$ induces a jump of $\hat{\kappa}e^{2\hat{\beta}}/\hat{\gamma} \doteq 9.2$ events/day. The rate at which clusters arise is about $365\hat{\mu} \doteq 1.8$ events/year, so each gives rise to a further 3.2 shocks on average.

The top panel of Figure 6.20 shows the fitted intensity $\hat{\lambda}_H(t)$, with the value of $\hat{\mu}$ and the mean intensity; note the logarithmic scale. The fitted value is initially low perhaps because of the lack of data before $t = 0$, and it would be preferable to use only a portion of the likelihood, as in Example 6.29. The lower panels show the cumulative intensity for the transformed process $\hat{\Lambda}_H(t_j)$, which would be a straight line of unit gradient if the model fitted perfectly. The cumulative intensity lies within overall 95% confidence limits and gives no evidence against the model. However the variance-time curve of the transformed times shows clear overdispersion relative to a Poisson process. The data include an unusual series of about 25 large earthquakes in November–December 1938, all occurring in the same region. When these are

removed, the remainder have variance-time curve falling within the Poisson limits and the model then seems adequate. ■

Exercises 6.5

Recall that
 $(1 + a/k)^k \rightarrow e^a$ as
 $k \rightarrow \infty$.

- 1 For a Poisson process on $[0, t_0]$ of constant rate λ , show directly that $N(t_0)$ has a Poisson distribution of mean λt_0 by showing that

$$\Pr\{N(t_0) = m\} \doteq \frac{k!}{m!(k-m)!} \{\lambda \delta t + o(\delta t)\}^m \{1 - \lambda \delta t + o(\delta t)\}^{k-m},$$

where $\delta t = t_0/k$, and letting $k \rightarrow \infty$.

- 2 Check that

$$\int_0^{t_0} dt_n \int_0^{t_n} dt_{n-1} \cdots \int_0^{t_3} dt_2 \int_0^{t_2} dt_1 \lambda(t_1) \cdots \lambda(t_n) e^{-\Lambda(t_0)}$$

equals (6.30).

- 3 Consider a Poisson process of intensity λ in the plane. Find the distribution of the area of the largest disk centred on one point but containing no other points.
- 4 Show that the time to the r th event in a Poisson process of rate λ has the gamma distribution.
- 5 If T is the time to the first event in a one-dimensional Poisson process of positive intensity $\lambda(t)$, show that $\Lambda(T)$ has a standard exponential distribution. Write down an algorithm to generate the points $0 < T_1 < \cdots < T_N < t_0$ of a Poisson process of rate $\lambda(t)$ on $[0, t_0]$. Test it.

Deletion of points of a
 process is known as
thinning.

- 6 Over the centuries natural disasters in a particular country have occurred as a Poisson process of rate $\lambda(t)$. Any disaster at time t is known to have occurred only with probability $\pi(t)$, due to the patchiness of historical records. If records of different disasters are preserved independently, show that the point process of known disasters is Poisson with intensity $\lambda(t)\pi(t)$.
- 7 Find sequences $\{a_m\} > 0$ and $\{b_m\}$ such that (6.33) holds in the following cases: (i) $1 - F(x) = e^{-x}$ for $x > 0$; (ii) the distribution has a power-law upper tail, $1 - F(x) \sim x^{-\gamma}$, $\gamma > 0$, with $x_0 = \infty$; and (iii) $F(x) = x$ for $0 \leq x \leq 1$. In each case give the value of κ and sketch the limiting distribution.
- 8 Let M_n be the maximum of the random sample $X_1, \dots, X_n$ from a distribution F , and suppose that the limit

$$\lim_{n \rightarrow \infty} \Pr\left(\frac{M_n - b_n}{a_n} \leq y\right)$$

is a nondegenerate distribution function, $H(y)$, for some sequences of constants $a_n > 0$ and b_n . Show that

$$\Pr\left(\frac{M_n - a_n}{b_n} \leq y\right) = \Pr\left(\frac{M_m - a_n}{b_n} \leq y\right)^l,$$

where $n = ml$, and deduce that H must be *max-stable*, that is, for any l there must exist constants c_l and d_l such that $H(y)^l = H(c_l + d_l y)$. Verify that the generalized extreme-value distribution (6.34) is *max-stable*.

- 9 Show that the Fisher information for an observation from (6.38) is

$$i(\sigma, \xi) = (1 + 2\xi)^{-1} \left(\frac{1}{(1 + \xi)^{-1}} - \frac{(1 + \xi)^{-1}}{2(1 + \xi)^{-1}} \right), \quad \xi > -1/2.$$

What happens if $\xi \leq -1/2$?

- 10 (a) If W follows (6.38) and $u > 0$, show that conditional on $W > u$, $W - u$ follows (6.38) with parameters ξ and $\sigma_u = \sigma + \xi u$. Show also that $E(W - u \mid W > u) = \sigma/(1 - \xi)$, provided $\xi < 1$. What happens if $\xi \geq 1$? And if $\xi \geq 1/2$?

- (b) Derive a standard error for (6.40). For what values of ξ is it valid? Explain the saw-tooth form of the mean residual life plot.
- (c) Discuss how confidence bands in plots of $\hat{\xi}_u$ and $\hat{\sigma}_u - \hat{\xi}_u u$ against u might be constructed.
- 11 By reparametrizing (6.38) in terms of $\zeta = \xi/\sigma$ and ξ , show how to obtain maximum likelihood estimates of ξ and σ based on a random sample $w_1, \dots, w_n$ from G , using only a one-dimensional maximization.

6.6 Bibliographic Notes

A useful general account of stochastic modelling dealing with several of the topics in this chapter is Isham (1991).

There are many books on Markov chains. Cox and Miller (1965), Grimmett and Stirzaker (2001) and Norris (1997) give standard accounts of their probabilistic aspects, while Billingsley (1961) describes inference for them. Guttorp (1995) has a nice blend of probabilistic and statistical considerations. Multi-state modelling, including the use of Markov processes, is discussed in Chapters 5 and 6 of Hougaard (2000). MacDonald and Zucchini (1997) and Künsch (2001) describe inference for hidden Markov processes. Prum *et al.* (1995) describe a systematic approach to finding words in DNA sequences, with further references to this area.

Markov random fields emerged around 1970 as a natural generalization of Markov chains to more complex phenomena, though the Ising and related models had been known to physicists since the 1920s. The key result relating Markov random fields and Gibbs distributions was proved in 1971 by J. M. Hammersley and P. Clifford but not published at that time; Clifford (1990) describes its history and some more recent ideas and gives their version of the proof. A simpler proof was given in the important paper of Besag (1974), which discusses a wide range of topics related to spatial modelling; see Smith (1997). Applications to image analysis were described in Geman and Geman (1984) and Besag (1986), which strongly influenced later work on image analysis; see for example Chellappa and Jain (1993). Applications to point processes are reviewed by Isham (1981), while Kinderman and Snell (1980) give a gentle introduction oriented towards problems of classical physics; see also Brémaud (1999). Sheehan (2000) and Thompson (2001) discuss applications in statistical genetics, with numerous further references.

Graphical models have played an increasingly important role in statistics since about 1980, though similar ideas were used in other fields decades earlier. Edwards (2000) gives an applied account of graphical models with many examples, and includes a description of the software package MIM with which certain families of models can be fitted. Lauritzen (1996) is more mathematical, with details of the necessary graph theory and its statistical application. Whittaker (1990) lies between the two, with a blend of applications and theory, while Cox and Wermuth (1996) give a general view of the subject with some substantial applications. All these books contain references to the primary literature. Those by Lauritzen and Cox and Wermuth describe graphs in which different types of edges appear; see also Wermuth and Lauritzen (1990) and Lauritzen and Richardson (2002).

Graphical representations of probabilistic expert systems are described by Lauritzen and Spiegelhalter (1988) and Spiegelhalter *et al.* (1993), from which Example 6.16 is taken. Pearl (1988), Neopolitan (1990), Almond (1995), Castillo *et al.* (1997), Cowell *et al.* (1999) and Jensen (2001) provide fuller accounts.

There are books on multivariate statistics at all levels and in all styles. Accounts of classical models for multivariate data are Anderson (1958), Mardia *et al.* (1979), and Seber (1985). Chatfield and Collins (1980) is more practical, but all predate the emergence of graphical Gaussian modelling. The bibliographic notes for Chapter 10 give references for discrete multivariate data.

Chatfield (1996), Diggle (1990) and Brockwell and Davis (1996) are standard elementary books on time series, while Brockwell and Davis (1991) is a more advanced treatment. Beran (1994) and Tong (1990) describe respectively series with long-range dependence and nonlinearity. With the growth of financial markets over the last two decades financial time series has become an area of major research effort summarized by Shephard (1996); for longer accounts see Gouriéroux (1997) and Tsay (2002). These references primarily describe modelling in the so-called time domain, in which relationships among the observations themselves are central, but a complementary approach based on frequency analysis is the main focus of Bloomfield (1976), Priestley (1981), Brillinger (1981), and Percival and Walden (1993). This second approach is particularly useful in physical applications.

The Poisson process is a fundamental stochastic model and its probabilistic aspects are described in any of the large number of excellent introductory books on stochastic processes; see for example Grimmett and Stirzaker (2001). There are also various more specialised accounts such as in Rolski *et al.* (1999). Accounts of point process theory are by Cox and Isham (1980) and Daley and Vere-Jones (1988). Cox and Lewis (1966) is a thorough account of inference for one-dimensional data, while spatial point processes are the focus of Diggle (1983). Karr (1991) gives a theoretical account of inference for point processes. Ripley (1981, 1988) and Cressie (1991) are more general accounts of the analysis of spatial data. Point processes based on notions allied to Markov random fields are reviewed by Isham (1981), and a fuller treatment is given by van Lieshout (2000).

Statistics of extremes may be said to have started with Fisher and Tippett (1928), but the first systematic book-length treatment of the subject was Gumbel (1958). Modern accounts from roughly the viewpoint taken here are Smith (1990) and Coles (2001), while Embrechts *et al.* (1997) is a systematic mathematical treatment emphasising applications in finance and insurance. The approach using point processes is described by Smith (1989a). Davison and Smith (1990) give a thorough treatment of threshold methods. Books on probabilistic aspects include Leadbetter *et al.* (1983) and Resnick (1987).

6.7 Problems

- 1 Dataframe `alofi` contains three-state data derived from daily rainfall over three years at Alofi in the Niue Island group in the Pacific Ocean. The states are 1 (no rain), 2 (up to

From	To			From	To			From	To		
	1	2	3		1	2	3		1	2	3
11	247	86	29	21	86	27	23	31	29	13	8
12	70	32	24	22	29	35	26	32	37	35	18
13	13	16	31	23	17	17	34	33	20	45	59

From	To			From	To			From	To			From	To		
	1	2	3		1	2	3		1	2	3		1	2	3
1	106	34	14	97	29	17	60	24	16	98	39	12	98	39	12
2	41	27	10	32	21	13	27	27	25	36	13	18	36	13	18
3	8	16	15	13	17	32	13	27	52	15	15	25	15	15	25

Table 6.10 Counts for rainfall data at Alofi (Avery and Henderson, 1999). States are 1 (no rain), 2 (up to 5mm rain) and 3 (over 5mm). Upper: transition counts for successive triplets for the entire data. Lower: transition counts for successive pairs for four sub-sequences of length 274.

5mm rain) and 3 (over 5mm). Triplets of transition counts for all 1096 observations are given in the upper part of Table 6.10; its lower part gives transition counts for successive pairs for sub-sequences 1–274, 275–548, 549–822 and 823–1096.

(a) The maximized log likelihoods for first-, second-, and third-order Markov chains fitted to the entire dataset are -1038.06 , -1025.10 , and -1005.56 . Compute the log likelihood for the zeroth-order model, and compare the four fits using likelihood ratio statistics and using AIC. Give the maximum likelihood estimates for the best-fitting model. Does it simplify to a varying-order chain?

(b) Matrices of transition counts $\{n_{irs}\}$ are available for m independent S -state chains with transition matrices $P_i = (p_{irs})$, $i = 1, \dots, m$. Show that the maximum likelihood estimates are $\hat{p}_{irs} = n_{irs}/n_{i\cdot}$, where $\cdot$ denotes summation over the corresponding index. Show that the maximum likelihood estimates under the simpler model in which $P_1 = \dots = P_m = (p_{rs})$ are $\hat{p}_{rs} = n_{rs}/n_{\cdot}$. Deduce that the likelihood ratio statistic to compare these models is $2 \sum_{i,r,s} n_{irs} \log(\hat{p}_{irs}/\hat{p}_{rs})$ and give its degrees of freedom.

(c) Consider the lower part of Table 6.10. Explain how to use the statistic from (b) to test for equal transition probabilities in each section, and hence check stationarity of the data.

- 2 The nematode *Steinernema feltiae* is a tiny worm used for biological control of mushroom fly larvae. Once one has found and penetrated a larva, it kills it by releasing bacteria, but death is not immediate and other nematodes may also penetrate the larva before it dies. In experiments to assess their effectiveness, m nematodes challenged a single healthy larva. Let $X_t \in \{0, \dots, m\}$ denote the number of nematodes that have invaded the larva at time t , and let $p_r(t) = \Pr(X_t = r)$, with initial condition $p_0(0) = 1$.

(a) If the invasion process is modelled as a continuous-time Markov process with transition probabilities independent of t , explain why we may write

$$\Pr(X_{t+\delta t} = r + 1 \mid X_t = r) = \lambda_r \delta t + o(\delta t), \quad t \geq 0, \quad r = 0, \dots, m - 1,$$

where $\lambda_m = 0$, and give an interpretation of λ_r . Deduce that

$$\frac{dp_0(t)}{dt} = -\lambda_0 p_0(t), \quad \frac{dp_{r+1}(t)}{dt} = -\lambda_{r+1} p_{r+1}(t) + \lambda_r p_r(t), \quad r = 0, \dots, m - 1.$$

If $\lambda_r = (m - r)\beta$ for some $\beta > 0$, verify that these equations have solution

$$p_r(t) = \binom{m}{r} \{1 - \exp(-\beta t)\}^r \exp(-\beta t)^{m-r},$$

and give its interpretation.

Table 6.11 Numbers of nematodes invading individual fly larvae for various initial numbers of challengers (Faddy and Fenlon, 1999).

Challengers m	Number of fly larvae with $r = 0, \dots, 10$ invading nematodes											Total
	0	1	2	3	4	5	6	7	8	9	10	
10	1	8	12	11	11	6	9	6	6	2	0	72
7	9	14	27	15	6	3	1	0				75
4	28	18	17	7	3							73
2	44	26	6									76
1	158	60										218

Table 6.12 Numbers of sites showing differences between introns of human and owl monkey insulin genes (Li, 1997, p. 83).

Human	Owl monkey			
	A	C	G	T
A	20	0	0	2
C	0	24	5	1
G	1	5	45	0
T	2	2	0	56

(b) A total of n independent experiments performed with $t = 1$ (in arbitrary units) gave data $(m_1, r_1), \dots, (m_n, r_n)$ shown in Table 6.11. Thus, for example, of the 72 larvae challenged by 10 nematodes, 1 was not penetrated, 8 were penetrated by just one nematode, 12 were penetrated by two nematodes, and so forth. Show that the corresponding log likelihood may be written as

$$\ell(\beta) = (s_m - s_r)\beta + s_r \log(1 - e^{-\beta}),$$

and deduce that β has maximum likelihood estimate $\hat{\beta} = \log\{s_m/(s_m - s_r)\}$ with standard error $[s_r/\{s_m(s_m - s_r)\}]^{1/2}$.

(c) Find the values of $\hat{\beta}$ and their standard errors for models in which the value of β is (i) the same for all m and (ii) different for each m . Discuss which fits the data better, given that the likelihood ratio statistic to compare them equals 11.2.

(d) A different model has $\lambda_r = (m - r) \exp(\gamma_0 + \gamma_1 r)$, so the larva's resistance to penetration changes each time it is invaded. What feature of Table 6.11 suggests that this model might be better? What difficulties would arise in fitting it?

- One way to estimate the evolutionary distance between species is to identify sections of their DNA which are similar and so must derive from a common ancestor species. If such sections differ at very few sites, the species are closely related and must have separated recently in the evolutionary past, but if the sections differ by more, the species are further apart. For example, data from the first introns of human and owl monkey insulin genes are in Table 6.12. The first row means that there are 20 sites with A on both genes, 0 with A on the human and C on the monkey, and so on. If all the data lay on the diagonal, this section would be identical in both species. Note that even if sites on both genes have the same base, there could have been changes such as (ancestor) A $\rightarrow$ G $\rightarrow$ T (human) and (ancestor) A $\rightarrow$ C $\rightarrow$ A $\rightarrow$ T (monkey).

Here is a (greatly simplified) model for evolutionary distance. We suppose that at a time t_0 in the past the two species we now see began to evolve away from a common ancestor species, which had a section of DNA of length n similar to those we now see. Each site on that section had one of the four bases A, C, G, or T, and for each species the base at each site has since changed according to a continuous-time Markov chain with infinitesimal

generator

$$G = \begin{pmatrix} -3\gamma & \gamma & \gamma & \gamma \\ \gamma & -3\gamma & \gamma & \gamma \\ \gamma & \gamma & -3\gamma & \gamma \\ \gamma & \gamma & \gamma & -3\gamma \end{pmatrix},$$

independent of other sites. That is, the rate at which one base changes into, or is *substituted* by, another is the same for any pair of bases.

In fact substitutions can be of various types, but we do not distinguish them here.

(a) Check that G has eigendecomposition

$$\frac{1}{4} \begin{pmatrix} 1 & -1 & -1 & -1 \\ 1 & -1 & -1 & 3 \\ 1 & -1 & 3 & -1 \\ 1 & 3 & -1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -4\gamma & 0 & 0 \\ 0 & 0 & -4\gamma & 0 \\ 0 & 0 & 0 & -4\gamma \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \end{pmatrix},$$

find its equilibrium distribution π , and show that the chain is reversible.

(b) Show that $\exp(tG)$ has diagonal elements $(1 + 3e^{-4\gamma t})/4$ and off-diagonal elements $(1 - e^{-4\gamma t})/4$. Use this and reversibility of the chain to explain why the likelihood for γ based on data like those above is proportional to

$$(1 + 3e^{-8\gamma t_0})^{n-R} (1 - e^{-8\gamma t_0})^R,$$

where R is the number of sites at which the two sections disagree. Hence find an estimate and standard error for γt_0 for the data above.

(c) Show that for each site, the probability of no substitution on either species in period t is $1 - \exp(-6\gamma t)$, deduce that substitutions occur as a Poisson process of rate 6γ , and hence show that the estimated mean number of substitutions per site for the data above is 0.120.

Discuss the fit of this model.

- 4 Let $Y_1, \dots, Y_n$ represent the trajectory of a stationary two-state discrete-time Markov chain, in which

$$\Pr(Y_j = a \mid Y_1, \dots, Y_{j-1}) = \Pr(Y_j = a \mid Y_{j-1} = b) = \theta_{ba}, \quad a, b = 1, 2;$$

note that $\theta_{11} = 1 - \theta_{12}$ and $\theta_{22} = 1 - \theta_{21}$, where θ_{12} and θ_{21} are the transition probabilities from state 1 to 2 and vice versa.

Show that the likelihood can be written in form $\theta_{12}^{n_{12}} (1 - \theta_{12})^{n_{11}} \theta_{21}^{n_{21}} (1 - \theta_{21})^{n_{22}}$, where n_{ab} is the number of $a \rightarrow b$ transitions in $y_1, \dots, y_n$. Find a minimal sufficient statistic for $(\theta_{12}, \theta_{21})$, the maximum likelihood estimates $\hat{\theta}_{12}$ and $\hat{\theta}_{21}$, and their asymptotic variances.

- 5 Let $Y_{(1)} < \dots < Y_{(n)}$ be the order statistics of a sample from the exponential density, $\lambda e^{-\lambda y}$, $y > 0$, $\lambda > 0$. Show that for $r = 2, \dots, n$,

$$\Pr(Y_{(r)} > y \mid Y_{(1)}, \dots, Y_{(r-1)}) = \exp\{-\lambda r(y - Y_{(r-1)})\}, \quad y > Y_{(r-1)},$$

and deduce that the order statistics from a general continuous distribution form a Markov process.

- 6 Let $\mathcal{G}$ denote an undirected graph with nodes $\mathcal{J}$ and for any $\mathcal{A} \subset \mathcal{J}$ let $\text{cl}(\mathcal{A})$ denote the set $\bigcup_{a \in \mathcal{A}} (\{a\} \cup \mathcal{N}_a)$. Then we can write the local, global and pairwise Markov properties as

(G) if $\mathcal{A}, \mathcal{B}, \mathcal{D}$ is a triple of disjoint sets such that $\mathcal{D}$ separates $\mathcal{A}$ from $\mathcal{B}$ in $\mathcal{G}$, then

$$Y_{\mathcal{A}} \perp Y_{\mathcal{B}} \mid Y_{\mathcal{D}};$$

(L) for any node a , $Y_a \perp Y_{\mathcal{J} - \text{cl}(\{a\})} \mid Y_{\mathcal{N}_a}$;

(P) if a, b are non-adjacent nodes, then $Y_a \perp Y_b \mid Y_{\mathcal{J} - \{a, b\}}$.

(a) Show that (G) $\Rightarrow$ (L) $\Rightarrow$ (P).

(b) We say that Y satisfies (F) if the density factorizes according to (6.14) and (6.15). Show that (F) $\Rightarrow$ (G). Interpret the Hammersley–Clifford theorem as showing that if in addition (6.12) holds, then (P) $\Rightarrow$ (F).

- 7 Consider a rectangular grid of pixels with a first-order neighbourhood structure, and denote its random variables by u_{ij} , $i, j = 1, \dots, m$. Suppose that the observed data are

$y_{ij} = u_{ij} + \varepsilon_{ij}$ where $\varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$. Thus the u_{ij} are observed with noise. Give the moral graph for the u_{ij} and y_{ij} . Hence show that the local characteristics $f(u_{ij} | y, u_{-ij})$ depends on the neighbouring u s and y_{ij} and find $f(u_{ij} | y, u_{-ij})$ when the u_{ij} follow an Ising model.

- 8 (a) Suppose that conditional on $U = u$, $Y \sim N_p(\mu, v u^{-1} \Omega)$, where $u \sim \chi_v^2$. Show that the marginal density of Y is multivariate t ,

$$f(y; \mu, \Omega) = \frac{\Gamma\left(\frac{p+v}{2}\right) |\Omega|^{-1/2}}{(\pi v)^{p/2} \Gamma\left(\frac{v}{2}\right)} \{1 + (y - \mu)^T \Omega^{-1} (y - \mu) / v\}^{-(p+v)/2},$$

and establish that $E(U | Y = y) = (v + p) / \{v + (y - \mu)^T \Omega^{-1} (y - \mu)\}$.

(b) Use this as the basis for an EM algorithm for estimation of μ and Ω , extending that of Problem 5.18.

(c) The density of Y is called elliptical because of the shape of its contours. Other such densities may be produced by supposing that $Y \sim N_p(\mu, u^{-1} \Omega)$ conditional on $U = u$ and letting $U \sim g$, where g has support in the positive half-line. What changes to the algorithm in (b) are then needed to produce an EM algorithm for estimation of μ and Ω ? (Section 5.5.2)

- 9 Show that the MA(1) models $Y_t = \varepsilon_t + \beta \varepsilon_{t-1}$ and $Y_t = \varepsilon_t + \beta^{-1} \varepsilon_{t-1}$ have the same correlations and deduce that they are indistinguishable from their correlograms alone. If $Y_t = (1 + \beta B) \varepsilon_t$ in terms of the backshift operator B , show that ε_t may be expressed as a linear combination of $Y_t, Y_{t-1}, \dots$ in which the infinite past has no effect only if $|\beta| < 1$. The ARMA process $a(B)Y_t = b(B)\varepsilon_t$ is said to be invertible if the zeros of the polynomial $b(z)$ all lie outside the unit disk. Show that the MA(1) process is invertible only if $|\beta| < 1$. Compare this with the condition for stationarity of the AR(1) model. Discuss.
- 10 Show that strict stationarity of a time series $\{Y_j\}$ means that for any r we have

$$\text{cum}(Y_{j_1}, \dots, Y_{j_r}) = \text{cum}(Y_0, \dots, Y_{j_r - j_1}) = \kappa^{j_2 - j_1, \dots, j_r - j_1},$$

say. Suppose that $\{Y_j\}$ is stationary with mean zero and that for each r it is true that $\sum_u |\kappa^{u_1, \dots, u_{r-1}}| = c_r < \infty$.

The r th cumulant of $T = n^{-1/2}(Y_1 + \dots + Y_n)$ is

$$\begin{aligned} \text{cum}\{n^{-1/2}(Y_1 + \dots + Y_n)\} &= n^{-r/2} \sum_{j_1, \dots, j_r} \text{cum}(Y_{j_1}, \dots, Y_{j_r}) \\ &= n^{-r/2} \sum_{j_1=1}^n \sum_{j_2, \dots, j_r} \kappa^{j_2 - j_1, \dots, j_r - j_1} \\ &= n \times n^{-r/2} \sum_{j_2, \dots, j_r} \kappa^{j_2 - j_1, \dots, j_r - j_1} \\ &\leq n^{1-r/2} \sum_{j_2, \dots, j_r} |\kappa^{j_2 - j_1, \dots, j_r - j_1}| \leq n^{1-r/2} c_r. \end{aligned}$$

Justify this reasoning, and explain why it suggests that T has a limiting normal distribution as $n \rightarrow \infty$, despite the dependence among the Y_j .

Obtain the cumulants of T for the MA(1) model, and convince yourself that your argument extends to the MA(q) model.

Can you extend the argument to arbitrary linear combinations of the Y_j ?

- 11 (a) Check that the Gumbel distribution arises from (6.34) in the limit as $\xi \rightarrow 0$.
 (b) Derive the densities for (6.34) and the Gumbel distribution, and plot them for $\xi = -1, -0.5, 0, 0.5$, and 1 . Which do you think is most plausible for extreme rainfall, for high tides, and for the fastest times to run a mile?
 (c) Write a function that generates random samples from (6.34) by inversion.
 (d) Show that the Gumbel plotting positions are $-\log[-\log\{1 - i/(n+1)\}]$ and use these and your simulation routine to see how easy it is to detect departures from $\xi = 0$ in random samples of size $n = 40$ with $\xi = -0.3, 0.3$. Try varying ξ and n , and write a brief account of your conclusions.

This condition applies to many common models, but excludes those where variables far apart are highly correlated.

- 12 Consider a stationary point process and denote the numbers of counts in successive intervals $(k\tau, (k+1)\tau]$ of length τ by N_k , where $k = \dots, -1, 0, 1, \dots$. Let $\text{var}(N_0) < \infty$ and set $\gamma_j = \text{cov}(N_0, N_j)$.
- (a) Show that $\{N_j\}$ is a stationary time series and deduce that

$$\text{var}\{N(m\tau)\} = m\gamma_0 + 2 \sum_{j=1}^{m-1} (m-j)\gamma_j, \quad m = 1, 2, \dots$$

Hence explain how the variance-time curve $V(t)$ for $t = \tau, 2\tau, \dots$ may be estimated using the empirical covariances $\hat{\gamma}_j$ of counts of data observed over $(0, t_0]$. Call the estimator $\hat{V}(t)$.

(b) If $k\tau = t_0$ and the data follow a Poisson process of rate λ , then

$$\text{E}(\hat{\gamma}_j) \doteq \begin{cases} (k-1)\lambda\tau/k, & j = 0, \\ 0, & \text{otherwise,} \end{cases} \quad \text{var}(\hat{\gamma}_j) = \begin{cases} \lambda\tau(2\lambda\tau + 1)/k + o(k^{-1}), & j = 0, \\ (\lambda\tau)^2/k + o(k^{-1}), & \text{otherwise,} \end{cases}$$

while $\text{cov}(\hat{\gamma}_i, \hat{\gamma}_j) = o(k^{-1})$ when $i \neq j$. Hence show that in this case $\text{E}\{\hat{V}(t)\} \doteq (1 - t/t_0)V(t)$ and

$$\text{var}\{\hat{V}(t)\} = \{2/3 + 4/(3m)\}(\lambda t)^2(t/t_0) + (\lambda t)(t/t_0) + o(\tau/t_0),$$

where $t = m\tau$.

(c) Explain the construction of the lower left panel of Figure 6.19.

- 13 Sampling of point processes is not straightforward. If the process is running already and sampling begins at an arbitrary time origin, then this origin is likely to fall into an interval that is longer than is typical, and this *length-biased sampling* has knock-on effects for subsequent intervals unless their lengths are independent. Suppose that a very long stretch of n intervals is available from a stationary process with mean interval length μ and marginal density $f(y)$ for times between events, into which the origin falls randomly. Of the total length $n\mu$ of the intervals, a length $nf(y) \times y$ will be taken by intervals of length y . Explain why the probability that the origin falls into one of these is $g(y)dy = nyf(y)dy/(n\mu)$, and hence show that the length of the selected interval has probability density g .

Now consider the *forward recurrence time* to the next event starting from the origin. The origin having fallen uniformly at random into an interval of length y , the conditional density of its position within that interval is y^{-1} . Show that the forward recurrence time has density

$$\int_x^\infty y^{-1}g(y)dy = \mu^{-1}\mathcal{F}(x),$$

where $\mathcal{F}$ is the survivor function of f , and find the density of the *backward recurrence time* to the point before the origin.

Show that in a homogeneous Poisson process of rate λ the interval into which the origin falls has density $\lambda^2 ye^{-\lambda y}$, $y > 0$, and that the forward and backward recurrence times are both exponential variables. Explain why these results are obvious intuitively.

- 14 A Poisson process of rate $\lambda(t)$ on the set $\mathcal{S} \subset \mathbb{R}^k$ is a collection of random points with the following properties (among others):
- the number of points $N_{\mathcal{A}}$ in a subset $\mathcal{A}$ of $\mathcal{S}$ has the Poisson distribution with mean $\Lambda(\mathcal{A}) = \int_{\mathcal{A}} \lambda(t) dt$;
 - given $N_{\mathcal{A}} = n$, the positions of the points are sampled randomly from the density $\lambda(t)/\int_{\mathcal{A}} \lambda(s) ds$, $t \in \mathcal{A}$.
- (a) Assuming that you have reliable generators of $U(0, 1)$ and Poisson variables, show how to generate the points of a Poisson process of constant rate λ on the interval $[0, t_0]$.
- (b) Let $t = (x, y) \in \mathbb{R}^2$, $\eta, \xi \in \mathbb{R}$, $\tau > 0$, $\lambda(x, y) = \tau^{-1} \{1 + \xi(y - \eta)/\tau\}^{-1/\xi - 1}$. Give an algorithm to generate realisations from the Poisson process with rate $\lambda(x, y)$ on

$$\mathcal{S} = \{(x, y) : 0 \leq x \leq 1, y \geq u, \lambda(x, y) > 0\}.$$

Table 6.13 Times (days) between successive failures of a piece of software developed as part of a large data system (Jelinski and Moranda, 1972). The software was released after the first 31 failures. The last three failures occurred after release. The data are to be read across rows.

9	12	11	4	7	2	5	8	5	7	1	6	1	9	4	1	3
3	6	1	11	33	7	91	2	1	87	47	12	9	135	258	16	35

- 15 Show that the likelihood for data $(t_1, y_1), \dots, (t_n, y_n)$ observed in $[0, t_0] \times [u, \infty)$ and with intensity (6.36) is

$$\prod_{j=1}^n \tau^{-1} \left(1 + \xi \frac{y_j - \eta}{\tau} \right)^{-1/\xi - 1} \times \exp \left\{ -t_0 \left(1 + \xi \frac{u - \eta}{\tau} \right)^{-1/\xi} \right\}.$$

Show that this may be reparametrized to give (6.39) and that this is the log likelihood corresponding to a decomposition

$$\Pr(N = n; \lambda) \times \prod_{j=1}^n g(w_j; \xi, \sigma).$$

Give the distributions of N , of the W_j , and of $Y = \max(W_1, \dots, W_N)$. Surprised?

- 16 A computer program has an unknown number of bugs m . Each bug causes the program to crash, and is then located and (instantaneously!) removed. If the times at which the m failures occur are independent exponential variables with common mean β^{-1} , and if m is Poisson with mean μ/β , then show that

$$\Pr\{N(t) = 0\} = \exp\{-\mu(1 - e^{-\beta t})/\beta\}, \quad t \geq 0.$$

(a) Deduce that the times of crashes follow a Poisson process of rate $\mu e^{-\beta t}$. Show that the likelihood when failures occur at times $0 \leq t_1 < \dots < t_n \leq t_0$ is

$$L(\mu, \beta) = \mu^n \exp \left\{ -\beta \sum_{j=1}^n t_j - \mu \beta^{-1} (1 - e^{-\beta t_0}) \right\},$$

and that this is an exponential family model.

- (b) Reliability growth occurs if $\beta > 0$. Show that a test for this may be based on the conditional distribution of $S = \sum T_j$ given that n failures have occurred in $[0, t_0]$, and that if $\beta = 0$, $E(S) = nt_0/2$ and $\text{var}(S) = nt_0^2/12$. Suggest how to perform such a test.
 (c) We now treat m as an unknown parameter and aim to estimate it. Show that

$$L(m, \beta) = \frac{m!}{(m-n)!} \beta^n \exp\{-\beta t_0(m + s/t_0 - n)\}, \quad \beta > 0, m = n, n+1, \dots,$$

and hence find the profile log likelihood $\ell_p(m)$ for m .

- (d) The code below plots $\ell_p(m)$ after the first r failures of the data in Table 6.13. Try varying r up to 30, and observe the shapes taken by the profile log likelihood.

```
y <- c(9,12,11,4,7,2,5,8,5,7,1,6,1,9,4,1,3,3,
6,1,11,33,7,91,2,1,87,47,12,9,135,258,16,35)
L <- function(m,n,s) lgamma(m+1)-lgamma(m-n+1) - n*log(m-n+s)
r <- 20 # just take data up to time of rth failure
y <- cumsum(y[1:r])
s <- sum(y)/y[r]
x <- r:(r+100)
plot(x,L(x,r,s)) # plot log likelihood
```

What problems do you see with likelihood inference for this model?

The software was released after 31 failures. Give the maximum likelihood estimate of m at that point, and its confidence interval, if possible.

(Section 6.5.1, Jelinski and Moranda, 1972)